# Spirit Gate — alterações de código

Relatório técnico da entrega de raças, domínios e launcher. Emitido em 21/10/2026.

## 1. Escopo e referência

Este relatório descreve o commit 07accf07e0a301207bf4ff253a222b65ac4095b9 ("adicao das racas") em comparação com seu pai f9b5de1eedb4c6f745d4767e0e75eb3413600149. O diff é extraído diretamente do Git, não reconstruído a partir da conversa.

Foram identificados 91 arquivos alterados: 81 arquivos de código, configuração, scripts e testes, mais 10 documentos e evidências. O inventário inclui arquivos adicionados, modificados e removidos. O anexo do PDF contém o diff integral dos 81 arquivos técnicos, com as linhas adicionadas e removidas e o contexto original.

O PDF de raças e domínios é o documento de design. Este é o documento de implementação. O patch separado preserva os mesmos bytes do diff para consulta por ferramentas. Linhas longas do anexo são quebradas apenas para caber na página.

## 2. Modelo de personagem e persistência

CharacterRaces centraliza Reaper, Human, Aetheri e Spirit e seus fundamentos iniciais. CharacterIdentity passa a validar raças atuais; a criação rejeita Guardian/Mystic como novas opções, enquanto o construtor permite sua normalização ao ler identidades antigas. O acesso race() foi acrescentado e o componente archetype permanece por compatibilidade do snapshot.

Domains cadastra dez caminhos com raça, descrição e técnicas para prévia: quatro de Reaper, dois humanos, dois de arco Aetheri e dois de trevas Spirit. DomainProgression mantém um único domínio, a escolha a partir do nível 15, XP em meias unidades, data do reset e conclusão da quest. Não permite alternância normal entre domínios.

O save V2 grava race explicitamente. LocalSaveStore aceita V1, migra Guardian para Reaper e Mystic para Human e registra a migração, preservando identidade, atributos-base, XP, itens, equipamento e quests. RpgSave valida compatibilidade racial e nível mínimo do domínio. CharacterSaves apresenta os nomes das raças no catálogo.

Antes da escolha, não se acumula XP de domínio. Depois dela, a implementação recebe metade da recompensa geral no domínio. A proporção foi confirmada para o mob de nível 1; seu uso em quests continua provisório. Existe armazenamento de XP de domínio, mas ainda não existe cálculo do nível de domínio nem requisitos finais por técnica.

## 3. Combate, progressão e reset

RpgWorld inicia sem domínio: Reaper recebe small_sword de poder adicional zero, Human/Aetheri começam com socos e Spirit com feixe luminoso básico. Qualquer raça pode optar por socos sem ganhar especialização marcial. Feixe básico de Spirit não concede domínio de luz.

O ataque básico de Aetheri passa a usar alcance de 280 pixels ao escolher bow_long e 140 em bow_short. Os demais ataques iniciais usam 82 pixels para Reaper, 60 para socos e 180 para Spirit. Esses valores são provisórios. As árvores e habilidades elementais finais não estão implementadas pelas prévias.

Shunpo ofensivo e defensivo exigem nível 20 em todas as raças e compartilham recarga. O ofensivo valida alvo visível, alcance e destino; o defensivo recua e cria uma janela de esquiva. O estado de recarga, vigor, direção e esquiva é preservado ao salvar. RpgSecondaryAbility foi adaptado aos metadados do Shunpo defensivo; os antigos ataques de arquétipo foram retirados do fluxo atual.

Parâmetros provisórios do Shunpo: custo 25 de vigor, recarga de 90 ticks, alcance ofensivo 240 pixels, recuo defensivo 70 pixels e janela de 6 ticks, com simulação de 30 Hz. Apenas o requisito de nível 20 está fechado pelo design.

EnemyService adiciona nível explícito, com compatibilidade para definições antigas. Os mobs atuais são tratados como nível 1 e recompensam 3 XP geral e 1,5 de domínio após a escolha. Níveis de mob acima de 1 são recusados enquanto suas recompensas não forem definidas. A curva geral continua provisória; cap=0 retira o teto de design, mantendo 1.000.000 XP como proteção técnica do save.

DomainResetPolicy define 35 Spirit Coins, equivalentes a R$ 35. DomainProgression exige quest concluída, saldo suficiente e um mês de calendário desde o último reset; zera o domínio e suas técnicas, preservando a base. A validação é uma regra de domínio, sem checkout ou débito financeiro real integrado. O cliente informa essa indisponibilidade. Conteúdo da quest e serviço de carteira continuam pendentes; o intervalo por personagem é uma interpretação provisória.

Foram corrigidos o evento de morte repetido em alvo já morto, a redução indevida de XP em recompensa e o alvo oculto mais próximo impedindo ataque a outro visível. A atribuição de XP geral/domínio valida as duas parcelas antes de alterar o estado, evitando atualização parcial em caso de overflow.

## 4. Interface e World Studio

CharacterCreation troca os dois arquétipos por quatro raças, apresenta o início como pequena alma/aprendiz e informa a ausência de domínio. CharacterSelection mostra raça, domínio e XP fracionária. RpgMenu remove a entrada de arena do menu atual sem remover o motor que sustenta outras ferramentas.

DomainPreview adiciona demonstrações Java2D animadas dos dez caminhos e lista técnicas de conceito. A consulta não concede XP nem desbloqueios. A escolha é habilitada apenas no nível 15 e sem domínio atual. Foi corrigido um problema de primeiro layout que podia ocultar a cena da prévia.

RpgLauncher integra prévias, escolha, salvamento e mensagens de reset, e atualiza a barra de ações para Shunpo e socos. RpgHud, RpgCombatStatus e GettingStarted refletem raça, domínio, XP e nível mínimo do Shunpo. RpgPlayerSprites, RpgVisualState e RpgCombatFeedback diferenciam feixes, arco e socos por eventos reais, sem criar regras de combate no desenho.

StudioExport e StudioPlaytest passam a criar Reaper. Foi removido o vínculo duplicado da tecla F no playtest. Editor, colisão, movimento e testes de mapa continuam preservados.

## 5. Launcher, distribuição e limpeza

Os cinco inicializadores Windows usam o fluxo compartilhado. compile-local.ps1 verifica JDK 17+, hashes de fontes/recursos/saídas e recompila para --release 17 quando necessário. A saída é isolada em server/target/launcher/classes. Uma falha de compilação impede o uso silencioso de classes antigas.

launch-local.ps1 seleciona cliente RPG, sprites, editor ou mapa. Os scripts portáteis de RPG, sprites e Studio compilam as fontes atuais e ajustam módulos do runtime ao JDK disponível. O empacotamento RPG aceita reutilização de runtime previamente validado por binários e módulos.

LauncherSelfCheck e a opção --verify-install verificam quatro raças, dez domínios, bloqueios iniciais e ciclo do save V2 sem janela nem personagens pessoais. run-local-tests.py executa JUnit Jupiter com dependências já presentes no cache, sem Maven ou rede. gerar_pdf_dominios.py gera o PDF de design.

Foram removidos KServer, Kclient, Network, DatabaseManager, Main, Archetypes e archetypes.properties, além das dependências Kryonet e PostgreSQL do POM. Permanecem o motor compartilhado, os testes úteis e o World Studio. A limpeza não apagou saves pessoais nem repositórios históricos de evidências.

O laboratório web anterior foi restrito ao proprietário por uma operação externa de Sites. Essa alteração de acesso está registrada na documentação, mas não é uma alteração de código no patch. O conector não oferecia exclusão definitiva.

## 6. Testes e evidências da entrega

Os registros da entrega em 21/10/2026 informam 191 fontes Java e 144 arquivos de teste compilados, 413 testes Java aprovados, 6 ignorados/abortados e zero falhas. A suíte Python registrou 30 testes aprovados. Esses números documentam a validação anterior; a geração deste PDF não reexecutou as suítes.

Os testes alterados e novos cobrem criação e seleção, prévias e layout, XP fracionária, bloqueio de escolha/troca, reset e calendário, preservação da base, migração V1/V2, Shunpo e recursos, alvo visível, evento de morte e compatibilidade com Studio. Os probes visuais e de jornada foram atualizados às novas raças e ações.

VALIDACAO-LAUNCHER-RACAS.json registra pacote 1.0.0-rc1 com Java integrado, 148 arquivos conferidos por tamanho/SHA-256, extração nova em caminho com espaços e retorno 0 do executável. O resultado informa quatro raças, dez domínios, saveSchema=2 e paymentEnabled=false. O compilador foi javac 24.0.2 com alvo Java 17.

Naquela sessão, a execução Java precisou ocorrer fora das restrições do sandbox por bloqueio de leitura de classpath/java.security. Isso não equivale a homologação em outro Windows nem a lançamento comercial.

## 7. Limites e próximos trabalhos

Ainda faltam o percurso jogável da pequena alma, árvores e skills finais, cálculo do nível de domínio, requisitos de técnicas, conteúdo da quest de reset, carteira e pagamentos reais, balanceamento e homologação de lançamento. As prévias não executam as skills finais no mundo de combate.

Controle poder atingir aliados da mesma party está definido no design, mas depende das skills e da integração multiplayer. A luz não ganha vantagem automática contra trevas. O cliente atual permanece local; o pacote validado é um candidato de desenvolvimento.

## 8. Inventário completo

A = adicionado; M = modificado; D = removido. + e - representam linhas do diff. Arquivos binários usam o marcador do Git. As dez entradas de documentação aparecem aqui por rastreabilidade; seu conteúdo completo não integra o anexo de código.

## Documentacao e evidencias — 10 arquivos

| Estado | Arquivo | + | - |
|---|---|---|---|
| M | README-PROTOTIPO.md | 4 | 358 |
| M | README-RPG.md | 12 | 4 |
| A | docs/AUDITORIA-PRE-CLASSES.md | 170 | 0 |
| A | docs/ENTREGA-RACAS-LAUNCHER.md | 97 | 0 |
| M | docs/EXPERIMENTO-NAVEGADOR.md | 2 | 0 |
| M | docs/ROTEIRO-RPG-OFFLINE.md | 16 | 19 |
| A | docs/SISTEMA-DE-RACAS-E-DOMINIOS.md | 195 | 0 |
| A | docs/Spirit_Gate_Racas_e_Dominios.pdf | 169 | 0 |
| A | docs/VALIDACAO-LAUNCHER-RACAS-CLI.json | 1 | 0 |
| A | docs/VALIDACAO-LAUNCHER-RACAS.json | 63 | 0 |

## Inicializadores Windows — 5 arquivos

| Estado | Arquivo | + | - |
|---|---|---|---|
| M | jogar-rpg-offline.cmd | 3 | 7 |
| M | rodar-prototipo.cmd | 5 | 32 |
| M | testar-mapa-rpg.cmd | 3 | 3 |
| M | testar-rpg-sprites.cmd | 3 | 9 |
| M | world-studio.cmd | 2 | 6 |

## Dependencias Maven — 1 arquivos

| Estado | Arquivo | + | - |
|---|---|---|---|
| M | server/pom.xml | 1 | 12 |

## Java de producao — 34 arquivos

| Estado | Arquivo | + | - |
|---|---|---|---|
| D | server/src/main/java/com/projetok/DatabaseManager.java | 0 | 30 |
| D | server/src/main/java/com/projetok/KServer.java | 0 | 71 |
| D | server/src/main/java/com/projetok/Kclient.java | 0 | 29 |
| D | server/src/main/java/com/projetok/Main.java | 0 | 11 |
| D | server/src/main/java/com/projetok/Network.java | 0 | 46 |
| M | server/src/main/java/com/projetok/domain/combat/CombatService.java | 4 | 1 |
| M | server/src/main/java/com/projetok/prototype/rpg/CharacterCreation.java | 51 | 34 |
| M | server/src/main/java/com/projetok/prototype/rpg/CharacterSelection.java | 13 | 11 |
| A | server/src/main/java/com/projetok/prototype/rpg/DomainPreview.java | 169 | 0 |
| M | server/src/main/java/com/projetok/prototype/rpg/GettingStarted.java | 1 | 1 |
| A | server/src/main/java/com/projetok/prototype/rpg/LauncherSelfCheck.java | 28 | 0 |
| M | server/src/main/java/com/projetok/prototype/rpg/RpgCombatFeedback.java | 18 | 3 |
| M | server/src/main/java/com/projetok/prototype/rpg/RpgCombatStatus.java | 3 | 3 |
| M | server/src/main/java/com/projetok/prototype/rpg/RpgHud.java | 2 | 1 |
| M | server/src/main/java/com/projetok/prototype/rpg/RpgLauncher.java | 24 | 5 |
| M | server/src/main/java/com/projetok/prototype/rpg/RpgMenu.java | 0 | 1 |
| M | server/src/main/java/com/projetok/prototype/rpg/RpgPlayerSprites.java | 4 | 4 |
| M | server/src/main/java/com/projetok/prototype/rpg/RpgVisualState.java | 1 | 1 |
| D | server/src/main/java/com/projetok/rpg/Archetypes.java | 0 | 21 |
| M | server/src/main/java/com/projetok/rpg/CharacterIdentity.java | 6 | 4 |
| A | server/src/main/java/com/projetok/rpg/CharacterRaces.java | 34 | 0 |
| M | server/src/main/java/com/projetok/rpg/CharacterSaves.java | 1 | 1 |
| A | server/src/main/java/com/projetok/rpg/DomainProgression.java | 83 | 0 |
| A | server/src/main/java/com/projetok/rpg/DomainResetPolicy.java | 14 | 0 |
| A | server/src/main/java/com/projetok/rpg/Domains.java | 43 | 0 |
| M | server/src/main/java/com/projetok/rpg/EnemyService.java | 6 | 3 |
| M | server/src/main/java/com/projetok/rpg/LocalSaveStore.java | 9 | 5 |
| M | server/src/main/java/com/projetok/rpg/RpgCombatEvent.java | 1 | 1 |
| M | server/src/main/java/com/projetok/rpg/RpgProgression.java | 9 | 5 |
| M | server/src/main/java/com/projetok/rpg/RpgSave.java | 2 | 1 |
| M | server/src/main/java/com/projetok/rpg/RpgSecondaryAbility.java | 4 | 7 |
| M | server/src/main/java/com/projetok/rpg/RpgWorld.java | 81 | 26 |
| M | server/src/main/java/com/projetok/studio/StudioExport.java | 1 | 1 |
| M | server/src/main/java/com/projetok/studio/StudioPlaytest.java | 3 | 3 |

## Recursos e regras — 4 arquivos

| Estado | Arquivo | + | - |
|---|---|---|---|
| D | server/src/main/resources/rpg/archetypes.properties | 0 | 2 |
| M | server/src/main/resources/rpg/content.properties | 1 | 0 |
| M | server/src/main/resources/rpg/items.properties | 1 | 0 |
| M | server/src/main/resources/rpg/progression.properties | 1 | 1 |

## Testes Java — 26 arquivos

| Estado | Arquivo | + | - |
|---|---|---|---|
| A | server/src/test/java/com/projetok/domain/combat/CombatServiceDeathTest.java | 31 | 0 |
| M | server/src/test/java/com/projetok/prototype/rpg/AmbientFeedbackTest.java | 1 | 1 |
| M | server/src/test/java/com/projetok/prototype/rpg/CharacterCreationTest.java | 11 | 5 |
| M | server/src/test/java/com/projetok/prototype/rpg/CharacterFactsTest.java | 1 | 1 |
| M | server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionTest.java | 2 | 2 |
| M | server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionVisualTest.java | 5 | 4 |
| M | server/src/test/java/com/projetok/prototype/rpg/CorujaoHudTest.java | 14 | 6 |
| A | server/src/test/java/com/projetok/prototype/rpg/DomainPreviewTest.java | 43 | 0 |
| M | server/src/test/java/com/projetok/prototype/rpg/GettingStartedTest.java | 2 | 1 |
| M | server/src/test/java/com/projetok/rpg/ArchetypesTest.java | 12 | 4 |
| M | server/src/test/java/com/projetok/rpg/CharacterIdentityTest.java | 2 | 2 |
| M | server/src/test/java/com/projetok/rpg/CorujaoAbilitiesTest.java | 29 | 18 |
| M | server/src/test/java/com/projetok/rpg/CorujaoCombatRegressionTest.java | 8 | 8 |
| A | server/src/test/java/com/projetok/rpg/DomainProgressionTest.java | 83 | 0 |
| M | server/src/test/java/com/projetok/rpg/DungeonCycleTest.java | 1 | 1 |
| M | server/src/test/java/com/projetok/rpg/EnemyRpgTest.java | 2 | 2 |
| M | server/src/test/java/com/projetok/rpg/EquipmentRpgTest.java | 1 | 1 |
| M | server/src/test/java/com/projetok/rpg/ItemCatalogTest.java | 1 | 1 |
| A | server/src/test/java/com/projetok/rpg/LegacyRaceMigrationTest.java | 30 | 0 |
| M | server/src/test/java/com/projetok/rpg/LocalSaveStoreTest.java | 1 | 1 |
| M | server/src/test/java/com/projetok/rpg/OfflineCycleTest.java | 1 | 1 |
| M | server/src/test/java/com/projetok/rpg/RpgProgressionTest.java | 8 | 2 |
| A | server/src/test/java/com/projetok/rpg/RpgRaceRuntimeTest.java | 76 | 0 |
| M | server/src/test/java/com/projetok/studio/StudioFinalRegressionTest.java | 1 | 1 |
| M | server/src/test/java/com/projetok/studio/StudioIntegrationTest.java | 1 | 1 |
| M | server/src/test/java/com/projetok/studio/StudioRuntimePlaytestTest.java | 1 | 1 |

## Ferramentas e empacotamento — 11 arquivos

| Estado | Arquivo | + | - |
|---|---|---|---|
| M | tools/SpiritGateCreationProbe.java | 3 | 3 |
| M | tools/SpiritGateFinalJourneyProbe.java | 8 | 6 |
| M | tools/SpiritGateSelectionProbe.java | 1 | 1 |
| M | tools/SpiritGateSelectionVisualProbe.java | 1 | 1 |
| M | tools/build-rpg-portable.ps1 | 33 | 6 |
| M | tools/build-rpg-sprites-portable.ps1 | 9 | 5 |
| M | tools/build-world-studio.ps1 | 7 | 3 |
| A | tools/compile-local.ps1 | 112 | 0 |
| A | tools/gerar_pdf_dominios.py | 100 | 0 |
| A | tools/launch-local.ps1 | 40 | 0 |
| A | tools/run-local-tests.py | 137 | 0 |

# Anexo — diff integral de código

Cada bloco reproduz o diff do Git. + adiciona; - remove; espaço indica contexto. As quebras de linhas longas são somente de apresentação. Consulte o arquivo .patch para os bytes originais.

### 01. [M] jogar-rpg-offline.cmd

```
diff --git a/jogar-rpg-offline.cmd b/jogar-rpg-offline.cmd
index 3f225ebef22bbae6ad3ca7b0a751eff5f8d4a3dd..674e25f62888bb63db7f1161f9d17c46a0621227 100644
--- a/jogar-rpg-offline.cmd
+++ b/jogar-rpg-offline.cmd
@@ -1,8 +1,4 @@
 @echo off
-setlocal
-cd /d "%~dp0"
-if exist "server\target\classes\com\projetok\prototype\rpg\SpiritGateEntry.class" (
-  start "Spirit Gate RPG offline" javaw -cp "server\target\classes" com.projetok.prototype.rpg.RpgLauncher
-  exit /b
-)
-start "Spirit Gate RPG offline" "%~dp0dist\RPG-Offline-0.66.1-rc1\SpiritGateRPG\SpiritGateRPG.exe"
+REM Alias mantido para atalhos existentes.
+call "%~dp0rodar-prototipo.cmd" %*
+exit /b %ERRORLEVEL%
```

### 02. [M] rodar-prototipo.cmd

```
diff --git a/rodar-prototipo.cmd b/rodar-prototipo.cmd
index 6c07cf28406a4f27510ed93cc2d267b50b4da707..124e08289033e5e53142b7ed02a643676d3ef6b6 100644
--- a/rodar-prototipo.cmd
+++ b/rodar-prototipo.cmd
@@ -1,34 +1,7 @@
 @echo off
-REM Spirit Gate - inicializador didatico da arena visual para Windows.
-REM Mantem a execucao dentro da pasta server onde esta o pom.xml.
 setlocal
-cd /d "%~dp0server"
-
-REM 1. Verifica se o Maven esta configurado no PATH.
-where mvn >nul 2>nul
-if errorlevel 1 (
-    echo ERRO: Maven nao encontrado. Execute mvn -version e revise o PATH.
-    pause
-    exit /b 1
-)
-
-REM 2. Executa os testes JUnit 5 antes de abrir a janela.
-echo [Spirit Gate] Compilando e executando testes...
-call mvn test
-if errorlevel 1 (
-    echo.
-    echo ERRO: Testes ou compilacao falharam. Veja o primeiro ERROR no terminal.
-    pause
-    exit /b 1
-)
-
-REM 3. Inicia o prototipo local (independente do banco de dados).
-echo.
-echo [Spirit Gate] Abrindo a arena 2D...
-call mvn exec:java -Dexec.mainClass=com.projetok.prototype.DebugLauncher
-if errorlevel 1 (
-    echo ERRO: Falha ao abrir a janela. Verifique a mensagem acima.
-    pause
-    exit /b 1
-)
-endlocal
+REM Entrada principal do cliente; compila fontes alteradas sem Maven ou rede.
+powershell.exe -NoProfile -ExecutionPolicy Bypass -File "%~dp0tools\launch-local.ps1" -Mode Rpg %*
+set "SPIRIT_GATE_EXIT=%ERRORLEVEL%"
+if not "%SPIRIT_GATE_EXIT%"=="0" pause
+exit /b %SPIRIT_GATE_EXIT%
```

### 03. [M] server/pom.xml

```
diff --git a/server/pom.xml b/server/pom.xml
index 826be1dc52e9bd0b49ca1fc45a97ecd7ef4a17c2..337b90ce1429ff7a00e9c22284391536182f6ddd 100644
--- a/server/pom.xml
+++ b/server/pom.xml
@@ -9,17 +9,6 @@
     <version>1.0-SNAPSHOT</version>

     <dependencies>
-        <dependency>
-            <groupId>com.esotericsoftware</groupId>
-            <artifactId>kryonet</artifactId>
-            <version>2.22.0-RC1</version>
-        </dependency>
-
-        <dependency>
-            <groupId>org.postgresql</groupId>
-            <artifactId>postgresql</artifactId>
-            <version>42.7.2</version>
-        </dependency>
         <!-- Dependências utilizadas somente pelos testes automatizados. -->
         <dependency>
             <groupId>org.junit.jupiter</groupId>
@@ -49,4 +38,4 @@
             </plugin>
         </plugins>
     </build>
-</project>
\ No newline at end of file
+</project>
```

### 04. [D] server/src/main/java/com/projetok/DatabaseManager.java

```
diff --git a/server/src/main/java/com/projetok/DatabaseManager.java
b/server/src/main/java/com/projetok/DatabaseManager.java
deleted file mode 100644
index c61c2138404e35f4a4408a6a1c1e3c0e3f1e53c5..0000000000000000000000000000000000000000
--- a/server/src/main/java/com/projetok/DatabaseManager.java
+++ /dev/null
@@ -1,30 +0,0 @@
-package com.projetok;
-
-import java.sql.Connection;
-import java.sql.DriverManager;
-import java.sql.SQLException;
-
-/**
- * Adaptador JDBC legado: dados de conexão são recebidos do ambiente, jamais do código.
- * O protótipo visual não usa PostgreSQL; este adaptador permanece para o login antigo.
- */
-public final class DatabaseManager {
-    private DatabaseManager() { }
-
-    /** Abre uma conexão JDBC utilizando credenciais externas ao repositório. */
-    public static Connection getConnection() throws SQLException {
-        String url = required("SPIRIT_DB_URL");
-        String username = required("SPIRIT_DB_USER");
-        String password = required("SPIRIT_DB_PASSWORD");
-        return DriverManager.getConnection(url, username, password);
-    }
-
-    /** Recusa configurações ausentes para evitar conexão inesperada com outro banco. */
-    private static String required(String name) throws SQLException {
-        String value = System.getenv(name);
-        if (value == null || value.isBlank()) {
-            throw new SQLException("Variavel de ambiente obrigatoria ausente: " + name);
-        }
-        return value;
-    }
-}
```

### 05. [D] server/src/main/java/com/projetok/KServer.java

```
diff --git a/server/src/main/java/com/projetok/KServer.java b/server/src/main/java/com/projetok/KServer.java
deleted file mode 100644
index ddd93e4e73307c964ccdb601428755b6e873c2d5..0000000000000000000000000000000000000000
--- a/server/src/main/java/com/projetok/KServer.java
+++ /dev/null
@@ -1,71 +0,0 @@
-package com.projetok;
-
-import java.io.IOException;
-import java.sql.PreparedStatement;
-import java.sql.ResultSet;
-import java.sql.SQLException;
-
-import com.esotericsoftware.kryonet.Connection; // Importante: Importar Connection da rede
-import com.esotericsoftware.kryonet.Listener;
-import com.esotericsoftware.kryonet.Server;
-
-public class KServer {
-
-    public static void main(String[] args) throws IOException {
-        // Teste de conexão com banco
-        try (java.sql.Connection conn = DatabaseManager.getConnection()) {
-            System.out.println("Conexão com o banco de dados estabelecida com sucesso!");
-        } catch (SQLException e) {
-            System.err.println("Erro ao conectar no banco: " + e.getMessage());
-        }
-
-        Server server = new Server();
-        Network.register(server);
-
-        Runtime.getRuntime().addShutdownHook(new Thread(() -> {
-            System.out.println("Parando servidor...");
-            server.stop();
-        }));
-
-        server.addListener(new Listener() {
-            @Override
-            public void received(Connection connection, Object object) {
-                if (object instanceof Network.LoginRequest) {
-                    Network.LoginRequest request = (Network.LoginRequest) object;

-
-                    // Aqui chamamos o método que está lá embaixo
-                    if (validarUsuario(request.username, request.password)) {
-                        System.out.println("Usuário " + request.username + " logou!");
-                        // connection.sendTCP(new Network.LoginSuccess()); // Adicione a classe LoginSuccess no
 Network
-                    } else {
-                        System.out.println("Login falhou para " + request.username);
-                    }
-                }
-            }
-        });
-
-        server.start();
-        server.bind(54555, 54777);
-        System.out.println("Spirit Gate Server iniciado com sucesso na porta 54555!");
-    }
-
-    // O método de validação fica aqui, limpo e organizado
-    public static boolean validarUsuario(String username, String password) {
-        String query = "SELECT password_hash FROM users WHERE username = ?";
-
-        try (java.sql.Connection conn = DatabaseManager.getConnection(); PreparedStatement pstmt =
conn.prepareStatement(query)) {
-
-            pstmt.setString(1, username);
-
-            try (ResultSet rs = pstmt.executeQuery()) {
-                if (rs.next()) {
-                    String hashNoBanco = rs.getString("password_hash");
-                    return password.equals(hashNoBanco);
-                }
-            }
-        } catch (SQLException e) {
-            System.err.println("Erro ao consultar o banco: " + e.getMessage());
-        }
```

```
-        return false;
-    }
-}
```

### 06. [D] server/src/main/java/com/projetok/Kclient.java

```
diff --git a/server/src/main/java/com/projetok/Kclient.java b/server/src/main/java/com/projetok/Kclient.java
deleted file mode 100644
index e4a9c64a6963af98d9a7e0aadd9948451ec1931f..0000000000000000000000000000000000000000
--- a/server/src/main/java/com/projetok/Kclient.java
+++ /dev/null
@@ -1,29 +0,0 @@
-package com.projetok;
-
-import java.io.IOException;
-
-import com.esotericsoftware.kryonet.Client;
-import com.esotericsoftware.kryonet.Listener;
-
-public class Kclient {
-
-    public static void main(String[] args) throws IOException {
-        Client client = new Client();
-        client.start();
-        Network.register(client);
-        client.connect(5000, "localhost", 54555, 54777);
-        client.addListener(new Listener() {
-            public void received(com.esotericsoftware.kryonet.Connection connection, Object object) {
-                if (object instanceof Network.LoginSuccess) {
-                    System.out.println("SUCESSO: O servidor aceitou o login!");
-                } else if (object instanceof Network.LoginFailed) {
-                    System.out.println("FALHA: " + ((Network.LoginFailed) object).motivo);
-                }
-            }
-        });
-        Network.LoginRequest login = new Network.LoginRequest();
-        login.username = "Charles";
-        login.password = "12345";
-        client.sendTCP(login);
-    }
-}
```

### 07. [D] server/src/main/java/com/projetok/Main.java

```
diff --git a/server/src/main/java/com/projetok/Main.java b/server/src/main/java/com/projetok/Main.java
deleted file mode 100644
index ebe9b151028e89a826f043e819e06a3e839ee491..0000000000000000000000000000000000000000
--- a/server/src/main/java/com/projetok/Main.java
+++ /dev/null
@@ -1,11 +0,0 @@
-package com.projetok;
-
-public class Main {
-
-    public static void main(String[] args) {
-        System.out.println("Hello world!");
-
-        /*PlayerConfiguration config = new PlayerConfiguration();
-meuHuman.levelUp(config);*/
-    }
-}
```

### 08. [D] server/src/main/java/com/projetok/Network.java

```
diff --git a/server/src/main/java/com/projetok/Network.java b/server/src/main/java/com/projetok/Network.java
deleted file mode 100644
index d31f311fa020766fefc230563c96213d8f61cc0b..0000000000000000000000000000000000000000
--- a/server/src/main/java/com/projetok/Network.java
+++ /dev/null
@@ -1,46 +0,0 @@
-package com.projetok;
-
-import com.esotericsoftware.kryonet.EndPoint;
-
-public class Network {
-
-    public static void register(EndPoint endPoint) {
-        endPoint.getKryo().register(LoginRequest.class);
-        endPoint.getKryo().register(LoginSuccess.class);
-        endPoint.getKryo().register(LoginFailed.class);
-        endPoint.getKryo().register(PlayerMovement.class);
-        endPoint.getKryo().register(ChatMessage.class);
-    }
-
-    public static class LoginRequest {
-
-        public String username;
-        public String password;
-    }
-
-    public static class PlayerMovement {
-
-        public float x, y;
-    }
-
-    public static class ChatMessage {
-
-        public String text;
-    }
-
-    public static class LoginSuccess {
-    }
-
-    public static class LoginFailed {

-
-        public String motivo;
-
-        public LoginFailed() {
-        }
-
-        public LoginFailed(String motivo) {
-            this.motivo = motivo;
-        }
-    }
-
-}
```

### 09. [M] server/src/main/java/com/projetok/domain/combat/CombatService.java

```
diff --git a/server/src/main/java/com/projetok/domain/combat/CombatService.java
b/server/src/main/java/com/projetok/domain/combat/CombatService.java
index 9566177c1f0327c29d1ed1a975173443e716986d..9ee5a45d2e2e72154debed89378371d560fee348 100644
--- a/server/src/main/java/com/projetok/domain/combat/CombatService.java
+++ b/server/src/main/java/com/projetok/domain/combat/CombatService.java
@@ -9,7 +9,7 @@ import com.projetok.domain.event.events.EntityDeathEvent;
 import java.util.Objects;

 /**
- * Único caminho de dano do jogo. Substitui Entity.currentDefense() + CombatCalculator +
+ * Caminho de dano das entidades de domínio usadas pela arena. Substitui Entity.currentDefense() +
CombatCalculator +
  * DefenseCalculator (que existiam soltos e desconectados no código original — nenhum dos
  * três realmente aplicava dano a um alvo) por um só lugar. Player ataca Mob, Mob ataca
  * Player, e futuramente Mob pode atacar Mob — todos passam por aqui, porque Player e Mob
@@ -24,6 +24,9 @@ public final class CombatService {
     }

     public <T extends GameEntity & Damageable> int attack(Attacker attacker, T target) {
+        Objects.requireNonNull(attacker, "attacker");
+        Objects.requireNonNull(target, "target");
+        if (target.isDead()) return 0;
         int rawDamage = attacker.computeOutgoingDamage();
         double affinity = target.race().affinityFor(attacker.outgoingDamageType());
         int finalDamage = (int) Math.round(rawDamage * affinity);
```

### 10. [M] server/src/main/java/com/projetok/prototype/rpg/CharacterCreation.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/CharacterCreation.java
b/server/src/main/java/com/projetok/prototype/rpg/CharacterCreation.java
index 03ea3e5f5a34bdd29eb33e22f3e721c9823f1d69..ae17c84d489b2e115eb99e960090b340401d8d9a 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/CharacterCreation.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/CharacterCreation.java
@@ -4,51 +4,64 @@ import com.projetok.rpg.*;
 import javax.swing.*;
 import java.awt.*;
 import java.nio.file.Path;
+import java.util.LinkedHashMap;
+import java.util.Map;
 import java.util.function.Consumer;

-/** Formulário de criação local; validação e atributos pertencem ao domínio. */
+/** Cria uma alma aprendiz: a raça é escolhida agora; o domínio, no nível 15. */
 final class CharacterCreation extends JPanel {
     final JTextField name = new JTextField(20);
-    final JRadioButton guardian = new JRadioButton("Guardian — Investida", true);
-    final JRadioButton mystic = new JRadioButton("Mystic — Cura");
+    final Map<String, JRadioButton> races = new LinkedHashMap<>();
     final JComboBox<String> appearance = new JComboBox<>(new String[]{"azure", "amber"});
     final JButton create = new CharacterSelection.SpiritButton("Criar personagem", true);
     final JButton back = new CharacterSelection.SpiritButton("Voltar", false);
+    final JButton techniques = new CharacterSelection.SpiritButton("Conhecer os caminhos", false);
     final JLabel error = CharacterSelection.label(" ", 12);
     final JLabel attributes = CharacterSelection.label(" ", 13);
+    final JLabel origin = CharacterSelection.label(" ", 12);
     final CharacterSelection.Preview preview = new CharacterSelection.Preview();

     CharacterCreation(RpgPlayerSprites sprites) {
         super(new BorderLayout(12, 12));
         setBackground(CharacterSelection.INK);
         setBorder(BorderFactory.createEmptyBorder(14, 14, 14, 14));
-        add(CharacterSelection.label("SPIRIT GATE | DESPERTAR", 22), BorderLayout.NORTH);
+        JPanel heading = new JPanel(new GridLayout(0, 1, 0, 3)); heading.setOpaque(false);
+        heading.add(CharacterSelection.label("SPIRIT GATE | DESPERTAR", 22));
+        heading.add(CharacterSelection.label("Uma pequena alma no início de seu caminho. Escolha sua raça.",
12));
+        add(heading, BorderLayout.NORTH);
         name.setText(System.getProperty("spiritgate.profile.name", ""));
         name.setBackground(new Color(12, 35, 51)); name.setForeground(Color.WHITE);
         name.setCaretColor(Color.WHITE); name.getAccessibleContext().setAccessibleName("Nome do personagem");
-        JPanel form = CharacterSelection.panel(new GridBagLayout());
+        JPanel fieldsColumn = new JPanel(); fieldsColumn.setOpaque(false);
+        fieldsColumn.setLayout(new BoxLayout(fieldsColumn, BoxLayout.Y_AXIS));
-        var constraints = new GridBagConstraints(); constraints.weightx=1; constraints.weighty=1;
-        constraints.anchor=GridBagConstraints.NORTHWEST; constraints.fill=GridBagConstraints.HORIZONTAL;
-        form.add(fieldsColumn,constraints);
```

```
         fieldsColumn.add(CharacterSelection.label("Nome (3–24 caracteres)", 13)); fieldsColumn.add(name);
-        ButtonGroup classes = new ButtonGroup(); classes.add(guardian); classes.add(mystic);
-        for (JRadioButton option : new JRadioButton[]{guardian, mystic}) {
-            option.setOpaque(false); option.setForeground(Color.WHITE); fieldsColumn.add(option);
+        fieldsColumn.add(Box.createVerticalStrut(8));
+        ButtonGroup options = new ButtonGroup();
+        for (var race : CharacterRaces.all()) {
+            JRadioButton option = new JRadioButton(race.name(), races.isEmpty());
+            option.setOpaque(false); option.setForeground(Color.WHITE);
+            option.getAccessibleContext().setAccessibleName("Raça " + race.name());
+            option.setToolTipText(race.description());
+            races.put(race.id(), option); options.add(option); fieldsColumn.add(option);
             option.addActionListener(e -> updatePreview());
         }
-        fieldsColumn.add(CharacterSelection.label("Aparência geométrica", 13)); fieldsColumn.add(appearance);
-        fieldsColumn.add(CharacterSelection.label("Sprite base compartilhado pelas classes", 11));
-        fieldsColumn.add(attributes);
-        fieldsColumn.add(preview.orientation());
-        for(Component field:fieldsColumn.getComponents()) {
-            if(field instanceof JComponent c) {c.setAlignmentX(Component.LEFT_ALIGNMENT);c.setMaximumSize(new
Dimension(Integer.MAX_VALUE,c.getPreferredSize().height));}
+        fieldsColumn.add(origin); fieldsColumn.add(techniques);
+        fieldsColumn.add(Box.createVerticalStrut(8));
+        fieldsColumn.add(CharacterSelection.label("Aparência", 13)); fieldsColumn.add(appearance);
+        fieldsColumn.add(attributes); fieldsColumn.add(preview.orientation());
+        for (Component field : fieldsColumn.getComponents()) {
+            if (field instanceof JComponent c) {
+                c.setAlignmentX(Component.LEFT_ALIGNMENT);
+                c.setMaximumSize(new Dimension(Integer.MAX_VALUE, c.getPreferredSize().height));
+            }
         }
         appearance.addActionListener(e -> updatePreview());
+        techniques.addActionListener(e -> DomainPreview.open(SwingUtilities.getWindowAncestor(this),
selectedRace(), 1, "none", null));
         JPanel center = new JPanel(new GridLayout(1, 2, 12, 0)); center.setOpaque(false);
-        JScrollPane fields = new JScrollPane(form); fields.setBorder(BorderFactory.createEmptyBorder());
+        JScrollPane fields = new JScrollPane(fieldsColumn);
fields.setBorder(BorderFactory.createEmptyBorder());
+        fields.getViewport().setBackground(CharacterSelection.INK);
         fields.setHorizontalScrollBarPolicy(JScrollPane.HORIZONTAL_SCROLLBAR_NEVER);
+        fields.getVerticalScrollBar().setUnitIncrement(20);
         center.add(fields); preview.sprites = sprites; center.add(preview); add(center);
         JPanel footer = new JPanel(new BorderLayout(4, 8)); footer.setOpaque(false);
         JPanel buttons = new JPanel(new GridLayout(1, 2, 8, 0)); buttons.setOpaque(false);
@@ -56,47 +69,51 @@ final class CharacterCreation extends JPanel {
         add(footer, BorderLayout.SOUTH); updatePreview();
     }

+    String selectedRace() {
+        return races.entrySet().stream().filter(e ->
e.getValue().isSelected()).map(Map.Entry::getKey).findFirst().orElseThrow();
+    }
+
     /** Consulta um mundo temporário, sem escrever saves ou alterar a sessão ativa. */
     void updatePreview() {
-        for(JRadioButton option:new JRadioButton[]{guardian,mystic}){
-            option.setOpaque(true);option.setBackground(option.isSelected()?new
Color(18,66,85):CharacterSelection.INK);
-            option.setBorderPainted(true);option.setBorder(BorderFactory.createCompoundBorder(BorderFactory.cre
ateLineBorder(option.isSelected()?CharacterSelection.CYAN:new
Color(40,103,125),option.isSelected()?2:1),BorderFactory.createEmptyBorder(8,6,8,6)));
-            var definition=Archetypes.get(option==guardian?"guardian":"mystic");
-            option.setToolTipText((option==guardian?"Combate corpo a corpo; Investida":"Combate à distância;
Cura")+" | Custo "+definition.cost()+" vigor | Recarga "+RpgCombatStatus.cooldown(definition.cooldown()));
-            option.setMaximumSize(new Dimension(Integer.MAX_VALUE,option.getPreferredSize().height));
-        }
-        var identity = CharacterIdentity.create("Prévia", guardian.isSelected() ? "guardian" : "mystic",
(String) appearance.getSelectedItem());
+        races.forEach((id, option) -> {
+            option.setOpaque(true); option.setBackground(option.isSelected() ? new Color(18, 66, 85) :
CharacterSelection.INK);
+            option.setBorderPainted(true);
+
option.setBorder(BorderFactory.createCompoundBorder(BorderFactory.createLineBorder(option.isSelected() ?
CharacterSelection.CYAN : new Color(40, 103, 125), option.isSelected() ? 2 : 1),
BorderFactory.createEmptyBorder(5, 6, 5, 6)));
+            option.setMaximumSize(new Dimension(Integer.MAX_VALUE, option.getPreferredSize().height));
+        });
```

```
+        var race = CharacterRaces.get(selectedRace());
+        origin.setText("<html>Aprendiz • " + race.startingAttackName() + "<br>Sem domínio • Escolha no nível
15</html>");
+        origin.setToolTipText(race.description());
+        origin.setMaximumSize(new Dimension(Integer.MAX_VALUE, origin.getPreferredSize().height));
+        var identity = CharacterIdentity.create("Prévia", race.id(), (String) appearance.getSelectedItem());
         var state = new RpgWorld(identity).save();
         preview.slot = new CharacterSaves.Slot(Path.of("preview.properties"), state, "", false);
         var view = RpgWorld.restore(state).snapshot();
         var combat = RpgCombatStatus.from(view);
         attributes.setText("<html>Nível " + view.level() + "<br>HP " + combat.hp() + " / " + combat.maxHp()
-                + "<br>Vigor " + combat.stamina()/10 + " / 100<br>Ataque " + view.attack() + " | Defesa " +
view.defense() + "</html>");
+                + "<br>Vigor " + combat.stamina() / 10 + " / 100<br>Ataque " + view.attack() + " | Defesa " +
view.defense() + "</html>");
         attributes.setMaximumSize(new Dimension(Integer.MAX_VALUE, attributes.getPreferredSize().height));
-        attributes.revalidate();
-        preview.repaint();
+        attributes.revalidate(); preview.repaint();
     }

-    /** Valida com o contrato real e só então entrega a identidade ao fluxo existente. */
     void submit(Consumer<CharacterIdentity> accepted) {
         try {
-            var identity = CharacterIdentity.create(name.getText(), guardian.isSelected() ? "guardian" :
"mystic", (String) appearance.getSelectedItem());
+            var identity = CharacterIdentity.create(name.getText(), selectedRace(), (String)
appearance.getSelectedItem());
             error.setText(" "); accepted.accept(identity);
         } catch (IllegalArgumentException failure) {
             error.setText(failure.getMessage()); error.setToolTipText(failure.getMessage());
name.requestFocusInWindow();
         }
     }

-    /** Abre diálogo modal com Enter, Escape e fechamento sem criação. */
     static void open(JFrame owner, RpgPlayerSprites sprites, Consumer<CharacterIdentity> accepted) {
         JDialog dialog = new JDialog(owner, "Criar personagem local", true);
         CharacterCreation form = new CharacterCreation(sprites);
         dialog.setDefaultCloseOperation(JDialog.DISPOSE_ON_CLOSE); dialog.setContentPane(form);
-        form.create.addActionListener(e -> form.submit(identity -> {dialog.dispose();
accepted.accept(identity);}));
+        form.create.addActionListener(e -> form.submit(identity -> { dialog.dispose();
accepted.accept(identity); }));
         form.back.addActionListener(e -> dialog.dispose());
         dialog.getRootPane().setDefaultButton(form.create);
         dialog.getRootPane().registerKeyboardAction(e -> dialog.dispose(), KeyStroke.getKeyStroke("ESCAPE"),
JComponent.WHEN_IN_FOCUSED_WINDOW);
-        dialog.setSize(Math.min(820, owner.getWidth()), Math.min(540, owner.getHeight()));
+        dialog.setSize(Math.min(820, owner.getWidth()), Math.min(620, owner.getHeight()));
         dialog.setLocationRelativeTo(owner); SwingUtilities.invokeLater(form.name::requestFocusInWindow);
dialog.setVisible(true);
     }
 }
```

### 11. [M] server/src/main/java/com/projetok/prototype/rpg/CharacterSelection.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/CharacterSelection.java
b/server/src/main/java/com/projetok/prototype/rpg/CharacterSelection.java
index 22a4dee988235472a9f577a3e6fd3456fe93d393..459e836c9be5b9d22667d9d2c703b5bbebd34725 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/CharacterSelection.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/CharacterSelection.java
@@ -13,7 +13,7 @@ final class CharacterSelection extends JPanel {
     final DefaultListModel<CharacterSaves.Slot> model=new DefaultListModel<>();
     final JList<CharacterSaves.Slot> list=new JList<>(model){@Override public boolean
getScrollableTracksViewportWidth(){return true;}};
     final JButton play=new SpiritButton("Jogar",true), create=new SpiritButton("Criar personagem",false),
back=new SpiritButton("Voltar",false), delete=new SpiritButton("Arquivar personagem",false);
-    final JButton arena=new SpiritButton("Arena histórica",false);
+    final JButton techniques=new SpiritButton("Conhecer os caminhos",false);
     final JLabel status=new JLabel("Carregando personagens locais...");
     final Preview preview=new Preview();
     final JLabel name=label("Seu próximo destino",22), details=label("Selecione um personagem",13);
@@ -45,27 +45,29 @@ final class CharacterSelection extends JPanel {
         JPanel center=new JPanel(new
GridLayout(1,2,12,0));center.setOpaque(false);center.add(roster);center.add(right);add(center);
         JPanel south=new JPanel(new BorderLayout(0,8));south.setOpaque(false);
         JPanel buttons=new JPanel(new GridLayout(0,2,8,6));buttons.setOpaque(false);
-        for(JButton button:new JButton[]{play,create,delete,arena,back})buttons.add(button);
+        for(JButton button:new JButton[]{play,create,delete,techniques,back})buttons.add(button);
         south.add(buttons);status.setForeground(CYAN);status.setFont(new
Font("SansSerif",Font.PLAIN,11));south.add(status,BorderLayout.SOUTH);add(south,BorderLayout.SOUTH);
-        play.setMnemonic('J');create.setMnemonic('C');back.setMnemonic('V');delete.setMnemonic('A');arena.setMn
emonic('H');
-        play.setEnabled(false);delete.setEnabled(false);create.setEnabled(false);
-        list.addListSelectionListener(e->{var slot=list.getSelectedValue();play.setEnabled(slot!=null&&slot.sta
te()!=null);delete.setEnabled(slot!=null&&!slot.backup());preview.slot=slot;updateSummary(slot);preview.repaint(
);});
+        play.setMnemonic('J');create.setMnemonic('C');back.setMnemonic('V');delete.setMnemonic('A');techniques.
setMnemonic('T');
+        play.setEnabled(false);delete.setEnabled(false);create.setEnabled(false);techniques.setEnabled(false);
+        list.addListSelectionListener(e->{var slot=list.getSelectedValue();play.setEnabled(slot!=null&&slot.sta
te()!=null);techniques.setEnabled(slot!=null&&slot.state()!=null);delete.setEnabled(slot!=null&&!slot.backup());
preview.slot=slot;updateSummary(slot);preview.repaint();});
+        techniques.addActionListener(e->{var
slot=list.getSelectedValue();if(slot!=null&&slot.state()!=null){var state=slot.state();DomainPreview.open(SwingU
tilities.getWindowAncestor(this),state.identity().race(),RpgWorld.restore(state).level(),DomainProgression.read(
state.flags()).domain(),null);}});
         list.getInputMap().put(KeyStroke.getKeyStroke("ENTER"),"play");list.getActionMap().put("play",new
AbstractAction(){public void actionPerformed(java.awt.event.ActionEvent e){play.doClick();}});
         getInputMap(WHEN_ANCESTOR_OF_FOCUSED_COMPONENT).put(KeyStroke.getKeyStroke("ESCAPE"),"back");getActionM
ap().put("back",new AbstractAction(){public void actionPerformed(java.awt.event.ActionEvent
e){back.doClick();}});
     }
     static JLabel label(String text,int size){JLabel l=new JLabel(text);l.setFont(new
Font(size>=20?"Serif":"SansSerif",size>=20?Font.BOLD:Font.PLAIN,size));l.setForeground(size>=20?new
Color(227,248,255):CYAN);return l;}
     static JPanel panel(LayoutManager layout){JPanel p=new JPanel(layout);p.setBackground(new
Color(5,19,33));p.setBorder(BorderFactory.createCompoundBorder(BorderFactory.createLineBorder(new
Color(40,103,125)),BorderFactory.createEmptyBorder(10,10,10,10)));return p;}
-    static String className(RpgSave s){return
s.identity().archetype().equals("guardian")?"Guardião":"Místico";}
-    static String describe(CharacterSaves.Slot slot){if(slot.state()==null)return "Save inválido |
"+slot.file().getFileName();var s=slot.state();return s.identity().name()+" | "+className(s)+" | Nível
"+RpgWorld.restore(s).level()+(slot.backup()?" | Backup recuperável":"");}
+    static String raceName(RpgSave s){return CharacterRaces.get(s.identity().race()).name();}
+    static String describe(CharacterSaves.Slot slot){if(slot.state()==null)return "Save inválido |
"+slot.file().getFileName();var s=slot.state();return s.identity().name()+" | "+raceName(s)+" | Nível
"+RpgWorld.restore(s).level()+(slot.backup()?" | Backup recuperável":"");}
     void updateSummary(CharacterSaves.Slot slot){
         name.setText(slot==null?"Seu próximo destino":slot.state()==null?"Arquivo
preservado":slot.state().identity().name());
-        details.setText(slot==null?"Selecione um personagem":slot.state()==null?"Save
inválido":className(slot.state())+"  •  Nível "+RpgWorld.restore(slot.state()).level());
+        details.setText(slot==null?"Selecione um personagem":slot.state()==null?"Save
inválido":raceName(slot.state())+"  •  Nível "+RpgWorld.restore(slot.state()).level());
         resources.setText(" ");style.setText(" ");
         if(slot!=null&&slot.state()!=null){
             var view=RpgWorld.restore(slot.state()).snapshot();var combat=RpgCombatStatus.from(view);
             resources.setText("HP "+combat.hp()+" / "+combat.maxHp()+"   •   Vigor "+combat.stamina()/10+" /
100");
-            var definition=Archetypes.get(slot.state().identity().archetype());
-            style.setText("<html>XP "+view.state().xp()+" • Ataque "+view.attack()+" • Defesa
```

```
"+view.defense()+"<br>"+(slot.state().identity().archetype().equals("guardian")?"Corpo a corpo •
Investida":"Combate à distância • Cura")+" • Custo "+definition.cost()+" vigor</html>");
+            var domain=DomainProgression.read(slot.state().flags());
+            String mastery=domain.domain().equals("none")?"Aprendiz • Domínio no nível
15":Domains.get(domain.domain()).name()+"<br>XP de domínio "+DomainPreview.formatXp(domain.xpHalfUnits());
+            style.setText("<html>XP "+view.state().xp()+" • Ataque "+view.attack()+" • Defesa
"+view.defense()+"<br>"+mastery+"</html>");
         }
         for(JLabel line:new JLabel[]{name,details,resources,style})line.setToolTipText(line.getText());
     }
@@ -87,7 +89,7 @@ final class CharacterSelection extends JPanel {
             slot=value;setBackground(selected?new Color(18,66,85):new Color(8,25,40));

setBorder(BorderFactory.createCompoundBorder(BorderFactory.createLineBorder(focus?Color.WHITE:selected?CYAN:new
Color(26,60,78),focus?2:1),BorderFactory.createEmptyBorder(5,6,5,6)));
             title.setText(value.state()==null?"Save inválido":value.state().identity().name());
-
subtitle.setText(value.state()==null?value.file().getFileName().toString():className(value.state())+" • Nível
"+RpgWorld.restore(value.state()).level());
+
subtitle.setText(value.state()==null?value.file().getFileName().toString():raceName(value.state())+" • Nível
"+RpgWorld.restore(value.state()).level());
             note.setText(value.backup()?"Backup recuperável":value.state()==null?"Arquivo
preservado":selected?"  Selecionado":"Personagem local");
             setToolTipText(value.problem().isEmpty()?describe(value):value.problem());return this;
         }
```

### 12. [A] server/src/main/java/com/projetok/prototype/rpg/DomainPreview.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/DomainPreview.java
b/server/src/main/java/com/projetok/prototype/rpg/DomainPreview.java
new file mode 100644
index 0000000000000000000000000000000000000000..8fde2df135fee6e1367f996417619db4ac2e5c3a
--- /dev/null
+++ b/server/src/main/java/com/projetok/prototype/rpg/DomainPreview.java
@@ -0,0 +1,169 @@
+package com.projetok.prototype.rpg;
+
+import com.projetok.rpg.*;
+import javax.swing.*;
+import java.awt.*;
+import java.awt.geom.Arc2D;
+import java.util.function.Consumer;
+
+/** Demonstrações visuais isoladas: não concedem XP, técnicas nem alteram o personagem. */
+final class DomainPreview extends JPanel {
+    final JComboBox<Domains.Definition> domains;
+    final DefaultListModel<String> techniqueModel = new DefaultListModel<>();
+    final JList<String> techniques = new JList<>(techniqueModel);
+    final JLabel status = CharacterSelection.label(" ", 12);
+    final JTextArea description = new JTextArea();
+    final JButton choose = new CharacterSelection.SpiritButton("Escolher domínio", true);
+    final JButton close = new CharacterSelection.SpiritButton("Voltar", false);
+    final TechniqueScene scene = new TechniqueScene();
+    private final boolean canChoose;
+
+    DomainPreview(String race, int level, String currentDomain, boolean selectionAllowed) {
+        super(new BorderLayout(10, 10)); setBackground(CharacterSelection.INK);
+        setBorder(BorderFactory.createEmptyBorder(14, 14, 14, 14));
+        canChoose = selectionAllowed && level >= DomainProgression.INITIATION_LEVEL &&
currentDomain.equals(DomainProgression.NONE);
+        JPanel heading = new JPanel(new GridLayout(0, 1, 0, 4)); heading.setOpaque(false);
+        heading.add(CharacterSelection.label("CAMINHOS | " + CharacterRaces.get(race).name(), 22));
+        heading.add(CharacterSelection.label("Prévias de conceito das técnicas em desenvolvimento", 12));
add(heading, BorderLayout.NORTH);
+        domains = new JComboBox<>(Domains.forRace(race).toArray(Domains.Definition[]::new));
+        domains.setRenderer(new DefaultListCellRenderer() {
+            @Override public Component getListCellRendererComponent(JList<?> list, Object value, int index,
boolean selected, boolean focus) {
+                return super.getListCellRendererComponent(list, value instanceof Domains.Definition d ?
d.name() : "", index, selected, focus);
+            }
+        });
+        domains.getAccessibleContext().setAccessibleName("Caminho para visualizar");
+        JPanel content = new JPanel(new BorderLayout(8, 8)); content.setOpaque(false); content.add(domains,
BorderLayout.NORTH);
+        JPanel previewArea = new JPanel(new BorderLayout(8, 8)); previewArea.setOpaque(false);
previewArea.add(scene);
```

```
+        techniques.setBackground(new Color(8, 25, 40)); techniques.setForeground(Color.WHITE);
+        techniques.setSelectionBackground(new Color(18, 66, 85));
techniques.setSelectionForeground(Color.WHITE);
+        techniques.setSelectionMode(ListSelectionModel.SINGLE_SELECTION);
+        techniques.getAccessibleContext().setAccessibleName("Técnica para visualizar");
+        techniques.setCellRenderer(new DefaultListCellRenderer() {
+            @Override public Component getListCellRendererComponent(JList<?> list, Object value, int index,
boolean selected, boolean focus) {
+                JLabel line = (JLabel) super.getListCellRendererComponent(list, value, index, selected, focus);
+                String text = value.toString(); int separator = text.indexOf(':');
+                line.setText(separator < 0 ? text : text.substring(0, separator)); line.setToolTipText(text);
return line;
+            }
+        });
+        JScrollPane techniqueList = new JScrollPane(techniques); techniqueList.setPreferredSize(new
Dimension(220, 72));
+        previewArea.add(techniqueList, BorderLayout.SOUTH); content.add(previewArea);
+        description.setEditable(false); description.setLineWrap(true); description.setWrapStyleWord(true);
+        description.setOpaque(false); description.setForeground(CharacterSelection.CYAN);
description.setFont(new Font("SansSerif", Font.PLAIN, 12));
+        description.setRows(4);
+        JScrollPane explanation = new JScrollPane(description);
explanation.setBorder(BorderFactory.createEmptyBorder());
+        explanation.setPreferredSize(new Dimension(400, 68));
explanation.getViewport().setBackground(CharacterSelection.INK);
+        content.add(explanation, BorderLayout.SOUTH); add(content);
+        JPanel footer = new JPanel(new BorderLayout(0, 8)); footer.setOpaque(false);
+        JPanel buttons = new JPanel(new GridLayout(1, 2, 8, 0)); buttons.setOpaque(false);
+        buttons.add(choose); buttons.add(close); footer.add(buttons, BorderLayout.NORTH);
+        status.setText(canChoose ? "Escolha permanente. Outro caminho exige reset de domínio."
+            : !currentDomain.equals(DomainProgression.NONE) ? "Domínio já escolhido. Para mudar, será
necessário um reset."
+            : "A iniciação e a escolha de domínio ficam disponíveis no nível 15.");
+        footer.add(status, BorderLayout.SOUTH); add(footer, BorderLayout.SOUTH);
+        choose.setEnabled(canChoose); choose.setVisible(selectionAllowed);
+        domains.addActionListener(e -> updateDomain());
+        techniques.addListSelectionListener(e -> updateTechnique()); updateDomain();
+    }
+
+    String selectedDomain() { return ((Domains.Definition) domains.getSelectedItem()).id(); }
+    boolean canChoose() { return canChoose; }
+    private void updateDomain() {
+        var domain = (Domains.Definition) domains.getSelectedItem();
+        scene.domain = domain.id(); techniqueModel.clear();
domain.techniquePreviews().forEach(techniqueModel::addElement);
+        if (!techniqueModel.isEmpty()) techniques.setSelectedIndex(0); scene.repaint();
+    }
+    private void updateTechnique() {
+        var domain = (Domains.Definition) domains.getSelectedItem();
+        scene.technique = Math.max(0, techniques.getSelectedIndex());
+        description.setText(domain.description() + "\n" + (techniques.getSelectedValue() == null ? "" :
techniques.getSelectedValue()));
+        scene.repaint();
+    }
+    static String formatXp(long halfUnits) { return halfUnits / 2 + (halfUnits % 2 == 0 ? "" : ",5"); }
+
+    static void open(Window owner, String race, int characterLevel, String currentDomain, Consumer<String>
selected) {
+        JDialog dialog = new JDialog(owner, "Spirit Gate | Iniciação", Dialog.ModalityType.APPLICATION_MODAL);
+        DomainPreview form = new DomainPreview(race, characterLevel, currentDomain, selected != null);
+        dialog.setContentPane(form); dialog.setDefaultCloseOperation(JDialog.DISPOSE_ON_CLOSE);
+        form.close.addActionListener(e -> dialog.dispose());
+        form.choose.addActionListener(e -> {
+            if (!form.canChoose || selected == null) return;
+            try { selected.accept(form.selectedDomain()); dialog.dispose(); }
+            catch (IllegalArgumentException | IllegalStateException failure) {
form.status.setText(failure.getMessage()); }
+        });
+        dialog.getRootPane().registerKeyboardAction(e -> dialog.dispose(), KeyStroke.getKeyStroke("ESCAPE"),
JComponent.WHEN_IN_FOCUSED_WINDOW);
+        Dimension screen = Toolkit.getDefaultToolkit().getScreenSize();
+        dialog.setSize(Math.min(780, screen.width - 40), Math.min(620, screen.height - 60));
+        dialog.setLocationRelativeTo(owner); dialog.setVisible(true);
+    }
+
+    static final class TechniqueScene extends JPanel {
+        String domain = "water"; int technique;
+        final Timer animation = new Timer(40, e -> { if (!SpiritTheme.reducedMotion) repaint(); });
```

```
+        TechniqueScene() { setPreferredSize(new Dimension(400, 220)); setMinimumSize(new Dimension(100, 80));
setBackground(new Color(5, 19, 33)); }
+        @Override public void addNotify() { super.addNotify(); animation.start(); }
+        @Override public void removeNotify() { animation.stop(); super.removeNotify(); }
+        @Override protected void paintComponent(Graphics original) {
+            super.paintComponent(original); Graphics2D g = (Graphics2D) original.create();
+            try {
+                g.setRenderingHint(RenderingHints.KEY_ANTIALIASING, RenderingHints.VALUE_ANTIALIAS_ON);
+                double phase = SpiritTheme.reducedMotion ? .55 : (System.nanoTime() / 1_000_000_000.0 % 2.4) /
2.4;
+                g.scale(getWidth() / 600.0, getHeight() / 260.0);
+                g.setColor(new Color(28, 64, 79)); for (int x = 0; x < 600; x += 40) g.drawLine(x, 0, x, 260);
+                for (int y = 0; y < 260; y += 40) g.drawLine(0, y, 600, y);
+                boolean closeRange = domain.equals("martial") || domain.equals("bow_short");
+                int source = closeRange ? 300 : 120, target = 450, y = 145;
+                actor(g, source, y, CharacterSelection.CYAN); actor(g, target, y, new Color(239, 150, 126));
+                g.setStroke(new BasicStroke(5, BasicStroke.CAP_ROUND, BasicStroke.JOIN_ROUND));
+                int projectile = source + (int) ((target - source) * phase);
+                switch (domain) {
+                    case "water" -> {
+                        g.setColor(new Color(81, 192, 255));
+                        g.draw(new Arc2D.Double(source - 40, y - 85, 145 + phase * 170, 150, -70 + phase * 40,
230, Arc2D.OPEN));
+                        if (technique == 0) g.drawOval(source - 30, y - 60, 60, 100);
+                    }
+                    case "thunder" -> {
+                        g.setColor(new Color(255, 229, 84)); int[] xs = {source, source + 90, source + 65,
projectile, projectile - 20, target};
+                        int[] ys = {y - 20, y - 35, y - 10, y - 40, y - 12, y - 25}; g.drawPolyline(xs, ys,
xs.length);
+                    }
+                    case "fire" -> {
+                        for (int i = 0; i < 9; i++) {
+                            int x = source + i * 36; int height = 20 + (int) (Math.sin(phase * 9 + i) * 12);
+                            g.setColor(new Color(255, 100 + i * 10, 40, 210)); g.fillPolygon(new int[]{x - 13,
x, x + 13}, new int[]{y + 15, y - height, y + 15}, 3);
+                        }
+                    }
+                    case "wind" -> {
+                        g.setColor(new Color(177, 248, 224)); for (int i = 0; i < 3; i++) g.drawArc(projectile
- 25 - i * 18, y - 65 + i * 20, 60, 75, -60, 140);
+                    }
+                    case "light" -> {
+                        g.setColor(new Color(255, 239, 170));
+                        if (technique == 0) { g.drawLine(source + 18, y - 20, projectile, y - 20);
g.fillOval(projectile - 8, y - 28, 16, 16); }
+                        else if (technique == 1) { g.drawLine(source - 17, y - 95, source + 17, y - 95);
g.drawLine(source, y - 112, source, y - 78); g.drawOval(source - 35, y - 60, 70, 110); }
+                        else g.drawRoundRect(source + 25, y - 65, 18, 110, 12, 12);
+                    }
+                    case "martial" -> {
+                        g.setColor(new Color(255, 209, 145)); for (int i = 0; i < 3; i++) g.fillOval(source +
35 + (int) (phase * 90), y - 40 + i * 20, 18, 18);
+                    }
+                    case "bow_long", "bow_short" -> {
+                        g.setColor(new Color(156, 218, 255)); g.drawArc(source - 4, y - 58, 38, 74, -80, 160);
g.drawLine(source + 18, y - 57, source + 18, y + 15);
+                        int count = domain.equals("bow_short") ? 3 : 1;
+                        for (int i = 0; i < count; i++) { int yy = y - 20 + (i - count / 2) * 22;
g.drawLine(projectile - 32, yy, projectile, yy); g.drawLine(projectile - 9, yy - 6, projectile, yy);
g.drawLine(projectile - 9, yy + 6, projectile, yy); }
+                    }
+                    case "darkness_debuff" -> {
+                        g.setColor(new Color(190, 120, 247)); g.drawOval(target - 35, y - 65, 70, 100);
+                        for (int i = 0; i < 4; i++) { int yy = y - 65 + i * 22; g.drawLine(target - 12, yy,
target + 12, yy + 15); g.drawLine(target + 12, yy, target - 12, yy + 15); }
+                    }
+                    case "darkness_control" -> {
+                        g.setColor(new Color(140, 95, 208, 100)); g.fillOval(target - 105, y - 20, 210, 75);
g.setColor(new Color(190, 120, 247));
+                        g.drawOval(target - 105, y - 20, 210, 75); for (int i = 0; i < 5; i++)
g.drawLine(target - 45 + i * 22, y - 70, target - 45 + i * 22, y + 25);
+                        actor(g, target + 75, y + 10, CharacterSelection.CYAN);
+                    }
+                    default -> { }
+                }
+            } finally { g.dispose(); }
+        }
```

```
+        private static void actor(Graphics2D g, int x, int y, Color color) {
+            g.setColor(color); g.fillOval(x - 10, y - 58, 20, 20); g.fillRoundRect(x - 14, y - 35, 28, 42, 9,
9);
+            g.setStroke(new BasicStroke(5)); g.drawLine(x - 7, y, x - 12, y + 25); g.drawLine(x + 7, y, x + 12,
 y + 25);
+        }
+    }
+}
```

### 13. [M] server/src/main/java/com/projetok/prototype/rpg/GettingStarted.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/GettingStarted.java
b/server/src/main/java/com/projetok/prototype/rpg/GettingStarted.java
index 97deacedff820f9073b77e99174cb62ba8e9da67..126332f98368ceda2eb70a0e487cccbbe1feb8d1 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/GettingStarted.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/GettingStarted.java
@@ -12,7 +12,7 @@ final class GettingStarted {
         StringBuilder text=new StringBuilder("BEM-VINDO AO SPIRIT GATE\n\nMovimentação: W A S D\n");
         for(String action:new
String[]{"attack","ability","potion","interact","inventory","journal","map","save","revive"})
             text.append(controls.key(action)).append(" —
").append(RpgLauncher.actionLabel(action)).append('\n');
-        return text.append("\nAproxime-se dos NPCs para interagir. As respostas aparecem no painel
inferior.\nEscape fecha o diálogo de NPC.\n\nVocê pode fechar esta ajuda e reabri-la pelo botão Ajuda.\nSalvar e
 voltar ao menu preserva o progresso antes de sair.").toString();
+        return text.append("\nVoce comeca como aprendiz, sem dominio. Reaper usa uma espada pequena; Human e
Aetheri usam socos; Spirit usa feixes de luz basicos.\n\nMobs e quests concedem experiencia. Um mob nivel 1
concede 3 XP de personagem e, apos escolher dominio, 1,5 XP de dominio.\n\nNo nivel 15, abra Dominios e tecnicas
 para visualizar e escolher um caminho. A escolha so muda com reset.\nShunpo ofensivo e defensivo exigem nivel
20 em todas as racas.\n\nAproxime-se dos NPCs para interagir. Escape fecha o dialogo.\nSalvar e voltar ao menu
preserva o progresso antes de sair.").toString();
     }
     /** Abre ajuda dispensável; o mundo não é modificado pela leitura. */
     static void open(JFrame owner,RpgControls controls) {
```

### 14. [A] server/src/main/java/com/projetok/prototype/rpg/LauncherSelfCheck.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/LauncherSelfCheck.java
b/server/src/main/java/com/projetok/prototype/rpg/LauncherSelfCheck.java
new file mode 100644
index 0000000000000000000000000000000000000000..3cdd63bd33ef6e5562f306ead26f7439f5a1205f
--- /dev/null
+++ b/server/src/main/java/com/projetok/prototype/rpg/LauncherSelfCheck.java
@@ -0,0 +1,28 @@
+package com.projetok.prototype.rpg;
+
+import com.projetok.rpg.*;
+import java.nio.file.*;
+import java.nio.charset.StandardCharsets;
+
+/** Verificacao do pacote sem abrir janelas e sem tocar em personagens pessoais. */
+final class LauncherSelfCheck {
+    private LauncherSelfCheck() { }
+    static void write(Path destination) {
+        try {
+            int races=0;
+            for(var race:CharacterRaces.all()) {
+                var world=new RpgWorld(CharacterIdentity.create("Verificacao",race.id(),"azure"));
+                if(world.domainProgression().chosen()||world.castAbility()||world.castSecondaryAbility())
+                    throw new IllegalStateException("Inicio de personagem invalido");
+                var state=world.save();
+                if(!RpgWorld.restore(LocalSaveStore.decode(LocalSaveStore.encode(state))).save().equals(state))
+                    throw new IllegalStateException("Falha no ciclo de save");
+                if(Domains.forRace(race.id()).size()<2)throw new IllegalStateException("Caminhos ausentes");
+                races++;
+            }
+            Path output=destination.toAbsolutePath();Files.createDirectories(output.getParent());
+            Files.writeString(output,"{\"ok\":true,\"races\":"+races+",\"domains\":"+Domains.all().size()
+                +",\"saveSchema\":2,\"paymentEnabled\":false}",StandardCharsets.UTF_8);
+        } catch(Exception error) { throw new IllegalStateException("Verificacao do launcher falhou",error); }
+    }
+}
```

### 15. [M] server/src/main/java/com/projetok/prototype/rpg/RpgCombatFeedback.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/RpgCombatFeedback.java
b/server/src/main/java/com/projetok/prototype/rpg/RpgCombatFeedback.java
index 2ae9a19e8ebc0f8ba9b4df041e7b4b735d4d636e..7a7de36a585496a17abbc713af04b0476cfad6ba 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/RpgCombatFeedback.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/RpgCombatFeedback.java
@@ -9,17 +9,26 @@ public final class RpgCombatFeedback {
     private final java.util.List<RpgCombatEvent> active=new ArrayList<>();
     private long sequence,tick=-1;
     private String region="";
+    private record Ray(float x,float y,boolean bow) { }
+    private final Map<Long,Ray> rays=new HashMap<>();
     /** Consome cada sequencia uma vez; ignora eventos antigos e limpa transicoes. */
     public void update(RpgWorld.View view) {
         if(tick>=0&&(view.ticks()<tick||!region.equals(view.state().region()))) {
-            active.clear();sequence=view.events().stream().mapToLong(RpgCombatEvent::sequence).max().orElse(0);
+
active.clear();rays.clear();sequence=view.events().stream().mapToLong(RpgCombatEvent::sequence).max().orElse(0);
         }
         tick=view.ticks();region=view.state().region();
         for(var event:view.events())if(event.sequence()>sequence) {
-            sequence=event.sequence();if(tick-event.tick()<30)active.add(event);
+            sequence=event.sequence();if(tick-event.tick()<30){
+                active.add(event);
+                var domain=DomainProgression.read(view.state().flags());
+                boolean bow=domain.domain().startsWith("bow_");
+                if(event.kind()==RpgCombatEvent.Kind.ATTACK&&(bow||view.identity().race().equals("spirit")))
+                    rays.put(event.sequence(),new Ray(view.x(),view.y(),bow));
+            }
         }
         active.removeIf(e->tick-e.tick()>=30);
         while(active.size()>64)active.remove(0);
+        rays.keySet().retainAll(active.stream().map(RpgCombatEvent::sequence).toList());
     }
     /** Copia imutavel para validacao e apresentacao. */
     public java.util.List<RpgCombatEvent> active(){return java.util.List.copyOf(active);}
@@ -31,9 +40,15 @@ public final class RpgCombatFeedback {
             for(var e:active) {
                 int age=(int)(tick-e.tick()),x=Math.round(e.x()),y=Math.round(e.y());
                 g.setColor(e.kind()==RpgCombatEvent.Kind.HEAL?new
Color(90,240,150):e.kind()==RpgCombatEvent.Kind.REJECT?new Color(230,210,150):new Color(255,120,105));
-                String text=switch(e.kind()){case HIT->"-"+e.amount();case HEAL->"+"+e.amount();case
KO->"Derrotado";case REJECT->"Indisponivel";case ATTACK->"";};
+                String text=switch(e.kind()){case HIT->"-"+e.amount();case HEAL->"+"+e.amount();case
KO->"Derrotado";case REJECT->"Indisponivel";case ATTACK,PUNCH->"";};
                 if(!text.isEmpty())g.drawString(text,x-12,y-50-age);
                 if(age<6&&e.kind()==RpgCombatEvent.Kind.HIT){g.setStroke(new
BasicStroke(2));g.drawOval(x-24,y-24,48,48);}
+                var ray=rays.get(e.sequence());
+                if(age<6&&ray!=null){
+                    g.setColor(ray.bow()?new Color(120,220,255,210):new Color(255,250,200,220));
+                    g.setStroke(new BasicStroke(ray.bow()?2:5));
+                    g.draw(new java.awt.geom.Line2D.Float(ray.x(),ray.y()-12,x,y-12));
+                }
             }
         } finally {g.dispose();}
     }
```

### 16. [M] server/src/main/java/com/projetok/prototype/rpg/RpgCombatStatus.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/RpgCombatStatus.java
b/server/src/main/java/com/projetok/prototype/rpg/RpgCombatStatus.java
index 1d15ddbf25790e1efa8cd549b21d0182bc665ac1..137f72c9367dab80500b1a6bd7ee032eab56eecc 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/RpgCombatStatus.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/RpgCombatStatus.java
@@ -7,14 +7,14 @@ public record RpgCombatStatus(int hp,int maxHp,int stamina,long attackTicks,long
     /** Le valores de save existentes; nao cria selecao de alvo nem recursos paralelos. */
     public static RpgCombatStatus from(RpgWorld.View v) {
         int stamina=(int)value(v,"stamina",1000);long attack=Math.max(0,value(v,"attack-until",0)-v.ticks());
-        long ability=Math.max(0,value(v,"ability-until",0)-v.ticks());var
definition=Archetypes.get(v.identity().archetype());
+        long ability=Math.max(0,value(v,"ability-until",0)-v.ticks());
         return new RpgCombatStatus(v.state().hp(),v.maxHp(),stamina,attack,ability,
-            v.state().hp()>0&&ability==0&&stamina>=definition.cost()*10&&(!definition.ability().equals("mend")|
|v.state().hp()<v.maxHp()),
+            v.level()>=RpgWorld.SHUNPO_LEVEL&&v.state().hp()>0&&ability==0&&stamina>=RpgWorld.SHUNPO_COST*10,
             v.state().hp()>0&&v.state().hp()*4<=v.maxHp());
     }
     private static long value(RpgWorld.View v,String prefix,long fallback){return
v.state().flags().stream().filter(f->f.startsWith(prefix+":"
)).mapToLong(f->Long.parseLong(f.substring(prefix.length()+1))).findFirst().orElse(fallback);}
     /** Duracao restante arredondada para decimos; usa 30 ticks por segundo. */
     public static String cooldown(long ticks){return ticks<=0?"Pronto":String.format(Locale.ROOT,"%.1f
s",ticks/30.0);}
     /** Resumo curto para janelas pequenas. */
-    public String text(){return "Golpe: "+cooldown(attackTicks)+" | Habilidade:
"+cooldown(abilityTicks)+(abilityReady?"":" (indisponivel)")+(lowHp?" | HP BAIXO":"");}
+    public String text(){return "Golpe: "+cooldown(attackTicks)+" | Shunpo:
"+cooldown(abilityTicks)+(abilityReady?"":" (indisponível)")+(lowHp?" | HP BAIXO":"");}
 }
```

### 17. [M] server/src/main/java/com/projetok/prototype/rpg/RpgHud.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/RpgHud.java
b/server/src/main/java/com/projetok/prototype/rpg/RpgHud.java
index ccf2ab5472836f43050d1a8040356317f0c0a8b0..43a4451a1c00fb772828a783d6879b8395763eb5 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/RpgHud.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/RpgHud.java
@@ -37,7 +37,8 @@ public final class RpgHud extends JPanel {
         previousXp=view.state().xp();combat.setText(values.text()+(view.ticks()<xpUntil?xpNotice:"")+" |
"+view.message());
         combat.setToolTipText(combat.getText());
         journal.setText(view.journal().stream().map(q->q.state()==QuestService.State.UNAVAILABLE?"Missão
indisponível":q.name()+" | "+q.state()+" | "+q.progress()+"/"+q.required()+" -
"+q.objective()).collect(java.util.stream.Collectors.joining("\n")));
-        status.setText("<html>"+view.identity().name()+" | Nível "+view.level()+" | XP "+view.state().xp()+" |
Moedas "+view.state().coins()+"<br>Ataque "+view.attack()+" | Defesa "+view.defense()+" |
"+view.state().region()+" | Atalhos na barra superior</html>");
+        var domain=DomainProgression.read(view.state().flags());
+        status.setText("<html>"+view.identity().name()+" |
"+CharacterRaces.get(view.identity().race()).name()+" | Nivel "+view.level()+" | XP "+view.state().xp()+" |
Moedas "+view.state().coins()+"<br>"+(domain.chosen()?Domains.get(domain.domain()).name()+" | XP de dominio
"+domain.xpText():"Aprendiz sem dominio - iniciacao no nivel 15")+" | Ataque "+view.attack()+" | Defesa
"+view.defense()+"</html>");
         if(!java.util.Objects.equals(last,view.dialogue())||!lastNpc.equals(view.activeNpc())) {
             last=view.dialogue();lastNpc=view.activeNpc();dialogue.removeAll();
             journal.getParent().getParent().setVisible(last==null);
```

### 18. [M] server/src/main/java/com/projetok/prototype/rpg/RpgLauncher.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/RpgLauncher.java
b/server/src/main/java/com/projetok/prototype/rpg/RpgLauncher.java
index dd06f1d32391e8683ba8a77542b1561e3a78436d..e6e480bbcd7f4e93915d5869ddf2edf173197658 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/RpgLauncher.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/RpgLauncher.java
@@ -9,6 +9,9 @@ import java.util.List;
 public final class RpgLauncher {
     private RpgLauncher() { }
     public static void main(String[] args) {
+        if(args.length==2&&args[0].equals("--verify-install")) {
+            LauncherSelfCheck.write(java.nio.file.Path.of(args[1]));return;
+        }
         if(args.length>0 && args[0].equals("--experimental-player-sprites")) {

System.setProperty(RpgPlayerSprites.ENABLED,"true");args=java.util.Arrays.copyOfRange(args,1,args.length);
        }
@@ -21,7 +24,7 @@ public final class RpgLauncher {
         if(name==null)name=JOptionPane.showInputDialog(null, "Nome do personagem (3 a 24 caracteres)", "Criar
personagem", JOptionPane.QUESTION_MESSAGE);
         if (name == null) return;
         com.projetok.rpg.CharacterIdentity identity;
-        try { identity = com.projetok.rpg.CharacterIdentity.create(name, "guardian", "azure"); }
+        try { identity = com.projetok.rpg.CharacterIdentity.create(name, "reaper", "azure"); }
         catch (IllegalArgumentException error) { JOptionPane.showMessageDialog(null, error.getMessage());
open(); return; }
         play(new com.projetok.rpg.RpgWorld(identity),new
com.projetok.rpg.LocalSaveStore(RpgMenu.dataHome().resolve("saves/characters/"+identity.id()+".properties")));
     }
@@ -63,8 +66,21 @@ public final class RpgLauncher {
         for(String action:new String[]{"attack","ability","potion","inventory","journal","map"}){JButton
button=new JButton(actionLabel(action)+" ["+controls[0].key(action)+"]");button.addActionListener(e->{audio.play
(RpgAudio.Cue.UI);actions.get(action).run();});toolbar.add(button);hotbar.put(action,button);}
         var
secondary=com.projetok.rpg.RpgSecondaryAbility.forArchetype(world.snapshot().identity().archetype());
         JButton secondaryButton=new JButton(secondary.name());
-        secondaryButton.setToolTipText("Custo "+secondary.cost()+" vigor; recarga compartilhada
"+RpgCombatStatus.cooldown(secondary.cooldown()));
+        secondaryButton.setToolTipText("Requer nivel 20. Custo "+secondary.cost()+" vigor; recarga
compartilhada "+RpgCombatStatus.cooldown(secondary.cooldown()));
+        hotbar.get("ability").setToolTipText("Shunpo ofensivo: requer nivel 20 e alvo visivel; recarga

compartilhada com a esquiva.");
         secondaryButton.addActionListener(e->runtime.command(w->w.activate("rpg.secondary")));toolbar.add(secon
daryButton);
+        JButton punchButton=new JButton("Soco desarmado");
+
punchButton.addActionListener(e->runtime.command(w->w.activate("rpg.unarmed")));toolbar.add(punchButton);
+        JButton domainsButton=new JButton("Dominios e tecnicas");
+        domainsButton.addActionListener(e->{runtime.input(0,0);var v=runtime.snapshot();
+            DomainPreview.open(frame,v.identity().race(),v.level(),com.projetok.rpg.DomainProgression.read(v.st
ate().flags()).domain(),
+
selected->runtime.action(w->w.chooseDomain(selected)).thenAccept(ok->SwingUtilities.invokeLater(()->{
+                    if(!ok)JOptionPane.showMessageDialog(frame,"Escolha recusada. Confira nivel, raca e dominio
 atual.");
+                    else runtime.save(store).whenComplete((unused,error)->SwingUtilities.invokeLater(()->JOptio
nPane.showMessageDialog(frame,
+                        error==null?"Dominio escolhido e salvo.":"Dominio escolhido. Falha ao salvar:
"+error.getMessage())));
+                }));});toolbar.add(domainsButton);
+        JButton resetButton=new JButton("Reset de dominio");
+        resetButton.addActionListener(e->JOptionPane.showMessageDialog(frame,world.resetRequirements(),"Reset
de dominio",JOptionPane.INFORMATION_MESSAGE));toolbar.add(resetButton);
         JButton configure=new JButton("Atalhos");configure.addActionListener(e->ControlsDialog.open(frame,contr
ols[0],controlsFile,next->{for(String key:controls[0].keys().values())for(boolean down:new boolean[]{true,false}
){panel.getInputMap(JComponent.WHEN_IN_FOCUSED_WINDOW).remove(KeyStroke.getKeyStroke((down?"pressed ":"released
")+key));panel.getActionMap().remove(key+down);}resetHeld(panel);controls[0]=next;bindActions.run();hotbar.forEa
ch((id,button)->button.setText(actionLabel(id)+" ["+next.key(id)+"]"));}));toolbar.add(configure);
         JButton settings=new CharacterSelection.SpiritButton("Opções",false);settings.addActionListener(e->Pref
erencesDialog.open(frame,preferences[0],preferencesFile,next->{preferences[0]=next;audio.preferences(next);audio
.play(RpgAudio.Cue.UI);}));toolbar.add(settings);
         JButton help=new CharacterSelection.SpiritButton("Ajuda",false);help.addActionListener(e->GettingStarte
d.open(frame,controls[0]));toolbar.add(help);
@@ -104,14 +120,17 @@ public final class RpgLauncher {
             if(view.state().region().equals(before.state().region())&&view.enemies().stream().mapToInt(com.proj
etok.rpg.EnemyService.View::hp).sum()<before.enemies().stream().mapToInt(com.projetok.rpg.EnemyService.View::hp)
.sum())audio.play(RpgAudio.Cue.HIT);
```

```
            if(view.ticks()/900>before.ticks()/900)audio.play(RpgAudio.Cue.AMBIENT);previous[0]=view;
            var keysNow=controls[0];
-           panel.setCustomHud(java.util.List.of("SPIRIT GATE | RPG OFFLINE",view.identity().name(),"Nivel
"+view.level()+" | XP "+view.state().xp(),"HP "+view.state().hp()+" / "+view.maxHp(),"Ataque "+view.attack()+" |
 Defesa "+view.defense(),"Moedas "+view.state().coins(),"Regiao:
"+view.state().region(),"["+keysNow.key("interact")+"] NPC / portal / objeto","["+keysNow.key("attack")+"]
Ataque basico","["+keysNow.key("ability")+"] Habilidade do arquetipo","["+keysNow.key("potion")+"] Pocao de
vida","["+keysNow.key("inventory")+"] Inventario / equipamentos","["+keysNow.key("journal")+"] Diario |
["+keysNow.key("map")+"] Mapa","["+keysNow.key("save")+"] Salvar progresso","["+keysNow.key("revive")+"]
Recuperar se morto"));
+           secondaryButton.setEnabled(RpgCombatStatus.from(view).abilityReady());
+           hotbar.get("ability").setEnabled(RpgCombatStatus.from(view).abilityReady());
+           var domain=com.projetok.rpg.DomainProgression.read(view.state().flags());
+           panel.setCustomHud(java.util.List.of("SPIRIT GATE",view.identity().name()+" |
"+com.projetok.rpg.CharacterRaces.get(view.identity().race()).name(),"Nivel "+view.level()+" | XP
"+view.state().xp(),"Dominio: "+(domain.chosen()?com.projetok.rpg.Domains.get(domain.domain()).name()+" | XP
"+domain.xpText():"Aprendiz - escolha no nivel 15"),"HP "+view.state().hp()+" / "+view.maxHp(),"Ataque
"+view.attack()+" | Defesa "+view.defense(),"Moedas "+view.state().coins(),"Regiao:
"+view.state().region(),"["+keysNow.key("interact")+"] NPC / portal / objeto","["+keysNow.key("attack")+"]
Ataque basico","["+keysNow.key("ability")+"] Shunpo (nivel 20)","["+keysNow.key("potion")+"] Pocao de
vida","["+keysNow.key("inventory")+"] Inventario / equipamentos","["+keysNow.key("journal")+"] Diario |
["+keysNow.key("map")+"] Mapa","["+keysNow.key("save")+"] Salvar progresso","["+keysNow.key("revive")+"]
Recuperar se morto"));

panel.setAmbientLight(com.projetok.rpg.EnvironmentCycle.light(view.ticks(),preferences[0].environment()));
            if(!displayedRegion[0].equals(view.state().region())){displayedRegion[0]=view.state().region();pane
l.setLocalMap(RpgMaps.region(displayedRegion[0]),false);}
            panel.repaint();hud.update(view);navigation.repaint();
        });
        frame.addWindowListener(new WindowAdapter() { public void windowClosed(WindowEvent e) { timer.stop(); r
untime.close();audio.close();java.awt.KeyboardFocusManager.getCurrentKeyboardFocusManager().removePropertyChange
Listener("activeWindow",audioFocus);if(System.getProperty("spiritgate.smoke.output")==null)RpgMenu.open(); } });
        frame.addWindowFocusListener(new WindowAdapter() { public void windowLostFocus(WindowEvent e)
{java.util.Arrays.fill(held, false);resetHeld(panel); runtime.input(0, 0); } });
-       for(java.awt.Component control:toolbar.getComponents())if(control instanceof JButton
button)button.setToolTipText(button.getText());
+       for(java.awt.Component control:toolbar.getComponents())if(control instanceof JButton
button&&button.getToolTipText()==null)button.setToolTipText(button.getText());
        runtime.start(); timer.start(); frame.setVisible(true);
        if(introduction)GettingStarted.open(frame,controls[0]);
        if(System.getProperty("spiritgate.smoke.output")!=null){Timer capture=new Timer(2000,e->{
@@ -149,5 +168,5 @@ public final class RpgLauncher {
     }
     /** Limpa teclas mantidas ao perder foco. */
     public static void resetHeld(JComponent component){Object
held=component.getClientProperty("rpg.held");if(held instanceof java.util.Set<?> keys)keys.clear();}
-    static String actionLabel(String action){return switch(action){case "attack"->"Golpe";case
"ability"->"Habilidade";case "potion"->"Pocao";case "inventory"->"Inventario";case "journal"->"Diario";case
"map"->"Mapa";case "interact"->"Interagir";case "save"->"Salvar";case "revive"->"Recuperar";default->action;};}
+    static String actionLabel(String action){return switch(action){case "attack"->"Ataque basico";case
"ability"->"Shunpo ofensivo";case "potion"->"Pocao";case "inventory"->"Inventario";case "journal"->"Diario";case
 "map"->"Mapa";case "interact"->"Interagir";case "save"->"Salvar";case "revive"->"Recuperar";default->action;};}
 }
```

### 19. [M] server/src/main/java/com/projetok/prototype/rpg/RpgMenu.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/RpgMenu.java
b/server/src/main/java/com/projetok/prototype/rpg/RpgMenu.java
index 0c3f1af7613795d212d93633004c92458841c088..a030236502cb31200915dbe1cef533cfc2721399 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/RpgMenu.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/RpgMenu.java
@@ -14,7 +14,6 @@ final class RpgMenu {
         var list=root.list;var status=root.status;
         CharacterSaves repository=new CharacterSaves(dataHome().resolve("saves/characters"));
         JButton create=root.create,resume=root.play;
-        root.arena.addActionListener(e->com.projetok.prototype.DebugLauncher.main(new String[0]));
         root.back.addActionListener(e->{frame.dispose();SpiritGateEntry.open();});
         root.delete.addActionListener(e->{
             var slot=list.getSelectedValue();if(slot==null||slot.backup())return;
```

### 20. [M] server/src/main/java/com/projetok/prototype/rpg/RpgPlayerSprites.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/RpgPlayerSprites.java
b/server/src/main/java/com/projetok/prototype/rpg/RpgPlayerSprites.java
index 7e64323ea73038027eefb77e796eb1566a64c112..6205a297390af4f7f526b452173f0301c605412a 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/RpgPlayerSprites.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/RpgPlayerSprites.java
@@ -22,7 +22,7 @@ public final class RpgPlayerSprites implements DebugPanel.ActorRenderer {
     private final RpgPlayerAnimation animation;
     private final double scale;
     private final RpgVisualState visual=new RpgVisualState();
-    private boolean mystic;
+    private boolean spirit;
     private RpgPlayerSprites(Map<String,Frame> frames,int frameMillis,double scale) {
         this.frames=Map.copyOf(frames);animation=new RpgPlayerAnimation(frameMillis);this.scale=scale;
     }
@@ -80,7 +80,7 @@ public final class RpgPlayerSprites implements DebugPanel.ActorRenderer {
         return new RpgPlayerSprites(frames,frameMillis,scale);
     }
     public double scale() { return scale; }
-    public RpgPlayerAnimation.Pose update(com.projetok.rpg.RpgWorld.View view) {
mystic=view.identity().archetype().equals("mystic");var
pose=animation.update(view);visual.update(view,pose.walking());return pose; }
+    public RpgPlayerAnimation.Pose update(com.projetok.rpg.RpgWorld.View view) {
spirit=view.identity().race().equals("spirit");var
pose=animation.update(view);visual.update(view,pose.walking());return pose; }
     /** Estado visual sem influencia nas regras do mundo. */
     public RpgVisualState.State visualState(){return visual.state();}
     public RpgPlayerAnimation.Pose pose() { return animation.pose(); }
@@ -113,9 +113,9 @@ public final class RpgPlayerSprites implements DebugPanel.ActorRenderer {
         try {
             int x=Math.round(actor.x()),y=Math.round(actor.y());
             if(visual.state()==RpgVisualState.State.ATTACK) {
-                effect.setColor(mystic?new Color(165,125,255):new Color(255,220,110));effect.setStroke(new
BasicStroke(3));
+                effect.setColor(spirit?new Color(165,125,255):new Color(255,220,110));effect.setStroke(new
BasicStroke(3));
                 int angle=switch(animation.pose().direction()){case FRONT->210;case BACK->30;case
LEFT->120;case RIGHT->300;};
-                if(mystic)effect.drawOval(x-32,y-45,64,64);else effect.drawArc(x-32,y-45,64,64,angle,120);
+                if(spirit)effect.drawOval(x-32,y-45,64,64);else effect.drawArc(x-32,y-45,64,64,angle,120);
             } else if(visual.state()==RpgVisualState.State.HIT) {
                 effect.setColor(new Color(255,100,100,190));effect.setStroke(new BasicStroke(2));
                 effect.drawOval(x-24,y-24,48,30);
```

### 21. [M] server/src/main/java/com/projetok/prototype/rpg/RpgVisualState.java

```
diff --git a/server/src/main/java/com/projetok/prototype/rpg/RpgVisualState.java
b/server/src/main/java/com/projetok/prototype/rpg/RpgVisualState.java
index e395efc5075c17c2f96a67c8c589b958cd90cbb4..badad2c87d7967b55fc732d39cf8fc08c649156f 100644
--- a/server/src/main/java/com/projetok/prototype/rpg/RpgVisualState.java
+++ b/server/src/main/java/com/projetok/prototype/rpg/RpgVisualState.java
@@ -21,7 +21,7 @@ public final class RpgVisualState {
         for(var event:view.events())if(event.sequence()>sequence) {
             sequence=event.sequence();
             if(event.kind()==RpgCombatEvent.Kind.HIT&&event.target()==1){state=State.HIT;until=event.tick()+6;}
-            else
if(event.kind()==RpgCombatEvent.Kind.ATTACK&&state!=State.HIT){state=State.ATTACK;until=event.tick()+9;}
+            else if((event.kind()==RpgCombatEvent.Kind.ATTACK||event.kind()==RpgCombatEvent.Kind.PUNCH)&&state!
=State.HIT){state=State.ATTACK;until=event.tick()+9;}
         }
         if(view.ticks()>=until&&state!=State.KO)state=moving?State.WALK:State.IDLE;
         return state;
```

### 22. [D] server/src/main/java/com/projetok/rpg/Archetypes.java

```
diff --git a/server/src/main/java/com/projetok/rpg/Archetypes.java
b/server/src/main/java/com/projetok/rpg/Archetypes.java
deleted file mode 100644
index c711c9e1cccfc1102e76b23461303a93c4736dd4..0000000000000000000000000000000000000000
--- a/server/src/main/java/com/projetok/rpg/Archetypes.java
+++ /dev/null
@@ -1,21 +0,0 @@
-package com.projetok.rpg;
-import java.io.*;
-import java.util.*;
-public final class Archetypes {
-    public record Definition(String id,int hp,int attack,String ability,int cost,int cooldown,int effect) {
-        public Definition { if(hp<1||hp>1000||attack<1||attack>100||!Set.of("dash","mend").contains(ability)||c
ost<1||cost>100||cooldown<1||cooldown>900||effect<1||effect>200)throw new IllegalArgumentException("Arquetipo
invalido"); }
-    }
-    private Archetypes() { }
-    public static Definition get(String id) {
-        if(!Set.of("guardian","mystic").contains(id))throw new IllegalArgumentException("Arquetipo
desconhecido");
-        return Builtin.VALUES.get(id);
-    }
-    private static final class Builtin {static final Map<String,Definition>
VALUES=Map.of("guardian",read("guardian"),"mystic",read("mystic"));}
-    private static Definition read(String id) {
-        try(InputStream in=Archetypes.class.getResourceAsStream("/rpg/archetypes.properties")) {
-            if(in==null)throw new IOException("Arquetipos ausentes"); Properties p=new Properties();p.load(in);
-            String[] v=p.getProperty(id).split("\\|");
-            return new Definition(id,Integer.parseInt(v[0]),Integer.parseInt(v[1]),v[2],Integer.parseInt(v[3]),
Integer.parseInt(v[4]),Integer.parseInt(v[5]));
-        }catch(IOException e){throw new IllegalStateException(e);}
-    }
-}
```

### 23. [M] server/src/main/java/com/projetok/rpg/CharacterIdentity.java

```
diff --git a/server/src/main/java/com/projetok/rpg/CharacterIdentity.java
b/server/src/main/java/com/projetok/rpg/CharacterIdentity.java
index f885a0357a3150c7ec522e37aa7a534a2aac37c4..896658426dd2cc90f9815cbebfc2f327335c04ff 100644
--- a/server/src/main/java/com/projetok/rpg/CharacterIdentity.java
+++ b/server/src/main/java/com/projetok/rpg/CharacterIdentity.java
@@ -11,12 +11,14 @@ public record CharacterIdentity(UUID id, String name, String archetype, String a
         Objects.requireNonNull(id);
         name = Normalizer.normalize(Objects.requireNonNull(name).strip(), Normalizer.Form.NFC);
         if (!name.matches("[\\p{L}][\\p{L}\\p{N} _-]{2,23}")) throw new IllegalArgumentException("Nome: 3 a 24
letras/numeros, espacos, _ ou -");
-        if (!java.util.Set.of("guardian", "mystic").contains(archetype)) throw new
IllegalArgumentException("Arquetipo desconhecido");
+        archetype = CharacterRaces.normalizeLegacy(archetype);
         if (!java.util.Set.of("azure", "amber").contains(appearance) || !"village".equals(initialRegion)
                 || baseHp < 1 || baseHp > 1000 || baseAttack < 1 || baseAttack > 100) throw new
IllegalArgumentException("Identidade invalida");
     }
-    public static CharacterIdentity create(String name, String archetype, String appearance) {
-        var definition=Archetypes.get(archetype);
-        return new CharacterIdentity(UUID.randomUUID(), name, archetype, appearance,
"village",definition.hp(),definition.attack());
+    /** Nome do componente mantido para compatibilidade do snapshot; o valor agora e uma raca. */
+    public String race() { return archetype; }
+    public static CharacterIdentity create(String name, String race, String appearance) {
+        var definition=CharacterRaces.get(race);
+        return new CharacterIdentity(UUID.randomUUID(), name, race, appearance,
"village",definition.hp(),definition.attack());
     }
 }
```

### 24. [A] server/src/main/java/com/projetok/rpg/CharacterRaces.java

```
diff --git a/server/src/main/java/com/projetok/rpg/CharacterRaces.java
b/server/src/main/java/com/projetok/rpg/CharacterRaces.java
new file mode 100644
index 0000000000000000000000000000000000000000..58a5f6c49b50c2476e6667b7dad5f62f6f1e91ee
--- /dev/null
+++ b/server/src/main/java/com/projetok/rpg/CharacterRaces.java
@@ -0,0 +1,34 @@
+package com.projetok.rpg;
+
+import java.util.List;
+import java.util.Objects;
+
+/** Origens disponiveis na criacao. Valores de combate sao provisórios e comuns a todas. */
+public final class CharacterRaces {
+    public record Definition(String id, String name, int hp, int attack, String startingAttack,
+                             String startingAttackName, String description) { }
+    private static final List<Definition> VALUES = List.of(
+        new Definition("reaper", "Reaper", 100, 18, "sword", "Espada pequena",
+            "Aprendiz de ceifador. Comeca com uma espada pequena, sem elemento."),
+        new Definition("human", "Human", 100, 18, "unarmed", "Socos comuns",
+            "Aprendiz humano. Comeca com socos comuns, sem luz ou especializacao marcial."),
+        new Definition("aetheri", "Aetheri", 100, 18, "unarmed", "Socos comuns",
+            "Aprendiz Aetheri. Comeca desarmado; os caminhos de arco chegam na iniciacao."),
+        new Definition("spirit", "Spirit", 100, 18, "beam", "Feixe espiritual luminoso",
+            "Espirito iniciante. Seus feixes luminosos basicos nao pertencem ao dominio Luz.")
+    );
+    private CharacterRaces() { }
+    public static List<Definition> all() { return VALUES; }
+    public static Definition get(String id) {
+        return VALUES.stream().filter(r -> r.id().equals(id)).findFirst()
+            .orElseThrow(() -> new IllegalArgumentException("Raca desconhecida: " + id));
+    }
+    /** Somente leitura de identidades antigas; a criacao exige um ID atual. */
+    public static String normalizeLegacy(String id) {
+        return switch (Objects.requireNonNull(id, "raca")) {
+            case "guardian" -> "reaper";
+            case "mystic" -> "human";
+            default -> get(id).id();
+        };
+    }
+}
```

### 25. [M] server/src/main/java/com/projetok/rpg/CharacterSaves.java

```
diff --git a/server/src/main/java/com/projetok/rpg/CharacterSaves.java
b/server/src/main/java/com/projetok/rpg/CharacterSaves.java
index 6324d86b97b3b8b84256606db8cd41104078c81d..7006296063890b0e6216af8c04f660aa7b7528e9 100644
--- a/server/src/main/java/com/projetok/rpg/CharacterSaves.java
+++ b/server/src/main/java/com/projetok/rpg/CharacterSaves.java
@@ -5,7 +5,7 @@ import java.util.*;
 /** Catalogo de saves locais, sem apagar entradas corrompidas. Chamar fora do EDT/tick. */
 public final class CharacterSaves {
     public record Slot(Path file,RpgSave state,String problem,boolean backup) {
-        public String toString(){return state==null?file.getFileName()+" - invalido:
"+problem:state.identity().name()+" | "+state.identity().archetype()+" | XP "+state.xp()+(backup?" | BACKUP
RECUPERAVEL":"");}
+        public String toString(){return state==null?file.getFileName()+" - invalido:
"+problem:state.identity().name()+" | "+CharacterRaces.get(state.identity().race()).name()+" | XP
"+state.xp()+(backup?" | BACKUP RECUPERAVEL":"");}
     }
     private final Path folder;
     public CharacterSaves(Path folder){this.folder=folder.toAbsolutePath();}
```

### 26. [A] server/src/main/java/com/projetok/rpg/DomainProgression.java

```
diff --git a/server/src/main/java/com/projetok/rpg/DomainProgression.java
b/server/src/main/java/com/projetok/rpg/DomainProgression.java
new file mode 100644
index 0000000000000000000000000000000000000000..f8e7a780fbb8315735bff1538dd3955be54e98cf
--- /dev/null
+++ b/server/src/main/java/com/projetok/rpg/DomainProgression.java
@@ -0,0 +1,83 @@
+package com.projetok.rpg;
+
+import java.time.LocalDate;
+import java.util.HashSet;
+import java.util.Objects;
+import java.util.Set;
+
+/** XP de dominio em meias unidades inteiras: 3 unidades representam exatamente 1,5 XP. */
+public record DomainProgression(String domain, long xpHalfUnits, LocalDate lastReset, boolean
resetQuestComplete) {
+    public static final String NONE = "none";
+    public static final int INITIATION_LEVEL = 15;
+    public static final int SHUNPO_LEVEL = 20;
+    public DomainProgression {
+        Objects.requireNonNull(domain, "dominio");
+        if (!domain.equals(NONE)) Domains.get(domain);
+        if (xpHalfUnits < 0 || domain.equals(NONE) && xpHalfUnits != 0)
+            throw new IllegalArgumentException("XP de dominio invalida");
+    }
+    public static DomainProgression initial() { return new DomainProgression(NONE, 0, null, false); }
+    public boolean chosen() { return !domain.equals(NONE); }
+    public double xp() { return xpHalfUnits / 2.0; }
+    public String xpText() { return (xpHalfUnits / 2) + (xpHalfUnits % 2 == 0 ? "" : ",5"); }
+    public DomainProgression choose(String race, int characterLevel, String selected) {
+        if (characterLevel < INITIATION_LEVEL) throw new IllegalStateException("Iniciacao disponivel no nivel
15");
+        if (chosen()) throw new IllegalStateException("Dominio ja escolhido; reset necessario");
+        if (!Domains.get(selected).race().equals(CharacterRaces.get(race).id()))
+            throw new IllegalArgumentException("Dominio indisponivel para esta raca");
+        return new DomainProgression(selected, 0, lastReset, resetQuestComplete);
+    }
+    /** Recompensas de quests e mobs usam metade da XP normal; sem dominio, nao acumulam maestria. */
+    public DomainProgression award(long characterXp) {
+        if (characterXp < 0) throw new IllegalArgumentException("Recompensa invalida");
+        return chosen() ? new DomainProgression(domain, Math.addExact(xpHalfUnits, characterXp), lastReset,
resetQuestComplete) : this;
+    }
+    public DomainProgression completeResetQuest() { return new DomainProgression(domain, xpHalfUnits,
lastReset, true); }
+    public LocalDate nextResetDate() { return lastReset == null ? null : lastReset.plusMonths(1); }
+    /** Valida tudo antes de retornar. O chamador debita cash somente apos o sucesso. */
+    public DomainProgression reset(LocalDate today, long cashBalance, DomainResetPolicy policy) {
+        Objects.requireNonNull(today); Objects.requireNonNull(policy);
+        if (cashBalance < 0) throw new IllegalArgumentException("Saldo de cash invalido");
+        if (!policy.enabled()) throw new IllegalStateException("Reset indisponivel");
+        if (!chosen()) throw new IllegalStateException("Personagem sem dominio");
+        if (!resetQuestComplete) throw new IllegalStateException("Conclua a quest de reset");
+        if (nextResetDate() != null && today.isBefore(nextResetDate())) throw new
IllegalStateException("Aguarde um mes desde o ultimo reset");
+        if (cashBalance < policy.cashCost()) throw new IllegalStateException("Cash insuficiente");
+        return new DomainProgression(NONE, 0, today, false);
+    }
+    public static DomainProgression read(Set<String> flags) {
+        String domain = NONE; long xp = 0; LocalDate reset = null; boolean quest = false;
+        Set<String> seen = new HashSet<>();
+        for (String flag : flags) {
+            String key = flag.substring(0, flag.indexOf(':') < 0 ? flag.length() : flag.indexOf(':'));
+            if (!Set.of("domain", "mastery-half", "reset-last", "reset-quest").contains(key)) continue;
+            if (!seen.add(key) || flag.indexOf(':') < 0) throw new IllegalArgumentException("Estado de dominio
duplicado/invalido");
+            String value = flag.substring(flag.indexOf(':') + 1);
+            switch (key) {
+                case "domain" -> domain = value;
+                case "mastery-half" -> xp = Long.parseLong(value);
+                case "reset-last" -> {
+                    try { reset = LocalDate.parse(value); }
+                    catch (java.time.DateTimeException invalid) { throw new IllegalArgumentException("Data de
reset invalida", invalid); }
+                }
```

```
+                case "reset-quest" -> {
+                    if (!value.equals("complete")) throw new IllegalArgumentException("Quest de reset
invalida");
+                    quest = true;
+                }
+                default -> throw new IllegalArgumentException("Estado de dominio invalido");
+            }
+        }
+        return new DomainProgression(domain, xp, reset, quest);
+    }
+    public void writeTo(Set<String> flags) {
+        flags.removeIf(f -> f.startsWith("domain:") || f.startsWith("mastery-half:") ||
f.startsWith("reset-last:") || f.startsWith("reset-quest:") || !chosen() && f.startsWith("domain-skill:"));
+        flags.add("domain:" + domain);
+        flags.add("mastery-half:" + xpHalfUnits);
+        if (lastReset != null) flags.add("reset-last:" + lastReset);
+        if (resetQuestComplete) flags.add("reset-quest:complete");
+    }
+    public void validateFor(CharacterIdentity identity, long characterXp) {
+        if (chosen() && (!Domains.get(domain).race().equals(identity.race()) ||
RpgProgression.builtin().level(characterXp) < INITIATION_LEVEL))
+            throw new IllegalArgumentException("Dominio incompativel com personagem");
+    }
+}
```

### 27. [A] server/src/main/java/com/projetok/rpg/DomainResetPolicy.java

```
diff --git a/server/src/main/java/com/projetok/rpg/DomainResetPolicy.java
b/server/src/main/java/com/projetok/rpg/DomainResetPolicy.java
new file mode 100644
index 0000000000000000000000000000000000000000..9cea589e26a574a12d57008754880131d410ad90
--- /dev/null
+++ b/server/src/main/java/com/projetok/rpg/DomainResetPolicy.java
@@ -0,0 +1,14 @@
+package com.projetok.rpg;
+
+/** Regra do reset: 35 Spirit Coins. Compra e validacao financeira pertencem ao servico online. */
+public record DomainResetPolicy(long cashCost) {
+    public DomainResetPolicy {
+        if (cashCost < -1 || cashCost == 0) throw new IllegalArgumentException("Custo de reset invalido");
+    }
+    public boolean enabled() { return cashCost > 0; }
+    public static DomainResetPolicy disabled() { return new DomainResetPolicy(-1); }
+    public static DomainResetPolicy builtin() {
+        String configured = System.getProperty("spiritgate.domainReset.cashCost");
+        return configured == null ? new DomainResetPolicy(35) : new
DomainResetPolicy(Long.parseLong(configured));
+    }
+}
```

### 28. [A] server/src/main/java/com/projetok/rpg/Domains.java

```
diff --git a/server/src/main/java/com/projetok/rpg/Domains.java
b/server/src/main/java/com/projetok/rpg/Domains.java
new file mode 100644
index 0000000000000000000000000000000000000000..32a504fb9f44d4b8126674636231289f9bb8efb2
--- /dev/null
+++ b/server/src/main/java/com/projetok/rpg/Domains.java
@@ -0,0 +1,43 @@
+package com.projetok.rpg;
+
+import java.util.List;
+
+/** Catalogo de caminhos e previews. As tecnicas abaixo ainda nao sao skills executaveis. */
+public final class Domains {
+    public record Definition(String id, String race, String name, String description,
+                             List<String> techniquePreviews) {
+        public Definition { techniquePreviews = List.copyOf(techniquePreviews); }
+    }
+    private static final List<Definition> VALUES = List.of(
+        new Definition("water", "reaper", "Agua", "Fluidez, aparos e contra-ataques.",
+            List.of("Aparo fluido: recebe o golpe e prepara uma resposta.", "Corte de mare: corte de espada com
 fluxo de agua.")),
+        new Definition("thunder", "reaper", "Trovao", "Investidas e golpes rapidos.",
+            List.of("Avanco eletrico: aproximacao veloz.", "Corte relampago: golpe rapido com abertura apos
errar.")),
+        new Definition("fire", "reaper", "Fogo", "Pressao, queimadura e controle de espaco.",
+            List.of("Corte ardente: pressiona o alvo com fogo.", "Rastro de brasas: ocupa uma faixa do
terreno.")),
+        new Definition("wind", "reaper", "Vento", "Alcance de cortes e reposicionamento.",
+            List.of("Corte de ar: projeta o alcance da espada.", "Passo de brisa: reposiciona o espadachim.")),
+        new Definition("light", "human", "Luz", "Ofensiva, protecao e cura; pode ferir qualquer raca.",
+            List.of("Pulso luminoso: atinge um adversario de qualquer origem.", "Cura canalizada: recuperacao
com tempo de execucao.", "Barreira de luz: protecao espiritual.")),
+        new Definition("martial", "human", "Marcial", "Evolucao avancada do combate desarmado.",
+            List.of("Sequencia corporal: combina golpes proximos.", "Interrupcao marcial: explora uma abertura
do adversario.")),
+        new Definition("bow_long", "aetheri", "Arco de longa distancia", "Precisao, tiros carregados e
manutencao de distancia.",
+            List.of("Disparo concentrado: prepara uma flecha precisa.", "Tiro distante: alcanca alvos
afastados.")),
+        new Definition("bow_short", "aetheri", "Arco de curta distancia", "Disparos rapidos e mobilidade sob
pressao.",
+            List.of("Rajada proxima: dispara em sequencia.", "Leque de flechas: cobre uma abertura proxima.")),
+        new Definition("darkness_debuff", "spirit", "Trevas: efeitos negativos", "Marcas, enfraquecimento e
dano ao longo do tempo.",
+            List.of("Marca sombria: prepara efeitos negativos.", "Desgaste espiritual: enfraquece o alvo ao
longo do tempo.")),
+        new Definition("darkness_control", "spirit", "Trevas: controle", "Lentidao, imobilizacoes breves e
deslocamento. Pode atingir aliados, inclusive da party.",
+            List.of("Zona sombria: reduz o movimento dentro da area.", "Prisão espiritual: restringe movimento
por um periodo breve. Aliados tambem podem ser atingidos."))
+    );
+    private Domains() { }
+    public static List<Definition> all() { return VALUES; }
+    public static List<Definition> forRace(String race) {
+        CharacterRaces.get(race);
+        return VALUES.stream().filter(d -> d.race().equals(race)).toList();
+    }
+    public static Definition get(String id) {
+        return VALUES.stream().filter(d -> d.id().equals(id)).findFirst()
+            .orElseThrow(() -> new IllegalArgumentException("Dominio desconhecido: " + id));
+    }
+}
```

### 29. [M] server/src/main/java/com/projetok/rpg/EnemyService.java

```
diff --git a/server/src/main/java/com/projetok/rpg/EnemyService.java
b/server/src/main/java/com/projetok/rpg/EnemyService.java
index d729cd2d0a52968f6e00c116c90eb43b5fe1d37f..beeee15e6b27bb19fc4364d5a82ace15d40e4c1a 100644
--- a/server/src/main/java/com/projetok/rpg/EnemyService.java
+++ b/server/src/main/java/com/projetok/rpg/EnemyService.java
@@ -6,7 +6,10 @@ import java.io.*;

 /** Populacao finita definida em conteudo; estado persistente de cada spawn. */
 public final class EnemyService {
-    public record Definition(String id,int hp,int attack,int speed,int aggro,int leash,int respawn) { }
+    public record Definition(String id,int hp,int attack,int speed,int aggro,int leash,int respawn,int level) {
+        public Definition { if(level!=1)throw new IllegalArgumentException("Recompensas de mobs acima do nivel
1 ainda nao definidas"); }
+        public Definition(String id,int hp,int attack,int speed,int aggro,int leash,int respawn)
{this(id,hp,attack,speed,aggro,leash,respawn,1);}
+    }
     public static final class Enemy extends GameEntity {
         final String spawn,region; final Definition definition; final Position home;
         int hp; long generation=1,readyTick,attackTick; String behavior="PATROL";
@@ -23,10 +26,10 @@ public final class EnemyService {
         this.geometry=geometry;
         Properties definitions=read("enemies"),spawns=new Properties();spawns.putAll(spawnData);
Map<String,Definition> types=new TreeMap<>();
         for(String id:definitions.stringPropertyNames()) {
-            String[] v=definitions.getProperty(id).split("\\|");if(v.length!=6)throw new
IllegalArgumentException("Inimigo invalido");
+            String[] v=definitions.getProperty(id).split("\\|");if(v.length!=6&&v.length!=7)throw new
IllegalArgumentException("Inimigo invalido");
             content.get("enemy",id); int[] n=Arrays.stream(v).mapToInt(Integer::parseInt).toArray();
             if(n[0]<1||n[0]>10000||n[1]<1||n[1]>100||n[2]<1||n[2]>200||n[3]<40||n[3]>800||n[4]<n[3]||n[4]>900||
n[5]<0||n[5]>9000)throw new IllegalArgumentException("Parametros PvE invalidos");
-            types.put(id,new Definition(id,n[0],n[1],n[2],n[3],n[4],n[5]));
+            types.put(id,new Definition(id,n[0],n[1],n[2],n[3],n[4],n[5],n.length==7?n[6]:1));
         }
         if(spawns.size()>64)throw new IllegalArgumentException("Populacao excessiva"); int entityId=100;
         for(String spawn:new TreeSet<>(spawns.stringPropertyNames())) {
```

### 30. [M] server/src/main/java/com/projetok/rpg/LocalSaveStore.java

```
diff --git a/server/src/main/java/com/projetok/rpg/LocalSaveStore.java
b/server/src/main/java/com/projetok/rpg/LocalSaveStore.java
index 8e2e553cae1d97be44b057d65818f7a2bf106dc0..bde2747793c147dfb069ae35f6eb04f3fe7598a1 100644
--- a/server/src/main/java/com/projetok/rpg/LocalSaveStore.java
+++ b/server/src/main/java/com/projetok/rpg/LocalSaveStore.java
@@ -37,8 +37,8 @@ public final class LocalSaveStore {
     public static byte[] encode(RpgSave state) throws IOException {
         Properties p=new Properties();
         var i=state.identity();
-        p.setProperty("schema","1"); p.setProperty("id",i.id().toString()); p.setProperty("name",i.name());
-        p.setProperty("archetype",i.archetype()); p.setProperty("appearance",i.appearance());
+        p.setProperty("schema","2"); p.setProperty("id",i.id().toString()); p.setProperty("name",i.name());
+        p.setProperty("race",i.race()); p.setProperty("appearance",i.appearance());
         p.setProperty("baseHp",""+i.baseHp()); p.setProperty("baseAttack",""+i.baseAttack());
         p.setProperty("region",state.region()); p.setProperty("x",""+state.x());
p.setProperty("y",""+state.y());
         p.setProperty("hp",""+state.hp()); p.setProperty("xp",""+state.xp());
p.setProperty("coins",""+state.coins()); p.setProperty("ticks",""+state.ticks());
@@ -46,7 +46,7 @@ public final class LocalSaveStore {
         state.equipment().forEach((k,v)->p.setProperty("equipment."+k,v));
         state.quests().forEach((k,v)->p.setProperty("quest."+k,""+v));
         state.flags().forEach(k->p.setProperty("flag."+k,"true"));
-        ByteArrayOutputStream out=new ByteArrayOutputStream(); p.store(out,"Spirit Gate offline V1"); return
out.toByteArray();
+        ByteArrayOutputStream out=new ByteArrayOutputStream(); p.store(out,"Spirit Gate offline V2 - race and
domain"); return out.toByteArray();
     }
     public static RpgSave decode(byte[] bytes) throws IOException {
         if(bytes.length>262144) throw new IOException("Save excessivo");
@@ -57,8 +57,11 @@ public final class LocalSaveStore {
         };
         p.load(new ByteArrayInputStream(bytes));
         try {
-            if(!"1".equals(take(p,"schema"))) throw new IllegalArgumentException("Schema de save
incompativel");
-            CharacterIdentity identity=new CharacterIdentity(UUID.fromString(take(p,"id")),take(p,"name"),take(
p,"archetype"),take(p,"appearance"),"village",Integer.parseInt(take(p,"baseHp")),Integer.parseInt(take(p,"baseAt
tack")));
+            String schema=take(p,"schema");
+            if(!Set.of("1","2").contains(schema)) throw new IllegalArgumentException("Schema de save
incompativel");
+            String previousRace=take(p,schema.equals("1")?"archetype":"race");
+            if(schema.equals("2"))CharacterRaces.get(previousRace);
+            CharacterIdentity identity=new CharacterIdentity(UUID.fromString(take(p,"id")),take(p,"name"),previ
ousRace,take(p,"appearance"),"village",Integer.parseInt(take(p,"baseHp")),Integer.parseInt(take(p,"baseAttack"))
);
             String region=take(p,"region"); float
x=Float.parseFloat(take(p,"x")),y=Float.parseFloat(take(p,"y"));
             int hp=Integer.parseInt(take(p,"hp")),coins=Integer.parseInt(take(p,"coins")); long
xp=Long.parseLong(take(p,"xp")),ticks=Long.parseLong(take(p,"ticks"));
             Map<String,Integer> inventory=new TreeMap<>(),quests=new TreeMap<>(); Map<String,String>
equipment=new TreeMap<>(); Set<String> flags=new TreeSet<>();
@@ -70,6 +73,7 @@ public final class LocalSaveStore {
                 else if(key.startsWith("flag.") && "true".equals(v)) flags.add(key.substring(5));
                 else throw new IllegalArgumentException("Campo desconhecido: "+key);
             }
+            if(schema.equals("1")&&!previousRace.equals(identity.race()))flags.add("migration:"+previousRace+"-
to-"+identity.race());
             return new RpgSave(identity,region,x,y,hp,xp,coins,inventory,equipment,quests,flags,ticks);
         } catch(NullPointerException | NumberFormatException e) { throw new IllegalArgumentException("Save
incompleto/corrompido",e); }
     }
```

### 31. [M] server/src/main/java/com/projetok/rpg/RpgCombatEvent.java

```
diff --git a/server/src/main/java/com/projetok/rpg/RpgCombatEvent.java
b/server/src/main/java/com/projetok/rpg/RpgCombatEvent.java
index 9eee18570b8cc4d734c2ffa560474495b2ec0d09..163bb9531c554997e87b584450326459501ec36d 100644
--- a/server/src/main/java/com/projetok/rpg/RpgCombatEvent.java
+++ b/server/src/main/java/com/projetok/rpg/RpgCombatEvent.java
@@ -3,5 +3,5 @@ package com.projetok.rpg;
 /** Evento efemero de combate confirmado pelo mundo; nao integra o formato de save. */
 public record RpgCombatEvent(long sequence,long tick,Kind kind,int target,float x,float y,int amount) {
     /** Natureza real da acao ou consequencia; REJECT nunca representa dano. */
-    public enum Kind { ATTACK, HIT, HEAL, KO, REJECT }
+    public enum Kind { ATTACK, PUNCH, HIT, HEAL, KO, REJECT }
 }
```

### 32. [M] server/src/main/java/com/projetok/rpg/RpgProgression.java

```
diff --git a/server/src/main/java/com/projetok/rpg/RpgProgression.java
b/server/src/main/java/com/projetok/rpg/RpgProgression.java
index 25c44bd895c5fe781fffdac7abc048c33979635d..a1c121b8223161955026c71f597bab3a928b84dc 100644
--- a/server/src/main/java/com/projetok/rpg/RpgProgression.java
+++ b/server/src/main/java/com/projetok/rpg/RpgProgression.java
@@ -1,16 +1,20 @@
 package com.projetok.rpg;
 import java.io.*;
 import java.util.Properties;
-/** XP cumulativa triangular; parametros RPG independentes da arena. */
+/** Curva provisoria cumulativa. cap=0 nao estabelece limite de nivel; 1M e limite tecnico de save. */
 public record RpgProgression(int cap,int step,int hpPerLevel,int attackPerLevel) {
-    public RpgProgression {
if(cap<2||cap>20||step<1||step>1000||hpPerLevel<0||hpPerLevel>100||attackPerLevel<0||attackPerLevel>20) throw
new IllegalArgumentException("Progressao invalida"); }
+    public RpgProgression { if(cap<0||cap==1||cap>1000||step<1||step>1000||hpPerLevel<0||hpPerLevel>100||attack
PerLevel<0||attackPerLevel>20) throw new IllegalArgumentException("Progressao invalida"); }
     public static RpgProgression builtin() {
         try(InputStream in=RpgProgression.class.getResourceAsStream("/rpg/progression.properties")) {
             if(in==null) throw new IOException("Progressao ausente"); Properties p=new Properties();p.load(in);
             return new RpgProgression(Integer.parseInt(p.getProperty("cap")),Integer.parseInt(p.getProperty("st
ep")),Integer.parseInt(p.getProperty("hpPerLevel")),Integer.parseInt(p.getProperty("attackPerLevel")));
         } catch(IOException e) { throw new IllegalStateException(e); }
     }
-    public long threshold(int level) { if(level<1||level>cap) throw new IllegalArgumentException("Nivel
invalido"); return (long)step*level*(level-1)/2; }
-    public int level(long xp) { if(xp<0||xp>1_000_000) throw new IllegalArgumentException("XP invalida"); int
level=1;while(level<cap&&xp>=threshold(level+1))level++;return level; }
-    public long award(long xp,long amount) { level(xp);if(amount<0||amount>1_000_000)throw new
IllegalArgumentException("Recompensa invalida");return Math.min(threshold(cap),xp+amount); }
+    public long threshold(int level) { if(level<1||cap>0&&level>cap) throw new IllegalArgumentException("Nivel
invalido"); return (long)step*level*(level-1)/2; }
+    public int level(long xp) { if(xp<0||xp>1_000_000) throw new IllegalArgumentException("XP invalida"); int
level=1;while((cap==0||level<cap)&&xp>=threshold(level+1))level++;return level; }
+    public long award(long xp,long amount) {
+        level(xp);if(amount<0||amount>1_000_000)throw new IllegalArgumentException("Recompensa invalida");
+        long ceiling=cap==0?1_000_000:Math.min(1_000_000,threshold(cap));
+        return Math.max(xp,Math.min(ceiling,xp+amount));
+    }
 }
```

### 33. [M] server/src/main/java/com/projetok/rpg/RpgSave.java

```
diff --git a/server/src/main/java/com/projetok/rpg/RpgSave.java
b/server/src/main/java/com/projetok/rpg/RpgSave.java
index 200e36204e92c94df77104f3602b34554eded0c1..1f8504e46604f91583441f3cc26144886025ee9f 100644
--- a/server/src/main/java/com/projetok/rpg/RpgSave.java
+++ b/server/src/main/java/com/projetok/rpg/RpgSave.java
@@ -2,7 +2,7 @@ package com.projetok.rpg;

 import java.util.*;

-/** Snapshot persistente V1, copias profundas de colecoes e limites verificaveis. */
+/** Snapshot persistente, copias profundas e limites tecnicos; dominio e XP exata ficam nas flags. */
 public record RpgSave(CharacterIdentity identity, String region, float x, float y, int hp,
                       long xp, int coins, Map<String,Integer> inventory, Map<String,String> equipment,
                       Map<String,Integer> quests, Set<String> flags, long ticks) {
@@ -19,5 +19,6 @@ public record RpgSave(CharacterIdentity identity, String region, float x, float
         equipment.forEach((slot,id)->{ if(!Set.of("weapon","torso","legs","accessory").contains(slot)) throw
new IllegalArgumentException("Slot invalido"); content.get("item",id); });
         quests.forEach((id,q)->{ content.get("quest",id); if(q<0||q>10000) throw new
IllegalArgumentException("Quest invalida"); });
         for(String flag:flags) if(!flag.matches("[a-zA-Z0-9_:.-]{1,100}")) throw new
IllegalArgumentException("Flag invalida");
+        DomainProgression.read(flags).validateFor(identity, xp);
     }
 }
```

### 34. [M] server/src/main/java/com/projetok/rpg/RpgSecondaryAbility.java

```
diff --git a/server/src/main/java/com/projetok/rpg/RpgSecondaryAbility.java
b/server/src/main/java/com/projetok/rpg/RpgSecondaryAbility.java
index 8672974f2a91cb3b61daa55c329f289923fbc181..342e86a1ff5b1de98c372ceac38eeeb7e532afbb 100644
--- a/server/src/main/java/com/projetok/rpg/RpgSecondaryAbility.java
+++ b/server/src/main/java/com/projetok/rpg/RpgSecondaryAbility.java
@@ -1,13 +1,10 @@
 package com.projetok.rpg;

-/** Habilidade adicional experimental; compartilha recarga persistente com a habilidade primaria. */
+/** Metadados da tecnica livre defensiva, compartilhada por todas as racas. */
 public record RpgSecondaryAbility(String name,int range,int bonus,int cost,int cooldown) {
-    /** Parametros conservadores dos dois arquetipos; custos em pontos de vigor. */
+    /** Compatibilidade de apresentacao; nao concede habilidades de dominio ao nascer. */
     public static RpgSecondaryAbility forArchetype(String id) {
-        return switch(id) {
-            case "guardian"->new RpgSecondaryAbility("Golpe de Impacto",82,6,20,120);
-            case "mystic"->new RpgSecondaryAbility("Pulso Arcano",110,4,25,150);
-            default->throw new IllegalArgumentException("Arquetipo desconhecido");
-        };
+        CharacterRaces.get(CharacterRaces.normalizeLegacy(id));
+        return new RpgSecondaryAbility("Shunpo defensivo",70,0,RpgWorld.SHUNPO_COST,RpgWorld.SHUNPO_COOLDOWN);
     }
 }
```

### 35. [M] server/src/main/java/com/projetok/rpg/RpgWorld.java

```
diff --git a/server/src/main/java/com/projetok/rpg/RpgWorld.java
b/server/src/main/java/com/projetok/rpg/RpgWorld.java
index f532427a5dc33ecd948c1750bf616de24735251b..1cdaf6dfba3d72c459c69421669a696eeb4667fd 100644
--- a/server/src/main/java/com/projetok/rpg/RpgWorld.java
+++ b/server/src/main/java/com/projetok/rpg/RpgWorld.java
@@ -29,26 +29,62 @@ public final class RpgWorld {
     private boolean transitioning;
     private int stamina=1000;
     private long abilityReadyTick;
+    private long evadeUntilTick;
+    private DomainProgression domain = DomainProgression.initial();
+    public static final int SHUNPO_LEVEL = 20;
+    public static final int SHUNPO_COST = 25, SHUNPO_COOLDOWN = 90;
     private float facingX=1,facingY;
+    public DomainProgression domainProgression() { return domain; }
+    public boolean chooseDomain(String id) {
+        if(hp==0||transitioning)return false;
+        try { domain=domain.choose(identity.race(),level(),id); }
+        catch(IllegalArgumentException|IllegalStateException rejected) { message=rejected.getMessage();return
false; }
+        message="Dominio escolhido: "+Domains.get(id).name();return true;
+    }
+    /** Dados de pagamento nao sao inventados pelo cliente offline. A regra pura de reset
+     * fica em DomainProgression; a execucao paga depende do servico de carteira. */
+    public String resetRequirements() {
+        return "Reset: 35 Spirit Coins (R$ 35), missao de reset e intervalo de um mes. "
+            +"Pagamento e missao ainda indisponiveis; nivel, itens e equipamentos serao preservados.";
+    }
+    public boolean canShunpo() { return level()>=SHUNPO_LEVEL; }
+    /** Shunpo ofensivo: aproxima de alvo visivel sem atravessar geometria. */
     public boolean castAbility() {
-        int beforeHp=hp;
-        var ability=Archetypes.get(identity.archetype());
-        if(hp==0||transitioning||ticks<abilityReadyTick||stamina<ability.cost()*10||ability.ability().equals("m
end")&&hp==maxHp())return false;
-        if(ability.ability().equals("dash"))
movement.moveContinuous(body,facingX*ability.effect(),facingY*ability.effect(),18,space);
-        else hp=Math.min(maxHp(),hp+ability.effect());
-        stamina-=ability.cost()*10;abilityReadyTick=ticks+ability.cooldown();
+        if(!shunpoReady())return false;
+        var position=body.getPosition();
+        var target=enemies.current(region).stream().filter(e->e.hp>0
+            &&(!e.definition.id().equals("warden")||bossUnlocked())
+            &&distance(position,e.getPosition())<=240&&lineClear(position,e.getPosition()))
+            .min(java.util.Comparator.comparingDouble(e->distance(position,e.getPosition()))).orElse(null);
+        if(target==null){message="Shunpo ofensivo exige um alvo visivel no alcance";return false;}
+        var
direction=movement.direction(position.x()-target.getPosition().x(),position.y()-target.getPosition().y());
+        if(direction.x()==0&&direction.y()==0)direction=new Position(-facingX,-facingY);
+        float x=target.getPosition().x()+direction.x()*42, y=target.getPosition().y()+direction.y()*42;
+        if(!new CollisionService().isFree(body,x,y,18,space)
+                ||!lineClear(position,new Position(x,y))){message="Destino do Shunpo bloqueado";return false;}
+        body.setPosition(new Position(x,y));facingX=-direction.x();facingY=-direction.y();
+        consumeShunpo();message="Shunpo ofensivo";return true;
+    }
+    private boolean shunpoReady() {
+        if(!canShunpo()){message="Shunpo exige nivel 20 de personagem";return false;}
+        return hp>0&&!transitioning&&ticks>=abilityReadyTick&&stamina>=SHUNPO_COST*10;
+    }
+    private void consumeShunpo() {
+        stamina-=SHUNPO_COST*10;abilityReadyTick=ticks+SHUNPO_COOLDOWN;
         flags.removeIf(f->f.startsWith("ability-until:"));flags.add("ability-until:"+abilityReadyTick);
-
if(hp>beforeHp)event(RpgCombatEvent.Kind.HEAL,1,body.getPosition().x(),body.getPosition().y(),hp-beforeHp);
-        message=ability.ability().equals("dash")?"Investida!":"Restauracao!";return true;
     }
     private String message="Bem-vindo a Aurora";
     public int level() { return progression.level(xp); }
     public int maxHp() { return identity.baseHp()+(level()-1)*progression.hpPerLevel(); }
     public void awardExperience(long amount) {
-        int previous=maxHp(); xp=progression.award(xp,amount);
+        int previous=maxHp();long nextXp=progression.award(xp,amount);
+        DomainProgression nextDomain=domain.award(amount);
+        xp=nextXp;domain=nextDomain;
         if(hp>0) hp+=maxHp()-previous;
     }
```

```
-    public void receiveDamage(int amount) { if(amount<0||amount>10000) throw new IllegalArgumentException("Dano
 invalido"); int before=hp;hp=Math.max(0,hp-amount);
+    public void receiveDamage(int amount) { if(amount<0||amount>10000) throw new IllegalArgumentException("Dano
 invalido");
+        if(amount>0&&hp>0&&ticks<evadeUntilTick){message="Shunpo: esquiva!";return;}
+        int before=hp;hp=Math.max(0,hp-amount);
         if(before>hp)event(RpgCombatEvent.Kind.HIT,1,body.getPosition().x(),body.getPosition().y(),before-hp);
         if(before>0&&hp==0)event(RpgCombatEvent.Kind.KO,1,body.getPosition().x(),body.getPosition().y(),0); }
     public boolean use(String id) {
@@ -60,7 +96,7 @@ public final class RpgWorld {
         message=item.name()+": vida recuperada"; return true;
     }
     public InventoryService inventory() { return inventory; }
-    public boolean activate(String action){boolean accepted=switch(action){case "rpg.attack"->attack();case
"rpg.ability"->castAbility();case "rpg.secondary"->castSecondaryAbility();case
"item.potion"->use("potion");default->throw new IllegalArgumentException("Acao de hotbar desconhecida");};if(!ac
cepted)event(RpgCombatEvent.Kind.REJECT,1,body.getPosition().x(),body.getPosition().y(),0);return accepted;}
+    public boolean activate(String action){boolean accepted=switch(action){case "rpg.attack"->attack();case
"rpg.unarmed"->punch();case "rpg.ability"->castAbility();case "rpg.secondary"->castSecondaryAbility();case
"item.potion"->use("potion");default->throw new IllegalArgumentException("Acao de hotbar desconhecida");};if(!ac
cepted)event(RpgCombatEvent.Kind.REJECT,1,body.getPosition().x(),body.getPosition().y(),0);return accepted;}
    public int attackPower() { return
identity.baseAttack()+(level()-1)*progression.attackPerLevel()+equipmentPower("weapon"); }
    public int defense() { return equipmentPower("torso")+equipmentPower("legs")+equipmentPower("accessory"); }
    private int equipmentPower(String slot) { return
equipment.containsKey(slot)?items.get(equipment.get(slot)).power():0; }
@@ -77,7 +113,7 @@ public final class RpgWorld {
         inventory.replace(copy); equipment.remove(slot); return true;
     }
     private final GameEntity body;
-    public RpgWorld() { this(CharacterIdentity.create("Viajante", "guardian", "azure")); }
+    public RpgWorld() { this(CharacterIdentity.create("Viajante", "reaper", "azure")); }
     public RpgWorld(CharacterIdentity identity) {this(identity,(RpgSceneSet)null);}
     /** Configuracao exclusiva de sessao sintetica; construtor padrao conserva a campanha. */
     public RpgWorld(CharacterIdentity identity,RpgSceneSet scenes) {
@@ -92,6 +128,7 @@ public final class RpgWorld {
         portals=scenes==null?new PortalService(content):new PortalService(content,scenes.portals());
         objects=scenes==null?new WorldObjects(content,items):new WorldObjects(content,items,scenes.objects());
         hp=identity.baseHp(); inventory.add("potion",3);
+        if(identity.race().equals("reaper"))equipment.put("weapon","small_sword");
         flags.add("visited:"+region);
     }
     private final MovementService movement = new MovementService(new CollisionService());
@@ -167,20 +204,31 @@ public final class RpgWorld {
         activeNpc=npc.get().id();questService.event("talk",activeNpc,activeNpc,ticks+1,quests,flags);questServi
ce.refresh(quests,flags,inventory,region);dialogueNode=npc.get().dialogue();message=npc.get().name()+": seja
bem-vindo!";return true;
     }
     public RpgWorld(CharacterIdentity identity,java.util.random.RandomGenerator random) {
this(identity);this.random=java.util.Objects.requireNonNull(random); }
-    /** Ataque basico conserva alcance, dano e recarga anteriores. */
-    public boolean attack() { return strike(identity.archetype().equals("mystic")?145:82,attackPower()); }
-    /** Acao adicional com custo e recarga compartilhada; nenhuma mudanca de formato de save. */
+    /** Ataque racial inicial; feixe de Spirit nao concede o dominio Luz. */
+    public int basicAttackRange() {
+        if(domain.domain().equals("bow_long"))return 280;
+        if(domain.domain().equals("bow_short"))return 140;
+        return switch(identity.race()){case "spirit"->180;case "reaper"->82;default->60;};
+    }
+    /** Qualquer raca pode optar por socos sem receber especializacao Marcial. */
+    public boolean punch() { return
strike(60,identity.baseAttack()+(level()-1)*progression.attackPerLevel(),true); }
+    public boolean attack() { return strike(basicAttackRange(),attackPower()); }
+    /** Shunpo defensivo com janela curta e recarga compartilhada. */
     public boolean castSecondaryAbility() {
-        var ability=RpgSecondaryAbility.forArchetype(identity.archetype());
-        if(hp==0||transitioning||ticks<abilityReadyTick||stamina<ability.cost()*10)return false;
-        if(!strike(ability.range(),attackPower()+ability.bonus()))return false;
-        stamina-=ability.cost()*10;abilityReadyTick=ticks+ability.cooldown();
-        flags.removeIf(f->f.startsWith("ability-until:"));flags.add("ability-until:"+abilityReadyTick);
-        message=ability.name()+"!";return true;
+        if(!shunpoReady())return false;
+        var position=body.getPosition();
+        movement.moveContinuous(body,-facingX*70,-facingY*70,18,space);
+        if(distance(position,body.getPosition())<1){message="Esquiva bloqueada";return false;}
+        consumeShunpo();evadeUntilTick=ticks+6;
+        flags.removeIf(f->f.startsWith("evade-until:"));flags.add("evade-until:"+evadeUntilTick);
+        message="Shunpo defensivo";return true;
```

```
     }
     private boolean strike(int range,int power) {
+        return strike(range,power,false);
+    }
+    private boolean strike(int range,int power,boolean unarmed) {
         if(hp==0||transitioning||ticks<attackReadyTick)return false;

-        EnemyService.Enemy target=enemies.current(region).stream().filter(e->e.hp>0&&(!e.definition.id().equals
("warden")||bossUnlocked())&&distance(body.getPosition(),e.getPosition())<= range)
+        EnemyService.Enemy target=enemies.current(region).stream().filter(e->e.hp>0&&(!e.definition.id().equals
("warden")||bossUnlocked())&&distance(body.getPosition(),e.getPosition())<=
range&&lineClear(body.getPosition(),e.getPosition()))

.min(java.util.Comparator.comparingDouble(e->distance(body.getPosition(),e.getPosition()))).orElse(null);
         if(target==null||!lineClear(body.getPosition(),target.getPosition()))return false;
         attackReadyTick=ticks+12;int previousHp=target.hp;target.hp=Math.max(0,target.hp-power);
@@ -190,12 +238,12 @@ public final class RpgWorld {

if(loot.award(target.spawn,target.generation,target.definition.id(),inventory,flags,random==null?new
java.util.Random(741L^((long)target.spawn.hashCode()<<16)^target.generation):random)) {
                 questService.event("kill",target.definition.id(),target.spawn,target.generation,quests,flags);
                 questService.refresh(quests,flags,inventory,region);
-
awardExperience(content.get("enemy",target.definition.id()).value());coins=Math.min(1_000_000,coins+5);
+                awardExperience(3);coins=Math.min(1_000_000,coins+5);
                 message="Derrotado: +XP, loot e moedas";
                 if(target.definition.id().equals("warden")){flags.add("boss-defeated");flags.add("dungeon-compl
eted");message="Santuario concluido! Volte a Ancia com a reliquia.";}
             }else{target.hp=1;target.readyTick=0;message="Libere espaco nas recompensas pendentes antes da
derrota";}
         }
-        event(RpgCombatEvent.Kind.ATTACK,target.getId(),target.getPosition().x(),target.getPosition().y(),0);
+        event(unarmed?RpgCombatEvent.Kind.PUNCH:RpgCombatEvent.Kind.ATTACK,target.getId(),target.getPosition().
x(),target.getPosition().y(),0);
         if(previousHp>target.hp)event(RpgCombatEvent.Kind.HIT,target.getId(),target.getPosition().x(),target.ge
tPosition().y(),previousHp-target.hp);
         if(target.hp==0)event(RpgCombatEvent.Kind.KO,target.getId(),target.getPosition().x(),target.getPosition
().y(),0);
         return true;
@@ -207,6 +255,7 @@ public final class RpgWorld {
     public void collectLoot() { loot.collect(inventory,flags);message="Recompensas pendentes recolhidas quando
houver espaco"; }
     private long ticks;
     public RpgSave save() {
+        domain.writeTo(flags);
         var persistent=new java.util.TreeSet<>(flags);persistent.removeIf(f->f.startsWith("stamina:")||f.starts
With("mob:")||f.startsWith("attack-until:")||f.startsWith("portal-until:")||f.startsWith("facing:"));persistent.
add("stamina:"+stamina);persistent.add("facing:"+facingX+":"+facingY);persistent.add("attack-until:"+attackReady
Tick);persistent.add("portal-until:"+portalReadyTick);persistent.addAll(enemies.saveFlags());
         return new RpgSave(identity,region,body.getPosition().x(),body.getPosition().y(),hp,xp,coins,inventory.
snapshot(),equipment,quests,persistent,ticks);
     }
@@ -217,10 +266,14 @@ public final class RpgWorld {
         world.inventory.replace(InventoryService.restore(world.items,20,state.inventory()));
         world.hp=state.hp(); world.xp=state.xp(); world.coins=state.coins(); world.region=state.region();
         for(var entry:state.equipment().entrySet())
if(!world.items.get(entry.getValue()).slot().equals(entry.getKey())) throw new
IllegalArgumentException("Equipamento incompativel no save");
-        world.equipment.putAll(state.equipment()); world.quests.putAll(state.quests());
world.flags.addAll(state.flags());
+        world.equipment.clear();world.equipment.putAll(state.equipment()); world.quests.putAll(state.quests());
 world.flags.addAll(state.flags());
+        world.domain=DomainProgression.read(state.flags());
+        if(!world.domain.domain().equals("none")&&!Domains.get(world.domain.domain()).race().equals(world.ident
ity.race()))throw new IllegalArgumentException("Dominio incompativel com a raca");
+        if(!world.domain.domain().equals("none")&&world.level()<15)throw new IllegalArgumentException("Dominio
antes do nivel 15");
         for(String flag:world.flags) if(flag.startsWith("potion-until:"))
world.potionReadyTick=Long.parseLong(flag.substring(13));
         for(String flag:world.flags) {
             if(flag.startsWith("ability-until:"))world.abilityReadyTick=Long.parseLong(flag.substring(14));
+            if(flag.startsWith("evade-until:"))world.evadeUntilTick=Long.parseLong(flag.substring(12));
             if(flag.startsWith("stamina:"))world.stamina=Integer.parseInt(flag.substring(8));
             if(flag.startsWith("attack-until:"))world.attackReadyTick=Long.parseLong(flag.substring(13));
             if(flag.startsWith("portal-until:"))world.portalReadyTick=Long.parseLong(flag.substring(13));
@@ -236,7 +289,9 @@ public final class RpgWorld {
         for(String flag:source) {
             String[] p=flag.split(":");String key=p[0];
             switch(p[0]) {
```

```
-                case "stamina","potion-until","ability-until","attack-until","portal-until"->{if(p.length!=2||L
ong.parseLong(p[1])<0||Long.parseLong(p[1])>1_000_009_000L)throw new IllegalArgumentException("Recurso/recarga
invalido");}
+                case "stamina","potion-until","ability-until","attack-until","portal-until","evade-until"->{if(
p.length!=2||Long.parseLong(p[1])<0||Long.parseLong(p[1])>1_000_009_000L)throw new
IllegalArgumentException("Recurso/recarga invalido");}
+                case "domain","mastery-half","reset-last","reset-quest","domain-skill"->{if(p.length!=2)throw
new IllegalArgumentException("Estado de dominio invalido");if(p[0].equals("domain-skill"))key+=":"+p[1];}
+                case
"migration"->{if(p.length!=2||!java.util.Set.of("guardian-to-reaper","mystic-to-human").contains(p[1]))throw new
 IllegalArgumentException("Migracao invalida");}
                 case "facing"->{if(p.length!=3)throw new IllegalArgumentException("Direcao invalida");}
                 case "mob"->{if(p.length!=7&&p.length!=9)throw new IllegalArgumentException("Estado de mob
invalido");key+=":"+p[1];}
                 case "pending"->{if(p.length!=3)throw new IllegalArgumentException("Loot pendente
invalido");items.get(p[1]);int n=Integer.parseInt(p[2]);if(n<1||n>9999)throw new IllegalArgumentException("Loot
pendente fora do limite");key+=":"+p[1];}
```

### 36. [M] server/src/main/java/com/projetok/studio/StudioExport.java

```
diff --git a/server/src/main/java/com/projetok/studio/StudioExport.java
b/server/src/main/java/com/projetok/studio/StudioExport.java
index b63585f2fc76387c4d019aff71dcd6efcc2cafba..b0c99a5454154db4ef65aa6e6adc9cfe833e593a 100644
--- a/server/src/main/java/com/projetok/studio/StudioExport.java
+++ b/server/src/main/java/com/projetok/studio/StudioExport.java
@@ -35,7 +35,7 @@ public final class StudioExport {
         }
         var result=new RpgSceneSet(maps.get(start).region(),geometry,enemies,npcs,objects,portals);
         // Construtores reais recusam referencias/posicoes incompatíveis antes de qualquer exportacao.
-        new RpgWorld(CharacterIdentity.create("Validador","guardian","azure"),result);return result;
+        new RpgWorld(CharacterIdentity.create("Validador","reaper","azure"),result);return result;
     }
     /** Cria diretorio novo e publica manifesto por ultimo; nunca sobrescreve projeto existente. */
     public static Path write(Path output,String start,Map<String,WorldDocument> all,AssetCatalog.Result
catalog)throws IOException {
```

### 37. [M] server/src/main/java/com/projetok/studio/StudioPlaytest.java

```
diff --git a/server/src/main/java/com/projetok/studio/StudioPlaytest.java
b/server/src/main/java/com/projetok/studio/StudioPlaytest.java
index deb5d0f224040b99613a84b79794b945f2698bc4..2f9bc638a253da117f094b3b7b872c5cc1f35f66 100644
--- a/server/src/main/java/com/projetok/studio/StudioPlaytest.java
+++ b/server/src/main/java/com/projetok/studio/StudioPlaytest.java
@@ -18,7 +18,7 @@ public final class StudioPlaytest {
         SwingUtilities.invokeLater(()->open(bundle,images,player,bundlePath));
     }
     static void open(StudioExport.Bundle bundle,AssetImages images,RpgPlayerSprites player,Path output) {
-        RpgRuntime runtime=new RpgRuntime(new
RpgWorld(CharacterIdentity.create("Playtest","guardian","azure"),bundle.scenes()));
+        RpgRuntime runtime=new RpgRuntime(new
RpgWorld(CharacterIdentity.create("Playtest","reaper","azure"),bundle.scenes()));
         var template=DebugWorld.demo(new java.util.Random(91)).snapshot();
         DebugPanel panel=new DebugPanel(()->{var v=runtime.snapshot();var
space=bundle.scenes().maps().get(v.state().region()).space();return
RpgLauncher.adapt(v,template,space.obstacles().stream().map(o->new
DebugSnapshot.Obstacle(o.x(),o.y(),o.width(),o.height())).toList());});panel.setPhysicalGeometryVisible(false);
         var render=new StudioMapRenderer(bundle,images,runtime::snapshot,player);var feedback=new
RpgCombatFeedback();var boss=new RpgBossOverlay();panel.setActorRenderer(render);panel.setWorldUnderlay(g->rende
r.layers(g,false));panel.setWorldOverlay(g->{render.layers(g,true);boss.draw(g);feedback.draw(g);});
@@ -26,9 +26,9 @@ public final class StudioPlaytest {
         var logs=new ArrayList<String>();logs.add("Inicio: "+bundle.start()+" | runtime real, sem
LocalSaveStore");boolean[] held=new boolean[4];String[] keys={"W","S","A","D"};
         var events=new StudioEvents();var interacted=new java.util.concurrent.atomic.AtomicBoolean();var
environment=new java.util.concurrent.atomic.AtomicReference<>(EnvironmentPreview.Mode.OFF);
         panel.setWorldOverlay(g->{render.layers(g,true);EnvironmentPreview.draw(g,new
Rectangle(0,0,960,640),environment.get(),System.nanoTime());boss.draw(g);feedback.draw(g);});
-        JPanel controls=new JPanel(new java.awt.BorderLayout());controls.add(banner);JPanel actions=new
JPanel(new FlowLayout(FlowLayout.RIGHT));JToggleButton geometry=new JToggleButton("Colisoes");geometry.addAction
Listener(e->panel.setPhysicalGeometryVisible(geometry.isSelected()));actions.add(geometry);JButton secondary=new
 JButton("Golpe de Impacto");secondary.addActionListener(e->runtime.command(w->w.activate("rpg.secondary")));act
ions.add(secondary);controls.add(actions,java.awt.BorderLayout.EAST);root.add(controls,java.awt.BorderLayout.NOR
TH);
+        JPanel controls=new JPanel(new java.awt.BorderLayout());controls.add(banner);JPanel actions=new
JPanel(new FlowLayout(FlowLayout.RIGHT));JToggleButton geometry=new JToggleButton("Colisoes");geometry.addAction
Listener(e->panel.setPhysicalGeometryVisible(geometry.isSelected()));actions.add(geometry);JButton secondary=new
 JButton("Shunpo defensivo (nivel 20)");secondary.addActionListener(e->runtime.command(w->w.activate("rpg.second
ary")));actions.add(secondary);controls.add(actions,java.awt.BorderLayout.EAST);root.add(controls,java.awt.Borde
rLayout.NORTH);
```

```
        for(int i=0;i<4;i++)for(boolean down:new boolean[]{true,false}){final int at=i;String
key=keys[i]+down;panel.getInputMap(JComponent.WHEN_IN_FOCUSED_WINDOW).put(KeyStroke.getKeyStroke((down?"pressed
":"released ")+keys[i]),key);panel.getActionMap().put(key,new AbstractAction(){public void
actionPerformed(ActionEvent
e){held[at]=down;runtime.input((held[3]?1:0)-(held[2]?1:0),(held[1]?1:0)-(held[0]?1:0));}});}
-       RpgLauncher.bind(panel,"SPACE",()->runtime.command(RpgWorld::attack));RpgLauncher.bind(panel,"Q",()->ru
ntime.command(RpgWorld::castAbility));RpgLauncher.bind(panel,"R",()->runtime.command(RpgWorld::returnHome));RpgL
auncher.bind(panel,"1",()->runtime.command(w->w.use("potion")));RpgLauncher.bind(panel,"F",()->runtime.action(Rp
gWorld::interact).thenAccept(ok->{if(!ok)synchronized(logs){if(logs.size()<1000)logs.add("Interacao recusada
pela simulacao");}}));
+       RpgLauncher.bind(panel,"SPACE",()->runtime.command(RpgWorld::attack));RpgLauncher.bind(panel,"Q",()->ru
ntime.command(RpgWorld::castAbility));RpgLauncher.bind(panel,"R",()->runtime.command(RpgWorld::returnHome));RpgL
auncher.bind(panel,"1",()->runtime.command(w->w.use("potion")));
        RpgLauncher.bind(panel,"F",()->runtime.action(RpgWorld::interact).thenAccept(ok->{if(ok)interacted.set(
true);else synchronized(logs){if(logs.size()<1000)logs.add("Interacao recusada pela simulacao");}}));
        long[] lastBlocked={-30};RpgWorld.View[] previous={runtime.snapshot()};
        javax.swing.Timer timer=new javax.swing.Timer(33,e->{var
v=runtime.snapshot();if(runtime.failure()!=null){((javax.swing.Timer)e.getSource()).stop();logs.add("Falha:
"+runtime.failure());JOptionPane.showMessageDialog(frame,"Simulacao interrompida: "+runtime.failure().getMessage
());frame.dispose();return;}if(player!=null)player.update(v);feedback.update(v);boss.update(v);hud.update(v);pan
el.setCustomHud(java.util.List.of("PLAYTEST WORLD STUDIO",render.document().name(),"Regiao:
"+v.state().region(),"Sem gravacao de personagem","Assets residentes: "+images.bytes()/1024+" KiB","[F]
NPC/objeto/portal","[R] Recuperar apos KO"));
```

### 38. [D] server/src/main/resources/rpg/archetypes.properties

```
diff --git a/server/src/main/resources/rpg/archetypes.properties
b/server/src/main/resources/rpg/archetypes.properties
deleted file mode 100644
index c91639e022dc475d12f8db5575e2d7395e0cbb25..0000000000000000000000000000000000000000
--- a/server/src/main/resources/rpg/archetypes.properties
+++ /dev/null
@@ -1,2 +0,0 @@
-guardian=140|18|dash|25|90|120
-mystic=100|24|mend|35|120|35
```

### 39. [M] server/src/main/resources/rpg/content.properties

```
diff --git a/server/src/main/resources/rpg/content.properties b/server/src/main/resources/rpg/content.properties
index 7d4f3186318610d41413ffc3c42551362b6f9545..23b2b43807bb9ed8edc2bd30c8f44a6ccb29dd45 100644
--- a/server/src/main/resources/rpg/content.properties
+++ b/server/src/main/resources/rpg/content.properties
@@ -4,6 +4,7 @@ region.forest=Bosque dos Ecos|forest|0
 region.ruins=Santuario Partido|ruins|0
 item.potion=Pocao de vida|village|12
 item.sword=Espada de aprendiz|village|35
+item.small_sword=Espada pequena|village|0
 item.robe=Manto do viajante|village|25
 item.crystal=Cristal do eco|forest|3
 item.relic=Reliquia do Guardiao|ruins|0
```

### 40. [M] server/src/main/resources/rpg/items.properties

```
diff --git a/server/src/main/resources/rpg/items.properties b/server/src/main/resources/rpg/items.properties
index 4fd88713f602d60b9bf1c39297b6a04e02210b25..0a433684f4d296f04a253dfdbf42a514272d50b9 100644
--- a/server/src/main/resources/rpg/items.properties
+++ b/server/src/main/resources/rpg/items.properties
@@ -1,6 +1,7 @@
 schema=1
 potion=Recupera 50 HP; recarga de 2 segundos|CONSUMABLE|COMMON|10|none|0|50|1|local
 sword=Arma de treino; +8 ataque|EQUIPMENT|UNCOMMON|1|weapon|8|0|1|local
+small_sword=Espada pequena inicial do aprendiz de ceifador|EQUIPMENT|COMMON|1|weapon|0|0|1|local
 robe=Manto resistente; +5 defesa|EQUIPMENT|COMMON|1|torso|5|0|1|local
 crystal=Material de ecos; pode ser vendido|MATERIAL|COMMON|20|none|0|0|1|local
 relic=Trofeu do Santuario; nao pode ser vendido|QUEST|RARE|1|none|0|0|1|local
```

### 41. [M] server/src/main/resources/rpg/progression.properties

```
diff --git a/server/src/main/resources/rpg/progression.properties
b/server/src/main/resources/rpg/progression.properties
index 15f3f2429bb90ef78a3c435f7d4464b0e4d7a53a..70123e400c67bc5d2f4150b84b60757505195c6a 100644
--- a/server/src/main/resources/rpg/progression.properties
+++ b/server/src/main/resources/rpg/progression.properties
@@ -1,4 +1,4 @@
-cap=20
+cap=0
 step=60
 hpPerLevel=12
 attackPerLevel=3
```

### 42. [A] server/src/test/java/com/projetok/domain/combat/CombatServiceDeathTest.java

```
diff --git a/server/src/test/java/com/projetok/domain/combat/CombatServiceDeathTest.java
b/server/src/test/java/com/projetok/domain/combat/CombatServiceDeathTest.java
new file mode 100644
index 0000000000000000000000000000000000000000..3ac66e76466602c710c9ac3bb64c975badd1f214
--- /dev/null
+++ b/server/src/test/java/com/projetok/domain/combat/CombatServiceDeathTest.java
@@ -0,0 +1,31 @@
+package com.projetok.domain.combat;
+
+import com.projetok.domain.entity.*;
+import com.projetok.domain.entity.capability.*;
+import com.projetok.domain.event.GameEventBus;
+import com.projetok.domain.event.events.EntityDeathEvent;
+import com.projetok.domain.race.*;
+import org.junit.jupiter.api.Test;
+import static org.junit.jupiter.api.Assertions.*;
+
+class CombatServiceDeathTest {
+    static class Target extends GameEntity implements Damageable {
+        int hp=1;
+        Target() { super(2,"Alvo",new Position(100,100)); }
+        public void receiveDamage(int damage) { hp=Math.max(0,hp-damage); }
+        public boolean isDead() { return hp==0; }
+        public Race race() { return RaceRegistry.withDefaults().get("human"); }
+    }
+    @Test void deathIsPublishedOnlyOnTheLiveToDeadTransition() {
+        var bus=new GameEventBus();int[] deaths={0};int[] calculations={0};
+        bus.subscribe(EntityDeathEvent.class,e->deaths[0]++);
+        var combat=new CombatService(bus);var target=new Target();
+        Attacker attacker=new Attacker() {
+            public int computeOutgoingDamage(){calculations[0]++;return 10;}
+            public DamageType outgoingDamageType(){return DamageType.PHYSICAL;}
+        };
+        assertEquals(10,combat.attack(attacker,target));
+        assertEquals(0,combat.attack(attacker,target));
+        assertEquals(1,deaths[0]);assertEquals(1,calculations[0]);
+    }
+}
```

### 43. [M] server/src/test/java/com/projetok/prototype/rpg/AmbientFeedbackTest.java

```
diff --git a/server/src/test/java/com/projetok/prototype/rpg/AmbientFeedbackTest.java
b/server/src/test/java/com/projetok/prototype/rpg/AmbientFeedbackTest.java
index 017127cbba8273f4dce30f0fd4b76f4a7c00f6f6..836ab937faa2374985ffd59e294cc3682bfd9955 100644
--- a/server/src/test/java/com/projetok/prototype/rpg/AmbientFeedbackTest.java
+++ b/server/src/test/java/com/projetok/prototype/rpg/AmbientFeedbackTest.java
@@ -5,7 +5,7 @@ import static org.junit.jupiter.api.Assertions.*;
 class AmbientFeedbackTest {
     @Test void repeatedSnapshotDoesNotReplayItemOrRejectedAbility(){
         var sounds=new java.util.ArrayList<RpgAudio.Cue>();var feedback=new AmbientFeedback(sounds::add);
-        var world=new RpgWorld(CharacterIdentity.create("Místico","mystic","azure"));feedback.update(world.snap
shot());assertTrue(sounds.isEmpty());
+        var world=new RpgWorld(CharacterIdentity.create("Humano","human","azure"));feedback.update(world.snapsh
ot());assertTrue(sounds.isEmpty());

world.inventory().add("sword",1);feedback.update(world.snapshot());assertEquals(1,sounds.size());for(int
i=0;i<30;i++)feedback.update(world.snapshot());assertEquals(1,sounds.size());
         assertFalse(world.activate("rpg.ability"));feedback.update(world.snapshot());assertEquals(2,sounds.size
());feedback.update(world.snapshot());assertEquals(2,sounds.size());
     }
```

### 44. [M] server/src/test/java/com/projetok/prototype/rpg/CharacterCreationTest.java

```
diff --git a/server/src/test/java/com/projetok/prototype/rpg/CharacterCreationTest.java
b/server/src/test/java/com/projetok/prototype/rpg/CharacterCreationTest.java
index 8b4a06f15460bd47134cd147dcef37f6a4277e64..86642e358456af2e0c0924af54a7e885579d5dbe 100644
--- a/server/src/test/java/com/projetok/prototype/rpg/CharacterCreationTest.java
+++ b/server/src/test/java/com/projetok/prototype/rpg/CharacterCreationTest.java
@@ -7,18 +7,24 @@ import static org.junit.jupiter.api.Assertions.*;
 import java.util.concurrent.atomic.AtomicReference;

 class CharacterCreationTest {
-    @Test void validationAndClassPreviewUseDomain() throws Exception {
+    @Test void validationAndRacePreviewStartWithoutDomain() throws Exception {
         SwingUtilities.invokeAndWait(() -> {
             var form = new CharacterCreation(null);
             var accepted = new AtomicReference<CharacterIdentity>();
             form.name.setText("!"); form.submit(accepted::set);
             assertNull(accepted.get()); assertFalse(form.error.getText().isBlank());
-            form.mystic.doClick(); form.appearance.setSelectedItem("amber");
-            assertEquals("mystic", form.preview.slot.state().identity().archetype());
-            assertEquals(Archetypes.get("mystic").hp(), form.preview.slot.state().hp());
+            assertEquals(4, form.races.size());
+            for (var race : CharacterRaces.all()) {
+                form.races.get(race.id()).doClick();
+                assertEquals(race.id(), form.preview.slot.state().identity().race());
+                assertEquals(race.hp(), form.preview.slot.state().hp());
+                assertFalse(DomainProgression.read(form.preview.slot.state().flags()).chosen());
+                assertTrue(form.origin.getText().contains(race.startingAttackName()));
+            }
+            form.races.get("aetheri").doClick(); form.appearance.setSelectedItem("amber");
             var previous=form.preview.direction;form.preview.orientation().doClick();assertNotEquals(previous,f
orm.preview.direction);
             form.name.setText("Heroi teste"); form.submit(accepted::set);
-            assertEquals("mystic", accepted.get().archetype());
+            assertEquals("aetheri", accepted.get().race());
             assertEquals("amber", accepted.get().appearance());
         });
     }
```

### 45. [M] server/src/test/java/com/projetok/prototype/rpg/CharacterFactsTest.java

```
diff --git a/server/src/test/java/com/projetok/prototype/rpg/CharacterFactsTest.java
b/server/src/test/java/com/projetok/prototype/rpg/CharacterFactsTest.java
index ab7009365d3b36f2decfc714602f9cedebd8b000..970059d7c6e60865f050741e7cb01588025b8ebd 100644
--- a/server/src/test/java/com/projetok/prototype/rpg/CharacterFactsTest.java
+++ b/server/src/test/java/com/projetok/prototype/rpg/CharacterFactsTest.java
@@ -10,7 +10,7 @@ import static org.junit.jupiter.api.Assertions.*;
 class CharacterFactsTest {
     @Test void extremeIdentityAndMissingSelectionRemainReadOnly() throws Exception {
         SwingUtilities.invokeAndWait(() -> {
-            var identity=new CharacterIdentity(UUID.randomUUID(),"Guardião
extremo","guardian","azure","village",1000,100);
+            var identity=new CharacterIdentity(UUID.randomUUID(),"Humano
extremo","human","azure","village",1000,100);
             var world=new RpgWorld(identity);var before=world.save();
             var ui=new CharacterSelection();ui.updateSummary(new
CharacterSaves.Slot(Path.of("synthetic.properties"),before,"",false));
             assertTrue(ui.resources.getText().contains("1000"));assertTrue(ui.style.getText().contains("Ataque
100"));
```

### 46. [M] server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionTest.java

```
diff --git a/server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionTest.java
b/server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionTest.java
index aebc937f15b9bdca5b0a3a6e3c39f81c61cc94d3..97fa3f6a114771719e6ceb1cac0c9ec2fc2d6d85 100644
--- a/server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionTest.java
+++ b/server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionTest.java
@@ -11,11 +11,11 @@ import static org.junit.jupiter.api.Assertions.*;
 class CharacterSelectionTest {
     @TempDir Path folder;
     @Test void loadingEmptyValidCorruptAndKeyboardActions() throws Exception {
-        var repo=new CharacterSaves(folder);var state=new
RpgWorld(CharacterIdentity.create("Viajante","mystic","amber")).save();new LocalSaveStore(repo.newFile(state.ide
ntity())).save(state);Files.writeString(folder.resolve("broken.properties"),"broken");var slots=repo.list();
+        var repo=new CharacterSaves(folder);var state=new
RpgWorld(CharacterIdentity.create("Viajante","human","amber")).save();new LocalSaveStore(repo.newFile(state.iden
tity())).save(state);Files.writeString(folder.resolve("broken.properties"),"broken");var slots=repo.list();
         SwingUtilities.invokeAndWait(()->{
             var view=new
CharacterSelection();assertFalse(view.play.isEnabled());assertFalse(view.create.isEnabled());
             view.loaded(java.util.List.of());assertTrue(view.create.isEnabled());assertTrue(view.status.getText
().contains("Nenhum"));
-            view.loaded(slots);int valid=slots.get(0).state()!=null?0:1;view.list.setSelectedIndex(valid);asser
tTrue(view.play.isEnabled());assertTrue(CharacterSelection.describe(slots.get(valid)).contains("Nível
1"));assertTrue(CharacterSelection.describe(slots.get(valid)).contains("Místico"));
+            view.loaded(slots);int valid=slots.get(0).state()!=null?0:1;view.list.setSelectedIndex(valid);asser
tTrue(view.play.isEnabled());assertTrue(CharacterSelection.describe(slots.get(valid)).contains("Nível
1"));assertTrue(CharacterSelection.describe(slots.get(valid)).contains("Human"));
             AtomicInteger played=new AtomicInteger(),back=new AtomicInteger();view.play.addActionListener(e->pl
ayed.incrementAndGet());view.back.addActionListener(e->back.incrementAndGet());view.list.getActionMap().get("pla
y").actionPerformed(null);view.getActionMap().get("back").actionPerformed(null);assertEquals(1,played.get());ass
ertEquals(1,back.get());
             view.list.setSelectedIndex(1-valid);assertFalse(view.play.isEnabled());view.list.getActionMap().get
("play").actionPerformed(null);assertEquals(1,played.get());
             view.preview.setSize(300,320);view.preview.paint(new
java.awt.image.BufferedImage(300,320,2).getGraphics());
```

### 47. [M] server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionVisualTest.java

```
diff --git a/server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionVisualTest.java
b/server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionVisualTest.java
index 5d25acc6b1b50cc151e132bc36e27dd95158783e..34a4f8204f019b755ac27dc18b25e535c56f82d1 100644
--- a/server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionVisualTest.java
+++ b/server/src/test/java/com/projetok/prototype/rpg/CharacterSelectionVisualTest.java
@@ -15,14 +15,15 @@ class CharacterSelectionVisualTest {

     @Test void cardsSummaryBackupAndActionsPreserveTheCatalogContract() throws Exception {
         var repo=new CharacterSaves(folder);
-        var state=new RpgWorld(CharacterIdentity.create("Sentinela","guardian","azure")).save();
+        var state=new RpgWorld(CharacterIdentity.create("Sentinela","reaper","azure")).save();
         Path file=repo.newFile(state.identity());var store=new LocalSaveStore(file);
         store.save(state);store.save(state);Files.writeString(file,"corrupt");
         var slots=repo.list();var
backup=slots.stream().filter(CharacterSaves.Slot::backup).findFirst().orElseThrow();
         SwingUtilities.invokeAndWait(()->{
             var view=new CharacterSelection();view.loaded(slots);view.list.setSelectedValue(backup,true);
             assertTrue(view.play.isEnabled());assertFalse(view.delete.isEnabled());
-
assertEquals("Sentinela",view.name.getText());assertTrue(view.details.getText().contains("Guardião"));
+
assertEquals("Sentinela",view.name.getText());assertTrue(view.details.getText().contains("Reaper"));
+            assertTrue(view.style.getText().contains("Domínio no nível 15"));
             var combat=RpgCombatStatus.from(RpgWorld.restore(state).snapshot());
             assertEquals("HP "+combat.hp()+" / "+combat.maxHp()+"   •   Vigor "+combat.stamina()/10+" /
100",view.resources.getText());
             var renderer=view.list.getCellRenderer();
@@ -33,7 +34,7 @@ class CharacterSelectionVisualTest {
             BufferedImage selectedImage=paint(selected,270,80);
             Component focused=renderer.getListCellRendererComponent(view.list,backup,1,true,true);
             assertNotEquals(selectedImage.getRGB(0,0),paint(focused,270,80).getRGB(0,0));
-            for(JButton button:new JButton[]{view.play,view.create,view.back,view.arena}){
+            for(JButton button:new JButton[]{view.play,view.create,view.back}){
                 AtomicInteger calls=new AtomicInteger();button.addActionListener(e->calls.incrementAndGet());bu
tton.doClick();assertEquals(1,calls.get());
             }

view.list.setSelectedValue(slots.stream().filter(s->s.state()==null).findFirst().orElseThrow(),true);
@@ -55,7 +56,7 @@ class CharacterSelectionVisualTest {
             for(int[] size:new int[][]{{1000,620},{640,480},{800,600}}){
```

```
                for(var skin:new RpgPlayerSprites[]{null,sprites}){
                    view.preview.sprites=skin;paint(view,size[0],size[1]);
-                   for(JComponent c:new
JComponent[]{view.play,view.create,view.delete,view.arena,view.back,view.preview,view.resources}){
+                   for(JComponent c:new
JComponent[]{view.play,view.create,view.delete,view.techniques,view.back,view.preview,view.resources}){
                        Rectangle bounds=SwingUtilities.convertRectangle(c.getParent(),c.getBounds(),view);
                        assertTrue(bounds.width>0&&bounds.height>0,bounds.toString());
                        assertTrue(new Rectangle(0,0,size[0],size[1]).contains(bounds),bounds.toString());
```

### 48. [M] server/src/test/java/com/projetok/prototype/rpg/CorujaoHudTest.java

```
diff --git a/server/src/test/java/com/projetok/prototype/rpg/CorujaoHudTest.java
b/server/src/test/java/com/projetok/prototype/rpg/CorujaoHudTest.java
index c08df23b4b0c53bd50a8ef603475e8a01d606998..9281037d76ea7e8786f8bd22ed61504472d59cfa 100644
--- a/server/src/test/java/com/projetok/prototype/rpg/CorujaoHudTest.java
+++ b/server/src/test/java/com/projetok/prototype/rpg/CorujaoHudTest.java
@@ -4,15 +4,23 @@ import org.junit.jupiter.api.Test;
 import static org.junit.jupiter.api.Assertions.*;
 class CorujaoHudTest {
     @Test void cooldownStaminaAndLowHpFollowWorldAndPersistedValues() {
-        var w=new RpgWorld();var
v=RpgCombatStatus.from(w.snapshot());assertEquals(1000,v.stamina());assertTrue(v.abilityReady());
-        assertTrue(w.castAbility());v=RpgCombatStatus.from(w.snapshot());assertEquals(750,v.stamina());assertEq
uals(90,v.abilityTicks());assertFalse(v.abilityReady());
+        var w=new RpgWorld();var
v=RpgCombatStatus.from(w.snapshot());assertEquals(1000,v.stamina());assertFalse(v.abilityReady());
+        assertFalse(w.castAbility());assertFalse(w.castSecondaryAbility());
+        w.awardExperience(RpgProgression.builtin().threshold(20));assertTrue(RpgCombatStatus.from(w.snapshot())
.abilityReady());
+        assertTrue(w.castSecondaryAbility());v=RpgCombatStatus.from(w.snapshot());assertEquals(750,v.stamina())
;assertEquals(90,v.abilityTicks());assertFalse(v.abilityReady());
         assertEquals("3.0 s",RpgCombatStatus.cooldown(v.abilityTicks()));
         assertEquals(v,RpgCombatStatus.from(RpgWorld.restore(w.save()).snapshot()));
-        w.receiveDamage(110);assertTrue(RpgCombatStatus.from(w.snapshot()).lowHp());
+        for(int tick=0;tick<6;tick++)w.tick(0,0);
+        w.receiveDamage(w.maxHp()-1);assertTrue(RpgCombatStatus.from(w.snapshot()).lowHp());
         w.receiveDamage(10000);v=RpgCombatStatus.from(w.snapshot());assertEquals(0,v.hp());assertFalse(v.abilit
yReady());assertFalse(v.lowHp());
     }
-    @Test void mysticFullHealthIsUnavailableAndCooldownNeverNegative() {
-        var w=new RpgWorld(CharacterIdentity.create("Teste","mystic","azure"));assertFalse(RpgCombatStatus.from
(w.snapshot()).abilityReady());
-        w.receiveDamage(10);assertTrue(RpgCombatStatus.from(w.snapshot()).abilityReady());assertEquals("Pronto"
,RpgCombatStatus.cooldown(-1));
+    @Test void everyRaceUnlocksShunpoAtLevelTwentyRegardlessOfHealth() {
+        for(var race:CharacterRaces.all()){
+            var w=new RpgWorld(CharacterIdentity.create("Teste",race.id(),"azure"));assertFalse(RpgCombatStatus
.from(w.snapshot()).abilityReady());
+            w.awardExperience(RpgProgression.builtin().threshold(19));assertFalse(RpgCombatStatus.from(w.snapsh
ot()).abilityReady());
+            w.awardExperience(RpgProgression.builtin().threshold(20)-w.save().xp());assertTrue(RpgCombatStatus.
from(w.snapshot()).abilityReady());
+            w.receiveDamage(10);assertTrue(RpgCombatStatus.from(w.snapshot()).abilityReady());
+        }
+        assertEquals("Pronto",RpgCombatStatus.cooldown(-1));
     }
 }
```

### 49. [A] server/src/test/java/com/projetok/prototype/rpg/DomainPreviewTest.java

```
diff --git a/server/src/test/java/com/projetok/prototype/rpg/DomainPreviewTest.java
b/server/src/test/java/com/projetok/prototype/rpg/DomainPreviewTest.java
new file mode 100644
index 0000000000000000000000000000000000000000..91a80668792252dea014e4a6bed784fad414bc02
--- /dev/null
+++ b/server/src/test/java/com/projetok/prototype/rpg/DomainPreviewTest.java
@@ -0,0 +1,43 @@
+package com.projetok.prototype.rpg;
+
+import com.projetok.rpg.*;
+import org.junit.jupiter.api.Test;
+import javax.swing.*;
+import java.awt.image.BufferedImage;
+import static org.junit.jupiter.api.Assertions.*;
+
+class DomainPreviewTest {
+    @Test void compatiblePathsAndPermanentSelectionRespectInitiation() throws Exception {
+        SwingUtilities.invokeAndWait(() -> {
+            for (var race : CharacterRaces.all()) {
+                var before = new DomainPreview(race.id(), 14, "none", true);
+                assertFalse(before.choose.isEnabled());
+                var available = new DomainPreview(race.id(), 15, "none", true);
+                assertTrue(available.choose.isEnabled());
+                assertEquals(Domains.forRace(race.id()).size(), available.domains.getItemCount());
+                available.setSize(640, 480); layout(available);
+                assertTrue(available.scene.getWidth() > 0);
+                assertTrue(available.scene.getHeight() > 80, "A descrição não pode ocultar a prévia inicial");
+                var chosen = new DomainPreview(race.id(), 30, Domains.forRace(race.id()).get(0).id(), true);
+                assertFalse(chosen.choose.isEnabled());
+                var browse = new DomainPreview(race.id(), 15, "none", false);
+                assertFalse(browse.choose.isVisible());
+                for (int i = 0; i < available.domains.getItemCount(); i++) {
+                    available.domains.setSelectedIndex(i);
+                    assertEquals(race.id(), Domains.get(available.selectedDomain()).race());
+                    assertFalse(available.description.getText().isBlank());
+                    available.scene.setSize(600, 260);
+                    var image = new BufferedImage(600, 260, BufferedImage.TYPE_INT_RGB);
+                    var graphics = image.createGraphics(); available.scene.paint(graphics); graphics.dispose();
+                    assertNotEquals(0, image.getRGB(120, 100));
+                }
+            }
+            assertEquals("1,5", DomainPreview.formatXp(3));
+            assertEquals("3", DomainPreview.formatXp(6));
+        });
+    }
+    private static void layout(java.awt.Container root) {
+        root.doLayout();
+        for (var child : root.getComponents()) if (child instanceof java.awt.Container nested) layout(nested);
+    }
+}
```

### 50. [M] server/src/test/java/com/projetok/prototype/rpg/GettingStartedTest.java

```
diff --git a/server/src/test/java/com/projetok/prototype/rpg/GettingStartedTest.java
b/server/src/test/java/com/projetok/prototype/rpg/GettingStartedTest.java
index 4b37679506670bbd04bb728f3be5a8226048a783..400d79b8d877e4bc0c12379ef60712c511c58380 100644
--- a/server/src/test/java/com/projetok/prototype/rpg/GettingStartedTest.java
+++ b/server/src/test/java/com/projetok/prototype/rpg/GettingStartedTest.java
@@ -6,6 +6,7 @@ class GettingStartedTest {
     @Test void helpReflectsCurrentBindingsWithoutChangingWorld() {
         var keys=new java.util.HashMap<>(RpgControls.DEFAULTS);keys.put("attack","Z");
         var world=new RpgWorld();var before=world.save();var text=GettingStarted.text(new RpgControls(keys));
-        assertTrue(text.contains("Z — Golpe"));assertFalse(text.contains("SPACE —
Golpe"));assertTrue(text.contains("W A S D"));assertEquals(before,world.save());
+        assertTrue(text.contains("Z — Ataque basico"));assertFalse(text.contains("SPACE — Ataque
basico"));assertTrue(text.contains("W A S D"));
+        assertTrue(text.contains("nivel 15"));assertTrue(text.contains("nivel
20"));assertEquals(before,world.save());
     }
 }
```

### 51. [M] server/src/test/java/com/projetok/rpg/ArchetypesTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/ArchetypesTest.java
b/server/src/test/java/com/projetok/rpg/ArchetypesTest.java
index 78b87e6b48ad814fa48f7e4efadbf4fc8920604e..7568da3d6f10fed178ef08ba5b18a2eb113c34f2 100644
--- a/server/src/test/java/com/projetok/rpg/ArchetypesTest.java
+++ b/server/src/test/java/com/projetok/rpg/ArchetypesTest.java
@@ -2,9 +2,17 @@ package com.projetok.rpg;
 import org.junit.jupiter.api.Test;
 import static org.junit.jupiter.api.Assertions.*;
 class ArchetypesTest {
-    @Test void guardianMovesMysticHealsAndCostsSurviveSave() {
-        var g=new RpgWorld(CharacterIdentity.create("Guardiao","guardian","amber"));float
x=g.snapshot().x();assertTrue(g.castAbility());assertTrue(g.snapshot().x()>x);assertFalse(g.castAbility());
-        var m=new RpgWorld(CharacterIdentity.create("Mistico","mystic","azure"));assertFalse(m.castAbility());m
.receiveDamage(60);assertTrue(m.castAbility());assertEquals(75,m.save().hp());
-
assertFalse(RpgWorld.restore(m.save()).castAbility());assertEquals(m.save(),RpgWorld.restore(m.save()).save());
+    @Test void racialCreationHasNoDomainOrFreeShunpo() {
+        for(var race:CharacterRaces.all()) {
+            var world=new RpgWorld(CharacterIdentity.create("Aprendiz",race.id(),"amber"));
+            assertEquals(race.id(),world.save().identity().race());
+            assertEquals(DomainProgression.NONE,world.domainProgression().domain());
+            assertEquals(0,world.domainProgression().xpHalfUnits());
+            assertFalse(world.castAbility());assertFalse(world.castSecondaryAbility());
+            assertEquals(race.id().equals("reaper"),world.save().equipment().containsKey("weapon"));
+            assertEquals(world.save(),RpgWorld.restore(world.save()).save());
+        }
+        assertThrows(IllegalArgumentException.class,()->CharacterIdentity.create("Antigo","guardian","azure"));
+        assertThrows(IllegalArgumentException.class,()->CharacterIdentity.create("Antigo","mystic","azure"));
     }
 }
```

### 52. [M] server/src/test/java/com/projetok/rpg/CharacterIdentityTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/CharacterIdentityTest.java
b/server/src/test/java/com/projetok/rpg/CharacterIdentityTest.java
index f98d252214b95d640e18291d2d300e9c72386a05..9d2b8c70a4dc097727036a39819f0218cd366fde 100644
--- a/server/src/test/java/com/projetok/rpg/CharacterIdentityTest.java
+++ b/server/src/test/java/com/projetok/rpg/CharacterIdentityTest.java
@@ -4,8 +4,8 @@ import static org.junit.jupiter.api.Assertions.*;
 class CharacterIdentityTest {
     @Test void validatesNamesAndRestoresStableIdentity() {
         for (String name : new String[]{"", "ab", "<script>", "A\nB", "a".repeat(25)})
-            assertThrows(IllegalArgumentException.class, () ->
CharacterIdentity.create(name,"guardian","azure"));
-        var original = CharacterIdentity.create("  Joao do Vale  ","guardian","azure");
+            assertThrows(IllegalArgumentException.class, () ->
CharacterIdentity.create(name,"reaper","azure"));
+        var original = CharacterIdentity.create("  Joao do Vale  ","reaper","azure");
         var restored = new CharacterIdentity(original.id(),original.name(),original.archetype(),original.appear
ance(),original.initialRegion(),original.baseHp(),original.baseAttack());
         assertEquals(original,restored);
         assertEquals("Joao do Vale",new RpgWorld(restored).snapshot().identity().name());
```

### 53. [M] server/src/test/java/com/projetok/rpg/CorujaoAbilitiesTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/CorujaoAbilitiesTest.java
b/server/src/test/java/com/projetok/rpg/CorujaoAbilitiesTest.java
index b8d00ecb676a25e4b87f03153e066a43eb682b24..1d4edf2dffcbe59df040e2ce9dd73f803658adc2 100644
--- a/server/src/test/java/com/projetok/rpg/CorujaoAbilitiesTest.java
+++ b/server/src/test/java/com/projetok/rpg/CorujaoAbilitiesTest.java
@@ -2,31 +2,42 @@ package com.projetok.rpg;
 import org.junit.jupiter.api.Test;
 import static org.junit.jupiter.api.Assertions.*;
 class CorujaoAbilitiesTest {
-    static RpgWorld forest(String archetype,float x) {
-        var s=new RpgWorld(CharacterIdentity.create("Teste",archetype,"azure")).save();
+    static RpgWorld forest(String race,float x,int level) {
+        var world=new RpgWorld(CharacterIdentity.create("Teste",race,"azure"));
+        world.awardExperience(RpgProgression.builtin().threshold(level));var s=world.save();
         return RpgWorld.restore(new RpgSave(s.identity(),"forest",x,360,s.hp(),s.xp(),s.coins(),s.inventory(),s
.equipment(),s.quests(),s.flags(),s.ticks()));
     }
-    @Test void bothAdditionalAbilitiesHaveExactCostSingleDamageAndPersistedCooldown() throws Exception {
-        for(String archetype:new String[]{"guardian","mystic"}) {
-            var w=forest(archetype,240);var ability=RpgSecondaryAbility.forArchetype(archetype);
-            int hp=w.snapshot().enemies().get(0).hp();assertTrue(w.castSecondaryAbility());
-            assertEquals(hp-w.attackPower()-ability.bonus(),w.snapshot().enemies().get(0).hp());
-            assertTrue(w.save().flags().contains("stamina:"+(1000-ability.cost()*10)));
-            assertFalse(w.castSecondaryAbility());assertFalse(w.castAbility());assertFalse(w.attack());
+    @Test void allRacesUnlockSharedShunpoAtExactlyLevel20WithoutDomain() throws Exception {
+        for(var race:CharacterRaces.all()) {
+            var
locked=forest(race.id(),180,19);assertFalse(locked.castAbility());assertFalse(locked.castSecondaryAbility());
+            var w=forest(race.id(),180,20);int enemyHp=w.snapshot().enemies().get(0).hp();
+            assertTrue(w.castAbility());assertEquals(238,w.snapshot().x(),0.01);
+            assertEquals(enemyHp,w.snapshot().enemies().get(0).hp());
+            assertTrue(w.save().flags().contains("stamina:750"));
+            assertFalse(w.castAbility());assertFalse(w.castSecondaryAbility());
+            assertFalse(w.domainProgression().chosen());
             var restored=RpgWorld.restore(LocalSaveStore.decode(LocalSaveStore.encode(w.save())));
-            assertEquals(w.save(),restored.save());assertFalse(restored.castSecondaryAbility());
-
assertEquals(1,w.snapshot().events().stream().filter(e->e.kind()==RpgCombatEvent.Kind.HIT).count());
+            assertEquals(w.save(),restored.save());assertFalse(restored.castAbility());assertFalse(restored.cas
tSecondaryAbility());
         }
     }
-    @Test void missesAndInsufficientResourcesDoNotConsumeAndRangesDiffer() {
-        var guardian=forest("guardian",180);assertFalse(guardian.castSecondaryAbility());
-        assertTrue(forest("mystic",180).castSecondaryAbility());
-        var s=guardian.save();assertTrue(s.flags().contains("stamina:1000"));
-        guardian.receiveDamage(10000);assertFalse(guardian.castSecondaryAbility());
-        var near=forest("guardian",240).save();var flags=new java.util.TreeSet<>(near.flags());
+    @Test void defensiveShunpoHasShortWindowAndResourcesPersist() throws Exception {
+        var world=forest("human",180,20);float x=world.snapshot().x();int hp=world.save().hp();
+        assertTrue(world.castSecondaryAbility());assertTrue(world.snapshot().x()<x);
+        world.receiveDamage(50);assertEquals(hp,world.save().hp());
+        var restored=RpgWorld.restore(LocalSaveStore.decode(LocalSaveStore.encode(world.save())));
+        restored.receiveDamage(50);assertEquals(hp,restored.save().hp());
+        for(int t=0;t<6;t++)restored.tick(0,0);
+        restored.receiveDamage(50);assertEquals(hp-50,restored.save().hp());
+        assertFalse(restored.castAbility());assertFalse(restored.castSecondaryAbility());
+        var near=forest("reaper",180,20).save();var flags=new java.util.TreeSet<>(near.flags());
         flags.removeIf(f->f.startsWith("stamina:"));flags.add("stamina:100");
         var tired=RpgWorld.restore(new RpgSave(near.identity(),near.region(),near.x(),near.y(),near.hp(),near.x
p(),near.coins(),near.inventory(),near.equipment(),near.quests(),flags,near.ticks()));
-        assertFalse(tired.castSecondaryAbility());assertEquals(near.hp(),tired.save().hp());
+        assertFalse(tired.castAbility());assertFalse(tired.castSecondaryAbility());assertEquals(near.hp(),tired
.save().hp());
         assertThrows(IllegalArgumentException.class,()->RpgSecondaryAbility.forArchetype("other"));
     }
+    @Test void missingTargetAndBlockedDestinationsDoNotSpendResources() {
+        var empty=new RpgWorld();empty.awardExperience(RpgProgression.builtin().threshold(20));
+        assertFalse(empty.castAbility());assertTrue(empty.save().flags().contains("stamina:1000"));
+        var dead=forest("reaper",180,20);dead.receiveDamage(10000);assertFalse(dead.castAbility());assertFalse(
dead.castSecondaryAbility());
+    }
 }
```

### 54. [M] server/src/test/java/com/projetok/rpg/CorujaoCombatRegressionTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/CorujaoCombatRegressionTest.java
b/server/src/test/java/com/projetok/rpg/CorujaoCombatRegressionTest.java
index b70ff0c331230fd6665dbdeb6deda83dbb85fe62..1ba69e411481cbba4ee79d0036247af9a0c852f5 100644
--- a/server/src/test/java/com/projetok/rpg/CorujaoCombatRegressionTest.java
+++ b/server/src/test/java/com/projetok/rpg/CorujaoCombatRegressionTest.java
@@ -6,16 +6,17 @@ class CorujaoCombatRegressionTest {
     private RpgWorld setup(String archetype,boolean boss) {
         var s=new RpgWorld(CharacterIdentity.create("Avaliacao",archetype,"azure")).save();var flags=new
TreeSet<String>();
         for(String f:s.flags()){if(boss&&f.startsWith("mob:ruins_guard_")){var
v=f.split(":");v[2]="0";f=String.join(":",v);}flags.add(f);}
-        return RpgWorld.restore(new RpgSave(s.identity(),boss?"ruins":"forest",boss?650:240,boss?320:360,s.hp()
,s.xp(),s.coins(),s.inventory(),s.equipment(),s.quests(),flags,s.ticks()));
+        return RpgWorld.restore(new RpgSave(s.identity(),boss?"ruins":"forest",boss?690:260,boss?320:360,s.hp()
,s.xp(),s.coins(),s.inventory(),s.equipment(),s.quests(),flags,s.ticks()));
     }
     @Test void deterministicEncountersMeasureBothArchetypesAndBossWithoutVisualSideEffects() throws Exception {
-        for(String archetype:new String[]{"guardian","mystic"})for(boolean boss:new boolean[]{false,true}) {
+        for(String archetype:new String[]{"reaper","human","aetheri","spirit"})for(boolean boss:new
boolean[]{false,true}) {
             var world=setup(archetype,boss);var control=RpgWorld.restore(world.save());
-            var feedback=new com.projetok.prototype.rpg.RpgCombatFeedback();int casts=0,elapsed=0;
+            var feedback=new com.projetok.prototype.rpg.RpgCombatFeedback();int elapsed=0;
             String target=boss?"sanctuary":"forest_a";
             while(world.snapshot().enemies().stream().filter(e->e.spawn().equals(target)).findFirst().orElseThr
ow().hp()>0&&elapsed<600) {
-                boolean used=world.castSecondaryAbility();assertEquals(used,control.castSecondaryAbility());
-                if(used)casts++;else assertEquals(world.attack(),control.attack());
+                assertFalse(world.castSecondaryAbility());assertFalse(control.castSecondaryAbility());
+                assertEquals(world.attack(),control.attack());
+                if(world.save().hp()<45){assertEquals(world.use("potion"),control.use("potion"));}
                 world.tick(0,0);control.tick(0,0);elapsed++;
                 feedback.update(world.snapshot());var image=new java.awt.image.BufferedImage(960,640,2);var
g=image.createGraphics();feedback.draw(g);g.dispose();
                 assertEquals(control.save(),world.save());assertTrue(world.save().hp()>0);
@@ -23,13 +24,12 @@ class CorujaoCombatRegressionTest {
             assertTrue(elapsed<600);assertTrue(world.save().xp()>0);
             int received=world.snapshot().events().stream().filter(e->e.kind()==RpgCombatEvent.Kind.HIT&&e.targ
et()==1).mapToInt(RpgCombatEvent::amount).sum();
             int dealt=world.snapshot().events().stream().filter(e->e.kind()==RpgCombatEvent.Kind.HIT&&e.target(
)!=1).mapToInt(RpgCombatEvent::amount).sum();
-            int spent=casts*RpgSecondaryAbility.forArchetype(archetype).cost();
-            System.out.println("CORUJAO_METRIC {\"archetype\":\""+archetype+"\",\"encounter\":\""+(boss?"boss":
"slime")+"\",\"ticks\":"+elapsed+",\"damageReceived\":"+received+",\"damageDealt\":"+dealt+",\"staminaSpent\":"+
spent+",\"secondaryUses\":"+casts+",\"survived\":true}");
+            assertTrue(received>=0);assertTrue(dealt>0);
             var saved=world.save();assertEquals(saved,RpgWorld.restore(LocalSaveStore.decode(LocalSaveStore.enc
ode(saved))).save());
         }
     }
     @Test void healingDefeatRecoveryAndResourcesRemainCompatible() throws Exception {
-        for(String archetype:new String[]{"guardian","mystic"}) {
+        for(String archetype:new String[]{"reaper","human","aetheri","spirit"}) {
             var
w=setup(archetype,false);w.receiveDamage(30);assertTrue(w.use("potion"));assertFalse(w.use("potion"));
             w.receiveDamage(10000);assertFalse(w.castSecondaryAbility());assertFalse(w.castAbility());assertFal
se(w.attack());

assertTrue(w.returnHome());assertEquals(w.maxHp(),w.save().hp());assertEquals("village",w.save().region());
```

### 55. [A] server/src/test/java/com/projetok/rpg/DomainProgressionTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/DomainProgressionTest.java
b/server/src/test/java/com/projetok/rpg/DomainProgressionTest.java
new file mode 100644
index 0000000000000000000000000000000000000000..5f39d0dffbaabc979c683cc050111162f721a0d5
--- /dev/null
+++ b/server/src/test/java/com/projetok/rpg/DomainProgressionTest.java
@@ -0,0 +1,83 @@
+package com.projetok.rpg;
+
+import java.time.LocalDate;
+import java.util.Set;
+import java.util.TreeSet;
+import org.junit.jupiter.api.Test;
+import static org.junit.jupiter.api.Assertions.*;
+
+class DomainProgressionTest {
+    @Test void choiceRequiresInitiationIsRacialAndCannotBeSwapped() {
+        for(var definition:Domains.all()) {
+            var empty=DomainProgression.initial();
+            assertThrows(IllegalStateException.class,()->empty.choose(definition.race(),14,definition.id()));
+            var chosen=empty.choose(definition.race(),15,definition.id());
+            assertEquals(definition.id(),chosen.domain());assertEquals(0,chosen.xpHalfUnits());
+            assertThrows(IllegalStateException.class,()->chosen.choose(definition.race(),30,definition.id()));
+            String wrongRace=definition.race().equals("spirit")?"human":"spirit";
+            assertThrows(IllegalArgumentException.class,()->empty.choose(wrongRace,15,definition.id()));
+        }
+    }
+    @Test void halfExperienceIsExactDoesNotAccumulateBeforeChoiceAndSurvivesPersistence() throws Exception {
+        var world=new RpgWorld();world.awardExperience(RpgProgression.builtin().threshold(15));
+        assertEquals(0,world.domainProgression().xpHalfUnits());assertTrue(world.chooseDomain("water"));
+        long xp=world.save().xp();world.awardExperience(3);
+        assertEquals(xp+3,world.save().xp());assertEquals(3,world.domainProgression().xpHalfUnits());
+
assertEquals(1.5,world.domainProgression().xp());assertEquals("1,5",world.domainProgression().xpText());
+        var loaded=RpgWorld.restore(LocalSaveStore.decode(LocalSaveStore.encode(world.save())));
+        assertEquals(world.save(),loaded.save());assertEquals(3,loaded.domainProgression().xpHalfUnits());
+        loaded.awardExperience(3);assertEquals(6,loaded.domainProgression().xpHalfUnits());
+        assertFalse(loaded.chooseDomain("thunder"));assertEquals("water",loaded.domainProgression().domain());
+    }
+    @Test void killingLevelOneMobAwardsThreeNormalAndOneAndHalfDomainXpOnlyOnce() {
+        var world=EnemyRpgTest.forest();world.awardExperience(RpgProgression.builtin().threshold(15));
+        assertTrue(world.chooseDomain("water"));long previous=world.save().xp();
+        assertTrue(world.attack());assertEquals(previous+3,world.save().xp());
+        assertEquals(3,world.domainProgression().xpHalfUnits());
+        for(int t=0;t<12;t++)world.tick(0,0);
+        assertFalse(world.attack());assertEquals(previous+3,world.save().xp());
+        assertEquals(3,world.domainProgression().xpHalfUnits());
+    }
+    @Test void resetCosts35NeedsQuestAndWaitsOneCalendarMonth() {
+        var policy=DomainResetPolicy.builtin();assertEquals(35,policy.cashCost());
+        var chosen=DomainProgression.initial().choose("reaper",15,"water").award(3);
+        var date=LocalDate.of(2028,1,31);
+        assertThrows(IllegalStateException.class,()->chosen.reset(date,35,policy));
+        var ready=chosen.completeResetQuest();
+        assertThrows(IllegalStateException.class,()->ready.reset(date,34,policy));
+        assertThrows(IllegalStateException.class,()->ready.reset(date,100,DomainResetPolicy.disabled()));
+        var reset=ready.reset(date,35,policy);assertFalse(reset.chosen());assertEquals(0,reset.xpHalfUnits());
+        assertFalse(reset.resetQuestComplete());assertEquals(LocalDate.of(2028,2,29),reset.nextResetDate());
+        var next=reset.choose("reaper",15,"fire").completeResetQuest();
+        assertThrows(IllegalStateException.class,()->next.reset(LocalDate.of(2028,2,28),35,policy));
+        assertFalse(next.reset(LocalDate.of(2028,2,29),35,policy).chosen());
+    }
+    @Test void resetClearsOnlyDomainSkillsAndPreservesBaseCharacterAndEquipment() {
+        var world=new RpgWorld();world.awardExperience(RpgProgression.builtin().threshold(20));
+        assertTrue(world.chooseDomain("water"));world.awardExperience(3);var before=world.save();
+        var flags=new TreeSet<>(before.flags());flags.add("domain-skill:water_aparo");
+        var reset=world.domainProgression().completeResetQuest().reset(LocalDate.of(2026,10,21),35,new
DomainResetPolicy(35));
+        reset.writeTo(flags);assertFalse(flags.contains("domain-skill:water_aparo"));
+        var after=new RpgSave(before.identity(),before.region(),before.x(),before.y(),before.hp(),before.xp(),b
efore.coins(),before.inventory(),before.equipment(),before.quests(),flags,before.ticks());
+        var restored=RpgWorld.restore(after);
+
assertEquals(before.identity(),restored.save().identity());assertEquals(before.xp(),restored.save().xp());
+        assertEquals(before.inventory(),restored.save().inventory());assertEquals(before.equipment(),restored.s
ave().equipment());
```

```
+        assertTrue(restored.canShunpo());assertFalse(restored.domainProgression().chosen());
+    }
+    @Test void corruptOrConflictingDomainStateIsRejected() {
+        for(Set<String> flags:java.util.List.of(Set.of("domain:fire","domain:water"),Set.of("domain:none","mast
ery-half:1"),Set.of("domain:unknown"),Set.of("mastery-half:-1"),Set.of("reset-last:2026-99-99"),Set.of("reset-qu
est:no")))
+            assertThrows(IllegalArgumentException.class,()->DomainProgression.read(flags));
+        var world=new RpgWorld();var before=world.save();
+        var flags=new TreeSet<>(before.flags());new DomainProgression("water",0,null,false).writeTo(flags);
+        assertThrows(IllegalArgumentException.class,()->new RpgSave(before.identity(),before.region(),before.x(
),before.y(),before.hp(),before.xp(),before.coins(),before.inventory(),before.equipment(),before.quests(),flags,
before.ticks()));
+    }
+    @Test void masteryOverflowCannotPartiallyAwardCharacterExperience() {
+        var world=new RpgWorld();world.awardExperience(RpgProgression.builtin().threshold(15));
+        assertTrue(world.chooseDomain("water"));var save=world.save();
+        var flags=new TreeSet<>(save.flags());new
DomainProgression("water",Long.MAX_VALUE,null,false).writeTo(flags);
+        var restored=RpgWorld.restore(new RpgSave(save.identity(),save.region(),save.x(),save.y(),save.hp(),sav
e.xp(),save.coins(),save.inventory(),save.equipment(),save.quests(),flags,save.ticks()));
+        var before=restored.save();
+        assertThrows(ArithmeticException.class,()->restored.awardExperience(3));
+        assertEquals(before,restored.save());
+    }
+}
```

### 56. [M] server/src/test/java/com/projetok/rpg/DungeonCycleTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/DungeonCycleTest.java
b/server/src/test/java/com/projetok/rpg/DungeonCycleTest.java
index 07126e81a881ecce4b9301b3b326a507407909fe..fcd2834d04c4fb5a9cdb18150c53a8233c72bbd2 100644
--- a/server/src/test/java/com/projetok/rpg/DungeonCycleTest.java
+++ b/server/src/test/java/com/projetok/rpg/DungeonCycleTest.java
@@ -3,7 +3,7 @@ import org.junit.jupiter.api.Test;
 import static org.junit.jupiter.api.Assertions.*;
 class DungeonCycleTest {
     static void fight(RpgWorld w,int ticks){for(int i=0;i<ticks;i++){w.attack();if(w.save().hp()<w.maxHp()-45)w
.use("potion");w.tick(0,0);}assertTrue(w.save().hp()>0);}
-    @org.junit.jupiter.params.ParameterizedTest
@org.junit.jupiter.params.provider.ValueSource(strings={"guardian","mystic"}) void
fullDungeonFromNewCharacterWithoutInjectedProgress(String archetype) {
+    @org.junit.jupiter.params.ParameterizedTest
@org.junit.jupiter.params.provider.ValueSource(strings={"reaper","human","aetheri","spirit"}) void
fullDungeonFromNewCharacterWithoutInjectedProgress(String archetype) {
         var w=new RpgWorld(CharacterIdentity.create("Expedicao",archetype,"amber"));w.interact();w.acceptQuest(
"first_steps");OfflineCycleTest.move(w,855,320);w.interact();fight(w,450);OfflineCycleTest.move(w,80,320);w.inte
ract();OfflineCycleTest.move(w,190,320);w.interact();assertTrue(w.completeQuest("first_steps"));assertTrue(w.acc
eptQuest("sanctuary"));
         OfflineCycleTest.move(w,330,440);w.harvest("starter_chest");assertTrue(w.equip("sword"));OfflineCycleTe
st.move(w,855,320);w.interact();OfflineCycleTest.move(w,855,320);assertTrue(w.interact());assertEquals("ruins",w
.save().region());
         OfflineCycleTest.move(w,270,420);fight(w,120);OfflineCycleTest.move(w,510,420);fight(w,160);OfflineCycl
eTest.move(w,670,320);fight(w,260);assertTrue(w.save().flags().contains("dungeon-completed"));assertEquals(1,w.i
nventory().count("relic"));
```

### 57. [M] server/src/test/java/com/projetok/rpg/EnemyRpgTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/EnemyRpgTest.java
b/server/src/test/java/com/projetok/rpg/EnemyRpgTest.java
index 78d88de6e907c9f5189cc6409d7a754c82042d8b..e4cb80ec2fdf76538c9c2e68dbcd6ac400cfda0d 100644
--- a/server/src/test/java/com/projetok/rpg/EnemyRpgTest.java
+++ b/server/src/test/java/com/projetok/rpg/EnemyRpgTest.java
@@ -6,8 +6,8 @@ class EnemyRpgTest {
     @Test void regionSpawnDeathRewardsAndRespawn() {
         assertTrue(new RpgWorld().snapshot().enemies().isEmpty());var
w=forest();assertEquals(3,w.snapshot().enemies().size());
         for(int i=0;i<3;i++){assertTrue(w.attack());for(int t=0;t<12;t++)w.tick(0,0);}
-        assertEquals(20,w.save().xp());assertEquals(5,w.save().coins());assertEquals(0,w.snapshot().enemies().g
et(0).hp());
-        assertFalse(w.attack());assertEquals(20,w.save().xp());
+        assertEquals(3,w.save().xp());assertEquals(5,w.save().coins());assertEquals(0,w.snapshot().enemies().ge
t(0).hp());
+        assertFalse(w.attack());assertEquals(3,w.save().xp());
         var restored=RpgWorld.restore(w.save());assertEquals(w.save(),restored.save());
         for(int t=0;t<250;t++)restored.tick(0,0);assertEquals(40,restored.snapshot().enemies().get(0).hp());
     }
```

### 58. [M] server/src/test/java/com/projetok/rpg/EquipmentRpgTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/EquipmentRpgTest.java
b/server/src/test/java/com/projetok/rpg/EquipmentRpgTest.java
index 5a7eeb28c45fc72a982aa3258cdbb6b67928c89e..a848382189aa94f66fb74da78e6f921520e2856b 100644
--- a/server/src/test/java/com/projetok/rpg/EquipmentRpgTest.java
+++ b/server/src/test/java/com/projetok/rpg/EquipmentRpgTest.java
@@ -7,7 +7,7 @@ class EquipmentRpgTest {
         assertFalse(w.equip("potion")); assertFalse(w.equip("sword"));
         w.inventory().add("sword",1); assertTrue(w.equip("sword")); assertEquals(base+8,w.attackPower());
         assertEquals(w.attackPower(),RpgWorld.restore(w.save()).attackPower());
-        assertTrue(w.inventory().add("robe",19)); var before=w.save();
+        assertTrue(w.inventory().add("robe",18)); var before=w.save();
         assertFalse(w.unequip("weapon")); assertEquals(before,w.save());
         w.inventory().remove("robe",1); assertTrue(w.unequip("weapon")); assertEquals(base,w.attackPower());
     }
```

### 59. [M] server/src/test/java/com/projetok/rpg/ItemCatalogTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/ItemCatalogTest.java
b/server/src/test/java/com/projetok/rpg/ItemCatalogTest.java
index b6adccd897283934b93466d09ba47ede4ebe35c3..d6386d59bed838a426f59bf9b81d79714e1201ef 100644
--- a/server/src/test/java/com/projetok/rpg/ItemCatalogTest.java
+++ b/server/src/test/java/com/projetok/rpg/ItemCatalogTest.java
@@ -3,7 +3,7 @@ import org.junit.jupiter.api.Test;
 import static org.junit.jupiter.api.Assertions.*;
 class ItemCatalogTest {
     @Test void categoryRestrictionsAreValidated() {
-        var catalog=new ItemCatalog(RpgContent.builtin()); assertEquals(5,catalog.all().size());
+        var catalog=new ItemCatalog(RpgContent.builtin()); assertEquals(6,catalog.all().size());
         assertEquals(50,catalog.get("potion").healing());
         assertThrows(IllegalArgumentException.class,()->new ItemCatalog.Item("bad","Name","Description",ItemCat
alog.Category.MATERIAL,ItemCatalog.Rarity.RARE,1,1,"weapon",0,0,1));
         assertThrows(IllegalArgumentException.class,()->catalog.get("missing"));
```

### 60. [A] server/src/test/java/com/projetok/rpg/LegacyRaceMigrationTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/LegacyRaceMigrationTest.java
b/server/src/test/java/com/projetok/rpg/LegacyRaceMigrationTest.java
new file mode 100644
index 0000000000000000000000000000000000000000..11b6901ee7db71d62c5d96f862d07d9c91e0c171
--- /dev/null
+++ b/server/src/test/java/com/projetok/rpg/LegacyRaceMigrationTest.java
@@ -0,0 +1,30 @@
+package com.projetok.rpg;
+
+import java.io.ByteArrayInputStream;
+import java.io.ByteArrayOutputStream;
+import java.util.Properties;
+import org.junit.jupiter.api.Test;
+import static org.junit.jupiter.api.Assertions.*;
+
+class LegacyRaceMigrationTest {
+    @Test void oldArchetypesMigrateWithoutLosingCharacterOrGrantingDomain() throws Exception {
+        for(String old:new String[]{"guardian","mystic"}) {
+            var world=new RpgWorld();world.awardExperience(500);
+            world.inventory().add("crystal",3);var before=world.save();
+            var properties=new Properties();properties.load(new
ByteArrayInputStream(LocalSaveStore.encode(before)));
+
properties.setProperty("schema","1");properties.remove("race");properties.setProperty("archetype",old);
+            properties.setProperty("baseHp",old.equals("guardian")?"140":"100");properties.setProperty("baseAtt
ack",old.equals("guardian")?"18":"24");
+            properties.remove("flag.domain:none");properties.remove("flag.mastery-half:0");
+            var bytes=new ByteArrayOutputStream();properties.store(bytes,"legacy fixture");
+            var migrated=LocalSaveStore.decode(bytes.toByteArray());
+            assertEquals(before.identity().id(),migrated.identity().id());assertEquals(before.identity().name()
,migrated.identity().name());
+            assertEquals(old.equals("guardian")?"reaper":"human",migrated.identity().race());
+            assertEquals(old.equals("guardian")?140:100,migrated.identity().baseHp());
+            assertEquals(old.equals("guardian")?18:24,migrated.identity().baseAttack());
+            assertEquals(before.xp(),migrated.xp());assertEquals(before.inventory(),migrated.inventory());
+            assertEquals(before.equipment(),migrated.equipment());assertFalse(DomainProgression.read(migrated.f
lags()).chosen());
+            var loaded=RpgWorld.restore(migrated).save();
+            assertEquals(loaded,LocalSaveStore.decode(LocalSaveStore.encode(loaded)));
+        }
+    }
```

```
+}
```

### 61. [M] server/src/test/java/com/projetok/rpg/LocalSaveStoreTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/LocalSaveStoreTest.java
b/server/src/test/java/com/projetok/rpg/LocalSaveStoreTest.java
index ca2442897aa45af8988c017d74425a62cc1592ec..288a7cf8bb0b75c1bd25add7c4f63dc919d6a74d 100644
--- a/server/src/test/java/com/projetok/rpg/LocalSaveStoreTest.java
+++ b/server/src/test/java/com/projetok/rpg/LocalSaveStoreTest.java
@@ -19,7 +19,7 @@ class LocalSaveStoreTest {
     @Test void rejectsCorruptionVersionAndDuplicateFields() throws Exception {
         byte[] good=LocalSaveStore.encode(new RpgWorld().save());
         String text=new String(good,java.nio.charset.StandardCharsets.ISO_8859_1);
-        for(String invalid:new
String[]{"garbage",text.replace("schema=1","schema=2"),text+"\nx=NaN\n",text.replace("x=140.0","x=NaN")})
+        for(String invalid:new
String[]{"garbage",text.replace("schema=2","schema=99"),text+"\nx=NaN\n",text.replace("x=140.0","x=NaN")})
             assertThrows(IllegalArgumentException.class,()->LocalSaveStore.decode(invalid.getBytes(java.nio.cha
rset.StandardCharsets.ISO_8859_1)));
     }
 }
```

### 62. [M] server/src/test/java/com/projetok/rpg/OfflineCycleTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/OfflineCycleTest.java
b/server/src/test/java/com/projetok/rpg/OfflineCycleTest.java
index 1728a17d70c1f4f0e9f2a72ff232f8614f270034..5a94ba09b18b89bc9928c50a4d9329afaaa5fd16 100644
--- a/server/src/test/java/com/projetok/rpg/OfflineCycleTest.java
+++ b/server/src/test/java/com/projetok/rpg/OfflineCycleTest.java
@@ -11,7 +11,7 @@ class OfflineCycleTest {
         fail("Nao chegou ao ponto "+x+","+y);
     }
     @Test void createQuestFightLootTradeEquipSaveAndResume() throws Exception {
-        var w=new RpgWorld(CharacterIdentity.create("Aurora
Teste","guardian","azure"));assertTrue(w.interact());assertTrue(w.acceptQuest("first_steps"));
+        var w=new RpgWorld(CharacterIdentity.create("Aurora
Teste","reaper","azure"));assertTrue(w.interact());assertTrue(w.acceptQuest("first_steps"));
         move(w,855,320);assertTrue(w.interact());assertEquals("forest",w.save().region());
         for(int i=0;i<800&&w.save().quests().get("first_steps")<2;i++){w.attack();w.tick(0,0);}
         assertEquals(2,w.save().quests().get("first_steps"));assertTrue(w.inventory().count("crystal")>=2);asse
rtTrue(w.save().hp()>0);
```

### 63. [M] server/src/test/java/com/projetok/rpg/RpgProgressionTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/RpgProgressionTest.java
b/server/src/test/java/com/projetok/rpg/RpgProgressionTest.java
index 6759f69b78f75a79b8f9ad1fb1579700f0b3a60f..353763337e2398c60079245b38a8bb7a60333299 100644
--- a/server/src/test/java/com/projetok/rpg/RpgProgressionTest.java
+++ b/server/src/test/java/com/projetok/rpg/RpgProgressionTest.java
@@ -2,11 +2,17 @@ package com.projetok.rpg;
 import org.junit.jupiter.api.Test;
 import static org.junit.jupiter.api.Assertions.*;
 class RpgProgressionTest {
-    @Test void multipleLevelsCapAndRestoration() {
+    @Test void multipleLevelsNoFinalCapAndRestoration() {
         var w=new RpgWorld();w.receiveDamage(20);int oldHp=w.save().hp();
         w.awardExperience(360);assertEquals(4,w.level());assertEquals(oldHp+36,w.save().hp());
         assertEquals(w.attackPower(),RpgWorld.restore(w.save()).attackPower());
-        w.awardExperience(1_000_000);assertEquals(20,w.level());
+        w.awardExperience(1_000_000);assertTrue(w.level()>20);
         var before=w.save();assertThrows(IllegalArgumentException.class,()->w.awardExperience(-1));assertEquals
(before,w.save());
     }
+    @Test void rewardNeverReducesPreviouslyAcceptedExperience() {
+        var legacyCapped=new RpgProgression(20,60,12,3);
+        assertEquals(50000,legacyCapped.award(50000,3));
+        assertEquals(50003,RpgProgression.builtin().award(50000,3));
+        assertEquals(1000000,RpgProgression.builtin().award(999999,3));
+    }
 }
```

### 64. [A] server/src/test/java/com/projetok/rpg/RpgRaceRuntimeTest.java

```
diff --git a/server/src/test/java/com/projetok/rpg/RpgRaceRuntimeTest.java
b/server/src/test/java/com/projetok/rpg/RpgRaceRuntimeTest.java
new file mode 100644
index 0000000000000000000000000000000000000000..1f68c3ab3147c645d50a75873566296f5f660c15
--- /dev/null
+++ b/server/src/test/java/com/projetok/rpg/RpgRaceRuntimeTest.java
@@ -0,0 +1,76 @@
+package com.projetok.rpg;
+
+import org.junit.jupiter.api.Test;
+import static org.junit.jupiter.api.Assertions.*;
+
+class RpgRaceRuntimeTest {
+    private static RpgWorld world(String race) {
+        return new RpgWorld(CharacterIdentity.create("Aprendiz",race,"azure"));
+    }
+    @Test void beginnersHaveRacialBasicsAndNoDomainOrShunpo() {
+        for(var race:CharacterRaces.all()) {
+            var w=world(race.id());
+            assertFalse(w.domainProgression().chosen());
+            assertFalse(w.castAbility());assertFalse(w.castSecondaryAbility());
+            assertEquals(1000,resource(w,"stamina"));
+            assertFalse(w.chooseDomain(Domains.forRace(race.id()).get(0).id()));
+            assertEquals(race.id().equals("reaper"),w.save().equipment().containsKey("weapon"));
+            assertEquals(race.id().equals("spirit")?180:race.id().equals("reaper")?82:60,w.basicAttackRange());
+        }
+    }
+    @Test void choiceAndFractionalMasteryPersistWithoutAllowingSwitch() throws Exception {
+        for(var race:CharacterRaces.all()) {
+            var w=world(race.id());w.awardExperience(RpgProgression.builtin().threshold(15));
+            assertEquals(0,w.domainProgression().xpHalfUnits());
+            var options=Domains.forRace(race.id());
+            assertTrue(w.chooseDomain(options.get(0).id()));
+            assertFalse(w.chooseDomain(options.get(1).id()));
+            w.awardExperience(3);
+            assertEquals(1.5,w.domainProgression().xp());
+            var loaded=RpgWorld.restore(LocalSaveStore.decode(LocalSaveStore.encode(w.save())));
+            assertEquals(w.save(),loaded.save());
+            assertEquals(w.domainProgression(),loaded.domainProgression());
+        }
+    }
+    @Test void oneLevelOneMobAwardsThreeAndOneAndAHalfMasteryOnce() {
+        var w=world("reaper");w.awardExperience(RpgProgression.builtin().threshold(15));
+        assertTrue(w.chooseDomain("water"));var s=w.save();
+        w=RpgWorld.restore(new RpgSave(s.identity(),"forest",240,360,s.hp(),s.xp(),s.coins(),s.inventory(),s.eq
uipment(),s.quests(),s.flags(),s.ticks()));
+        assertTrue(w.attack());assertEquals(s.xp()+3,w.save().xp());
+        assertEquals(1.5,w.domainProgression().xp());
+        assertFalse(w.attack());assertEquals(s.xp()+3,w.save().xp());
+    }
+    @Test void shunpoLevelTwentyIsSharedAndDefensiveWindowSurvivesSave() {
+        for(var race:CharacterRaces.all()) {
+            var w=world(race.id());w.awardExperience(RpgProgression.builtin().threshold(20));
+            assertTrue(w.canShunpo());assertTrue(w.castSecondaryAbility());
+            int hp=w.save().hp();w.receiveDamage(10);assertEquals(hp,w.save().hp());
+            w=RpgWorld.restore(w.save());w.receiveDamage(10);assertEquals(hp,w.save().hp());
+            assertFalse(w.castAbility());assertFalse(w.castSecondaryAbility());
+            for(int i=0;i<6;i++)w.tick(0,0);
+            w.receiveDamage(10);assertEquals(hp-10,w.save().hp());
+        }
+    }
+    @Test void hiddenNearestEnemyDoesNotBlockAVisibleEnemyInRange() {
+        var s=world("spirit").save();
+        var w=RpgWorld.restore(new RpgSave(s.identity(),"forest",360,180,s.hp(),s.xp(),s.coins(),s.inventory(),
s.equipment(),s.quests(),s.flags(),s.ticks()));
+        var enemies=w.snapshot().enemies();String hidden=enemies.get(0).spawn(),visible=enemies.get(1).spawn();
+        var flags=new java.util.TreeSet<String>();
+        for(String flag:w.save().flags()) {
+            String[] p=flag.split(":");
+            if(p[0].equals("mob")&&p[1].equals(hidden)){p[3]="520";p[4]="200";flag=String.join(":",p);}
+            else if(p[0].equals("mob")&&p[1].equals(visible)){p[3]="360";p[4]="350";flag=String.join(":",p);}
+            else if(p[0].equals("mob")&&p[1].equals(enemies.get(2).spawn())){p[2]="0";flag=String.join(":",p);}
+            flags.add(flag);
+        }
+        s=w.save();w=RpgWorld.restore(new RpgSave(s.identity(),s.region(),s.x(),s.y(),s.hp(),s.xp(),s.coins(),s
.inventory(),s.equipment(),s.quests(),flags,s.ticks()));
```

```
+        int hiddenHp=w.snapshot().enemies().get(0).hp(),visibleHp=w.snapshot().enemies().get(1).hp();
+        assertTrue(w.attack());
+        assertEquals(hiddenHp,w.snapshot().enemies().get(0).hp());
+        assertEquals(visibleHp-w.attackPower(),w.snapshot().enemies().get(1).hp());
+    }
+    private static int resource(RpgWorld w,String prefix) {
+        return w.save().flags().stream().filter(f->f.startsWith(prefix+":"))
+            .mapToInt(f->Integer.parseInt(f.substring(prefix.length()+1))).findFirst().orElseThrow();
+    }
+}
```

### 65. [M] server/src/test/java/com/projetok/studio/StudioFinalRegressionTest.java

```
diff --git a/server/src/test/java/com/projetok/studio/StudioFinalRegressionTest.java
b/server/src/test/java/com/projetok/studio/StudioFinalRegressionTest.java
index 5eaf9b85988dafce732e527f4e9219cb7aaf7ba8..6e2c1777897d2e0ebd32d8c57babb93489266fc9 100644
--- a/server/src/test/java/com/projetok/studio/StudioFinalRegressionTest.java
+++ b/server/src/test/java/com/projetok/studio/StudioFinalRegressionTest.java
@@ -11,5 +11,5 @@ class StudioFinalRegressionTest {
     @Test void randomizedEditsUndoAndPersistenceKeepOriginals()throws Exception{var
original=WorldDocument.empty("randomized",512,512);var d=original;var h=new EditorHistory(d);Random r=new
Random(114);for(int i=0;i<500;i++){int x=r.nextInt(16)*32,y=r.nextInt(16)*32;var before=d;d=i%4==0?EditorOperati
ons.erase(d,"ground",x,y):EditorOperations.pencil(d,EditorOperationsTest.brush(),x,y,true);h.commit(d);if(!befor
e.equals(d)){assertEquals(before,h.undo());assertEquals(d,h.redo());}if(i%20==0)assertEquals(d,DocumentStore.dec
ode(DocumentStore.encode(d)));}DocumentStore.save(root.resolve("mapa.sgmap"),d);assertEquals(d,DocumentStore.loa
d(root.resolve("mapa.sgmap")));assertTrue(original.elements().isEmpty());}
     @Test void forgedImageDimensionsAreRejectedBeforeDecode()throws Exception{Path
p=Files.createDirectories(root.resolve("assets")).resolve("image.png");ImageIO.write(new
BufferedImage(32,32,2),"png",p.toFile());var a=AssetCatalog.scan(root,null).assets().get(0);var forged=new Asset
Catalog.Asset(a.id(),a.path(),a.file(),a.sha(),8192,8192,true,a.category(),a.thumbnail(),a.indexedAt());assertTh
rows(java.io.IOException.class,()->new AssetImages().load(forged));}
     @Test void invalidAnimationDoesNotBreakCanvasAndUtf8CorruptionIsRejected()throws Exception{var
e=EditorOperationsTest.brush();var bad=new WorldDocument.Element(e.id(),e.kind(),e.asset(),e.sha(),e.crop(),0,0,
32,32,e.layer(),true,false,"","FRONT",Map.of("animation.frames","invalid"));var
d=WorldDocument.empty("mapa",128,128).withElements(List.of(bad));var canvas=new WorldCanvas(()->d,new
AssetImages());canvas.setSize(300,300);var image=new BufferedImage(300,300,2);var g=image.createGraphics();asser
tDoesNotThrow(()->canvas.paint(g));g.dispose();assertThrows(java.io.IOException.class,()->DocumentStore.decode(n
ew byte[]{(byte)0xff}));assertTrue(StudioValidator.validate(Map.of(d.id(),d),new AssetCatalog.Result(List.of(),L
ist.of(),List.of()),false).stream().anyMatch(p->p.severity()==StudioProblem.Severity.ERROR));}
-    @Test void realWorldUsesConfiguredNpcDialogueAndEventsKeepPortalGuards(){var d=StudioExample.document();var
 npc=d.elements().get(1);var edited=new WorldDocument.Element(npc.id(),npc.kind(),npc.asset(),npc.sha(),npc.crop
(),npc.x(),npc.y(),npc.width(),npc.height(),npc.layer(),true,false,npc.reference(),npc.direction(),Map.of("npc.d
ialogue","elder.help","npc.name","Guia"));d=d.withElements(List.of(d.elements().get(0),edited));var world=new co
m.projetok.rpg.RpgWorld(com.projetok.rpg.CharacterIdentity.create("Teste","guardian","azure"),StudioExport.scene
s(d.id(),Map.of(d.id(),d)));assertTrue(world.interact());assertEquals("elder.help",world.dialogue().id());assert
False(world.travel(StudioExport.key("portal_",d.id()+"/portal")));}
+    @Test void realWorldUsesConfiguredNpcDialogueAndEventsKeepPortalGuards(){var d=StudioExample.document();var
 npc=d.elements().get(1);var edited=new WorldDocument.Element(npc.id(),npc.kind(),npc.asset(),npc.sha(),npc.crop
(),npc.x(),npc.y(),npc.width(),npc.height(),npc.layer(),true,false,npc.reference(),npc.direction(),Map.of("npc.d
ialogue","elder.help","npc.name","Guia"));d=d.withElements(List.of(d.elements().get(0),edited));var world=new co
m.projetok.rpg.RpgWorld(com.projetok.rpg.CharacterIdentity.create("Teste","reaper","azure"),StudioExport.scenes(
d.id(),Map.of(d.id(),d)));assertTrue(world.interact());assertEquals("elder.help",world.dialogue().id());assertFa
lse(world.travel(StudioExport.key("portal_",d.id()+"/portal")));}

 }
```

### 66. [M] server/src/test/java/com/projetok/studio/StudioIntegrationTest.java

```
diff --git a/server/src/test/java/com/projetok/studio/StudioIntegrationTest.java
b/server/src/test/java/com/projetok/studio/StudioIntegrationTest.java
index 57ed5708a494dbc2e48b906660cc131f9d8d86c7..997f3ad23f0e12afa98630ee47e60c4d38120cb6 100644
--- a/server/src/test/java/com/projetok/studio/StudioIntegrationTest.java
+++ b/server/src/test/java/com/projetok/studio/StudioIntegrationTest.java
@@ -8,6 +8,6 @@ import static org.junit.jupiter.api.Assertions.*;
 class StudioIntegrationTest {
     @TempDir Path folder;
     static WorldDocument map(String id,String region){var d=WorldDocument.empty(id,960,640);var spawn=new
WorldDocument.Element("player",WorldDocument.Kind.PLAYER,"","",new
WorldDocument.Crop(0,0,32,32),140,320,32,32,"interaction",true,false,"","FRONT",Map.of());return new
WorldDocument(1,id,id,region,960,640,32,d.layers(),List.of(spawn),List.of(),List.of(),List.of(),Map.of());}
-    @Test void exportedCollisionAndPortalRunInRealWorldWithoutChangingLegacy()throws Exception{var
a=map("novo","village").withHitboxes(List.of(new
WorldDocument.Hitbox("wall",200,280,32,100,true,"collision"))).withPortals(List.of(new
WorldDocument.Portal("link",140,320,"bosque",140,320,"","FRONT")));var b=map("bosque","forest");var
path=StudioExport.write(folder.resolve("export"),a.id(),Map.of(a.id(),a,b.id(),b),new
AssetCatalog.Result(List.of(),List.of(),List.of()));var bundle=StudioExport.load(path);var w=new
RpgWorld(CharacterIdentity.create("Teste","guardian","azure"),bundle.scenes());for(int i=0;i<25;i++)w.tick(1,0);
assertTrue(w.snapshot().x()<=182.01);assertTrue(w.travel(StudioExport.key("portal_",a.id()+"/link")));assertEqua
ls("forest",w.save().region());assertEquals(140,w.snapshot().x());var legacy=new
RpgWorld();assertEquals(140,legacy.snapshot().x());assertTrue(legacy.snapshot().enemies().isEmpty());}
+    @Test void exportedCollisionAndPortalRunInRealWorldWithoutChangingLegacy()throws Exception{var
a=map("novo","village").withHitboxes(List.of(new
WorldDocument.Hitbox("wall",200,280,32,100,true,"collision"))).withPortals(List.of(new
WorldDocument.Portal("link",140,320,"bosque",140,320,"","FRONT")));var b=map("bosque","forest");var
path=StudioExport.write(folder.resolve("export"),a.id(),Map.of(a.id(),a,b.id(),b),new
AssetCatalog.Result(List.of(),List.of(),List.of()));var bundle=StudioExport.load(path);var w=new
RpgWorld(CharacterIdentity.create("Teste","reaper","azure"),bundle.scenes());for(int i=0;i<25;i++)w.tick(1,0);as
sertTrue(w.snapshot().x()<=182.01);assertTrue(w.travel(StudioExport.key("portal_",a.id()+"/link")));assertEquals
("forest",w.save().region());assertEquals(140,w.snapshot().x());var legacy=new
RpgWorld();assertEquals(140,legacy.snapshot().x());assertTrue(legacy.snapshot().enemies().isEmpty());}
     @Test void unsupportedSizeAndTamperedBundleAreRejected()throws Exception{var a=map("novo","village");var
out=StudioExport.write(folder.resolve("ok"),a.id(),Map.of(a.id(),a),new AssetCatalog.Result(List.of(),List.of(),
List.of()));Files.writeString(out.resolve("novo.sgmap"),"alterado");assertThrows(java.io.IOException.class,()->S
tudioExport.load(out));var big=WorldDocument.empty("grande",1920,1280);assertThrows(java.io.IOException.class,()
->StudioExport.write(folder.resolve("bad"),big.id(),Map.of(big.id(),big),new
AssetCatalog.Result(List.of(),List.of(),List.of())));assertFalse(Files.exists(folder.resolve("bad")));}
 }
```

### 67. [M] server/src/test/java/com/projetok/studio/StudioRuntimePlaytestTest.java

```
diff --git a/server/src/test/java/com/projetok/studio/StudioRuntimePlaytestTest.java
b/server/src/test/java/com/projetok/studio/StudioRuntimePlaytestTest.java
index aa3507d612ebf3eb0a1946cd7286943b419bada9..e067a2c043297fe40f685b83f7b9b915218b42cf 100644
--- a/server/src/test/java/com/projetok/studio/StudioRuntimePlaytestTest.java
+++ b/server/src/test/java/com/projetok/studio/StudioRuntimePlaytestTest.java
@@ -6,5 +6,5 @@ import java.util.*;
 import java.util.concurrent.TimeUnit;
 import static org.junit.jupiter.api.Assertions.*;
 class StudioRuntimePlaytestTest {
-    @Test void playtestAdvancesRealRuntimeAndClosesPendingActions()throws Exception{var
d=StudioIntegrationTest.map("novo","village");var scenes=StudioExport.scenes(d.id(),Map.of(d.id(),d));try(var
runtime=new RpgRuntime(new
RpgWorld(CharacterIdentity.create("Teste","guardian","azure"),scenes))){runtime.input(1,0);runtime.start();long
deadline=System.nanoTime()+TimeUnit.SECONDS.toNanos(3);while(runtime.snapshot().ticks()<4&&System.nanoTime()<dea
dline)Thread.sleep(5);assertTrue(runtime.snapshot().x()>140);assertNull(runtime.failure());var
action=runtime.action(w->w.attack());runtime.close();assertNotNull(action.get(1,TimeUnit.SECONDS));}}
+    @Test void playtestAdvancesRealRuntimeAndClosesPendingActions()throws Exception{var
d=StudioIntegrationTest.map("novo","village");var scenes=StudioExport.scenes(d.id(),Map.of(d.id(),d));try(var
runtime=new RpgRuntime(new
RpgWorld(CharacterIdentity.create("Teste","reaper","azure"),scenes))){runtime.input(1,0);runtime.start();long de
adline=System.nanoTime()+TimeUnit.SECONDS.toNanos(3);while(runtime.snapshot().ticks()<4&&System.nanoTime()<deadl
ine)Thread.sleep(5);assertTrue(runtime.snapshot().x()>140);assertNull(runtime.failure());var
action=runtime.action(w->w.attack());runtime.close();assertNotNull(action.get(1,TimeUnit.SECONDS));}}
 }
```

### 68. [M] testar-mapa-rpg.cmd

```
diff --git a/testar-mapa-rpg.cmd b/testar-mapa-rpg.cmd
index 74f86c4328417dd674637a19a419546c615b32b1..25157725ac8b34df02577af544840bfbf207ddf3 100644
--- a/testar-mapa-rpg.cmd
+++ b/testar-mapa-rpg.cmd
@@ -1,8 +1,8 @@
 @echo off
 setlocal
-cd /d "%~dp0"
 if "%~1"=="" (
   echo Uso: testar-mapa-rpg.cmd "pasta-exportada"
-  exit /b 1
+  exit /b 2
 )
-java -cp "server\target\classes" com.projetok.studio.StudioPlaytest "%~1" "%~dp0."
+powershell.exe -NoProfile -ExecutionPolicy Bypass -File "%~dp0tools\launch-local.ps1" -Mode Map -MapFolder
"%~1"
+exit /b %ERRORLEVEL%
```

### 69. [M] testar-rpg-sprites.cmd

```
diff --git a/testar-rpg-sprites.cmd b/testar-rpg-sprites.cmd
index f5a5cd0a1f72341b1893fab2d70359d6f9d325c5..1bb66f61643ddfb0fc9a37596c51be569772e963 100644
--- a/testar-rpg-sprites.cmd
+++ b/testar-rpg-sprites.cmd
@@ -1,17 +1,11 @@
 @echo off
 setlocal
-cd /d "%~dp0"
-if not defined SPIRIT_GATE_SPRITE_SCALE set "SPIRIT_GATE_SPRITE_SCALE=0.5"
-if not defined SPIRIT_GATE_FRAME_MS set "SPIRIT_GATE_FRAME_MS=250"
-if not exist "server\target\classes\com\projetok\prototype\rpg\RpgPlayerSprites.class" (
-  echo Compile primeiro: python tools/qualidade.py --offline --task 079-local
-  exit /b 1
-)
 if "%~1"=="--geometrico" (
-  java -cp "server\target\classes" com.projetok.prototype.rpg.RpgLauncher
+  powershell.exe -NoProfile -ExecutionPolicy Bypass -File "%~dp0tools\launch-local.ps1" -Mode Sprites
-Geometric
 ) else if "%~1"=="" (
-  java "-Dspiritgate.rpg.spriteScale=%SPIRIT_GATE_SPRITE_SCALE%"
"-Dspiritgate.rpg.spriteFrameMillis=%SPIRIT_GATE_FRAME_MS%" -cp "server\target\classes"
com.projetok.prototype.rpg.RpgLauncher --experimental-player-sprites
+  powershell.exe -NoProfile -ExecutionPolicy Bypass -File "%~dp0tools\launch-local.ps1" -Mode Sprites
 ) else (
   echo Uso: testar-rpg-sprites.cmd [--geometrico]
   exit /b 2
 )
+exit /b %ERRORLEVEL%
```

### 70. [M] tools/SpiritGateCreationProbe.java

```
diff --git a/tools/SpiritGateCreationProbe.java b/tools/SpiritGateCreationProbe.java
index cc0302820b1175bec77cc1524c163486319e095e..4323722913119d0122832ad6e221625a7e9b6dcc 100644
--- a/tools/SpiritGateCreationProbe.java
+++ b/tools/SpiritGateCreationProbe.java
@@ -31,7 +31,7 @@ public final class SpiritGateCreationProbe {
             var form=(CharacterCreation)target.getContentPane();
             form.name.setText("!");form.create.doClick();
             if(form.error.getText().isBlank()||accepted.get()!=null)throw new AssertionError("Validação");
-            form.mystic.doClick();form.appearance.setSelectedItem("amber");
+            form.races.get("aetheri").doClick();form.appearance.setSelectedItem("amber");
         });
         for(var size:new int[][]{{820,540},{800,600},{640,480}}) {
             SwingUtilities.invokeAndWait(() -> {
@@ -44,8 +44,8 @@ public final class SpiritGateCreationProbe {
         SwingUtilities.invokeAndWait(() -> {
             var form=(CharacterCreation)target.getContentPane();form.name.setText("Heroi
sintético");form.create.doClick();
         });
-        if(accepted.get()==null||!accepted.get().archetype().equals("mystic"))throw new
AssertionError("Criação");
-        Files.writeString(output.resolve("probe.txt"),"PASS: diálogo modal real, nome inválido preserva
formulário, Mystic/amber, fallback e criação válida. Capturas printAll; não homologam monitor físico. Nenhum
save escrito.\n");
+        if(accepted.get()==null||!accepted.get().race().equals("aetheri"))throw new AssertionError("Criação");
+        Files.writeString(output.resolve("probe.txt"),"PASS: diálogo modal real, nome inválido preserva
formulário, Aetheri/amber, fallback e criação válida sem domínio. Capturas printAll; não homologam monitor
físico. Nenhum save escrito.\n");
         SwingUtilities.invokeAndWait(() -> {for(Window window:Window.getWindows())window.dispose();});
     }
 }
```

### 71. [M] tools/SpiritGateFinalJourneyProbe.java

```
diff --git a/tools/SpiritGateFinalJourneyProbe.java b/tools/SpiritGateFinalJourneyProbe.java
index 414dd0a2f9b518cb9e5d0372e75bbee95ec96e31..df0173df2c4de9c24a9d918b37783df3e25b5032 100644
--- a/tools/SpiritGateFinalJourneyProbe.java
+++ b/tools/SpiritGateFinalJourneyProbe.java
@@ -23,7 +23,7 @@ public final class SpiritGateFinalJourneyProbe {
         SwingUtilities.invokeLater(SpiritGateEntry::open);waitFor(()->SpiritGateCorujao3Probe.frame("Spirit
Gate | Portal de entrada")!=null);
         JFrame entry=SpiritGateCorujao3Probe.frame("Spirit Gate | Portal de entrada");SwingUtilities.invokeAndW
ait(()->SpiritGateWindowsProbe.find(entry,JTextField.class,null).setText("Perfil sintético"));click(entry,"Entra
r");waitFor(()->menu()!=null);waitFor(()->SpiritGateWindowsProbe.find(menu(),JButton.class,"Criar
personagem").isEnabled());
         click(menu(),"Criar personagem");waitFor(()->SpiritGateWindowsProbe.dialog("Criar personagem
local")!=null);JDialog creation=SpiritGateWindowsProbe.dialog("Criar personagem local");
-        SwingUtilities.invokeAndWait(()->{var
form=(CharacterCreation)creation.getContentPane();form.name.setText("Jornada
sintética");form.guardian.doClick();form.create.doClick();});
+        SwingUtilities.invokeAndWait(()->{var
form=(CharacterCreation)creation.getContentPane();form.name.setText("Jornada
sintética");form.races.get("reaper").doClick();form.create.doClick();});
         waitFor(()->SpiritGateCorujao3Probe.frame("Spirit Gate | RPG
offline")!=null);game=SpiritGateCorujao3Probe.frame("Spirit Gate | RPG
offline");waitFor(()->SpiritGateWindowsProbe.dialog("Primeiros
passos")!=null);SpiritGateCorujao3Probe.closeHelp();
         key("W");click(game,"Minimapa");click(game,"Minimapa");click(game,"Mapa
[M]");waitFor(()->SpiritGateWindowsProbe.dialog("Diario e exploracao")!=null);JDialog
diary=SpiritGateWindowsProbe.dialog("Diario e
exploracao");capture(diary,"01-mapa");SwingUtilities.invokeAndWait(()->diary.setVisible(false));
         saveMenu();var repository=new CharacterSaves(data.resolve("saves/characters"));var
slot=repository.list().get(0);var store=new LocalSaveStore(slot.file());var
state=store.load();if(state.y()>=320)throw new AssertionError("Movimento não confirmado no save");
@@ -34,12 +34,14 @@ public final class SpiritGateFinalJourneyProbe {
         click(game,"Inventario [I]");waitFor(()->SpiritGateWindowsProbe.dialog("Inventario e
equipamento")!=null);JDialog inventory=SpiritGateWindowsProbe.dialog("Inventario e equipamento");SwingUtilities.
invokeAndWait(()->{lists(inventory).get(0).setSelectedIndex(1);SpiritGateWindowsProbe.find(inventory,JButton.cla
ss,"Equipar").doClick();});
         waitFor(()->{var model=lists(inventory).get(1).getModel();for(int
i=0;i<model.getSize();i++)if(model.getElementAt(i).toString().startsWith("weapon: Espada"))return true;return
false;});capture(inventory,"03-equipamento");SwingUtilities.invokeAndWait(()->inventory.setVisible(false));
         click(game,"Diario [J]");capture(diary,"04-diario");SwingUtilities.invokeAndWait(()->diary.setVisible(f
alse));saveMenu();state=store.load();

-        // Fixture de combate: só muda região/posição, mantendo inventário e moedas verificados.
-        store.save(new RpgSave(state.identity(),"forest",240,360,state.hp(),state.xp(),state.coins(),state.inve
ntory(),state.equipment(),state.quests(),state.flags(),state.ticks()));reloadMenu();openGame();
```

```
-        for(int i=0;i<6;i++){click(game,"Golpe [SPACE]");Thread.sleep(450);}click(game,"Habilidade
[Q]");capture(game,"05-combate");
+        // Fixture de combate: posiciona no bosque e concede nível 20 para validar Shunpo.
+        long combatStartXp=RpgProgression.builtin().threshold(20);
+        store.save(new RpgSave(state.identity(),"forest",240,360,state.hp(),combatStartXp,state.coins(),state.i
nventory(),state.equipment(),state.quests(),state.flags(),state.ticks()));reloadMenu();openGame();
+        click(game,RpgLauncher.actionLabel("ability")+" [Q]");
+        for(int i=0;i<6;i++){click(game,RpgLauncher.actionLabel("attack")+"
[SPACE]");Thread.sleep(450);}capture(game,"05-combate");
         click(game,"Opções");waitFor(()->SpiritGateWindowsProbe.dialog("Opções do Spirit Gate")!=null);JDialog
options=SpiritGateWindowsProbe.dialog("Opções do Spirit Gate");SwingUtilities.invokeAndWait(()->{SpiritGateWindo
wsProbe.find(options,JSlider.class,null).setValue(55);SpiritGateWindowsProbe.find(options,JButton.class,"Aplicar
").doClick();});waitFor(()->!options.isDisplayable());saveMenu();
-        var
saved=store.load();if(saved.xp()<=0||!saved.equipment().containsValue("sword")||saved.coins()<94)throw new
AssertionError("Combate/equipamento/moedas não
persistiram");if(RpgPreferences.load(data.resolve("saves/preferences.properties")).volume()!=55)throw new
AssertionError("Opções não persistiram");
+        var saved=store.load();if(saved.xp()<=combatStartXp||!saved.equipment().containsValue("sword")||saved.c
oins()<94)throw new AssertionError("Combate/equipamento/moedas não
persistiram");if(RpgPreferences.load(data.resolve("saves/preferences.properties")).volume()!=55)throw new
AssertionError("Opções não persistiram");
         openGame();capture(game,"06-continuar");saveMenu();var continued=store.load();if(!continued.identity().
equals(saved.identity())||continued.xp()!=saved.xp()||!continued.equipment().equals(saved.equipment()))throw new
 AssertionError("Continuação alterou progresso");
-        Files.writeString(output.resolve("probe.txt"),"PASS: entrada/perfil, criação Guardian, ajuda
dispensada, ação de movimento, mapa/minimapa, NPC comerciante, compra/venda, inventário/equipamento, missão
consultada, inimigo derrotado com XP, habilidade, opções volume 55, salvar/seleção/continuar íntegros. Mesmo
personagem criado pela UI. Duas fixtures explícitas reposicionam comércio e combate e fornecem saldo/espada
sintéticos; não constituem teste de progressão natural da campanha. Ações Swing, não teclado físico. Dados:
"+data+"\n");SwingUtilities.invokeAndWait(()->{for(Window window:Window.getWindows())window.dispose();});
+        Files.writeString(output.resolve("probe.txt"),"PASS: entrada/perfil, criação Reaper aprendiz, ajuda,
movimento, mapa/minimapa, comércio, inventário/equipamento, diário, derrota de inimigo com ganho de XP, Shunpo,
opções e continuação preservada. Fixtures explícitas fornecem saldo/espada, posição de combate e nível 20 para
testar Shunpo; não validam progressão natural da campanha. Ações Swing, sem teclado físico. Dados:
"+data+"\n");SwingUtilities.invokeAndWait(()->{for(Window window:Window.getWindows())window.dispose();});
     }

 }
```

### 72. [M] tools/SpiritGateSelectionProbe.java

```
diff --git a/tools/SpiritGateSelectionProbe.java b/tools/SpiritGateSelectionProbe.java
index 1646e11cec52d1557a912e9b21dee799837523e0..b376a6aae558f5b2af029b2b6800367dd03d7a29 100644
--- a/tools/SpiritGateSelectionProbe.java
+++ b/tools/SpiritGateSelectionProbe.java
@@ -26,7 +26,7 @@ public final class SpiritGateSelectionProbe {
         SwingUtilities.invokeAndWait(SpiritGateEntry::open);waitFor(()->frame("Spirit Gate | Portal de
entrada")!=null);var entry=frame("Spirit Gate | Portal de entrada");capture(entry,"01-entrada");
         final JFrame
initialEntry=entry;SwingUtilities.invokeAndWait(()->{find(initialEntry,JTextField.class,null).setText("Perfil de
 teste");find(initialEntry,JCheckBox.class,null).setSelected(true);});click(entry,"Entrar");waitFor(()->frame("S
pirit Gate | Inicio")!=null);JFrame menu=frame("Spirit Gate | Inicio");final JFrame
emptyMenu=menu;waitFor(()->find(emptyMenu,JButton.class,"Criar
personagem").isEnabled());capture(menu,"02-vazio");
         click(menu,"Criar personagem");waitFor(()->java.util.Arrays.stream(Window.getWindows()).anyMatch(w->w
instanceof JDialog&&w.isShowing()));
-        SwingUtilities.invokeLater(()->{for(Window w:Window.getWindows())if(w instanceof JDialog
d&&d.isShowing()){find(d,JTextField.class,null).setText("Heroi UI");JOptionPane
pane=find(d,JOptionPane.class,null);pane.setValue(JOptionPane.OK_OPTION);}});
+        SwingUtilities.invokeLater(()->{for(Window w:Window.getWindows())if(w instanceof JDialog
d&&d.isShowing()){find(d,JTextField.class,null).setText("Heroi UI");find(d,JButton.class,"Criar
personagem").doClick();}});
         waitFor(()->frame("Spirit Gate | RPG offline")!=null);JFrame game=frame("Spirit Gate | RPG
offline");capture(game,"03-personagem-criado");click(game,"Salvar e voltar ao menu");waitFor(()->frame("Spirit
Gate | Inicio")!=null);menu=frame("Spirit Gate | Inicio");final JFrame
returned=menu;waitFor(()->find(returned,JButton.class,"Jogar").isEnabled());capture(menu,"04-selecao-retorno");
         click(menu,"Jogar");waitFor(()->frame("Spirit Gate | RPG offline")!=null);game=frame("Spirit Gate | RPG
 offline");click(game,"Salvar e voltar ao menu");waitFor(()->frame("Spirit Gate | Inicio")!=null);
         menu=frame("Spirit Gate | Inicio");click(menu,"Voltar");waitFor(()->frame("Spirit Gate | Portal de
entrada")!=null);entry=frame("Spirit Gate | Portal de entrada");final JFrame
remembered=entry;waitFor(()->find(remembered,JTextField.class,null).getText().equals("Perfil de
teste"));capture(entry,"05-voltar-perfil");
```

### 73. [M] tools/SpiritGateSelectionVisualProbe.java

```
diff --git a/tools/SpiritGateSelectionVisualProbe.java b/tools/SpiritGateSelectionVisualProbe.java
index bc04e094d1ff9745eb4d15f74bc367a5372a047e..6cb399d74a7f64d35485ef871cb19b1edc72c355 100644
--- a/tools/SpiritGateSelectionVisualProbe.java
+++ b/tools/SpiritGateSelectionVisualProbe.java
@@ -26,7 +26,7 @@ public final class SpiritGateSelectionVisualProbe {
         SwingUtilities.invokeAndWait(SpiritGateEntry::open);waitFor(()->frame("Spirit Gate | Portal de
entrada")!=null);var entry=frame("Spirit Gate | Portal de entrada");capture(entry,"01-entrada");
         final JFrame
initialEntry=entry;SwingUtilities.invokeAndWait(()->{find(initialEntry,JTextField.class,null).setText("Perfil de
 teste");find(initialEntry,JCheckBox.class,null).setSelected(true);});click(entry,"Entrar");waitFor(()->frame("S
pirit Gate | Inicio")!=null);JFrame menu=frame("Spirit Gate | Inicio");final JFrame
emptyMenu=menu;waitFor(()->find(emptyMenu,JButton.class,"Criar
personagem").isEnabled());capture(menu,"02-vazio");
         click(menu,"Criar personagem");waitFor(()->java.util.Arrays.stream(Window.getWindows()).anyMatch(w->w
instanceof JDialog&&w.isShowing()));
-        SwingUtilities.invokeLater(()->{for(Window w:Window.getWindows())if(w instanceof JDialog
d&&d.isShowing()){find(d,JTextField.class,null).setText("Heroi UI");JOptionPane
pane=find(d,JOptionPane.class,null);pane.setValue(JOptionPane.OK_OPTION);}});
+        SwingUtilities.invokeLater(()->{for(Window w:Window.getWindows())if(w instanceof JDialog
d&&d.isShowing()){find(d,JTextField.class,null).setText("Heroi UI");find(d,JButton.class,"Criar
personagem").doClick();}});
         waitFor(()->frame("Spirit Gate | RPG offline")!=null);JFrame game=frame("Spirit Gate | RPG
offline");capture(game,"03-personagem-criado");click(game,"Salvar e voltar ao menu");waitFor(()->frame("Spirit
Gate | Inicio")!=null);menu=frame("Spirit Gate | Inicio");final JFrame
returned=menu;waitFor(()->find(returned,JButton.class,"Jogar").isEnabled());capture(menu,"04-selecao-retorno");
         click(menu,"Jogar");waitFor(()->frame("Spirit Gate | RPG offline")!=null);game=frame("Spirit Gate | RPG
 offline");click(game,"Salvar e voltar ao menu");waitFor(()->frame("Spirit Gate | Inicio")!=null);
         menu=frame("Spirit Gate | Inicio");click(menu,"Voltar");waitFor(()->frame("Spirit Gate | Portal de
entrada")!=null);entry=frame("Spirit Gate | Portal de entrada");final JFrame
remembered=entry;waitFor(()->find(remembered,JTextField.class,null).getText().equals("Perfil de
teste"));capture(entry,"05-voltar-perfil");
```

### 74. [M] tools/build-rpg-portable.ps1

```
diff --git a/tools/build-rpg-portable.ps1 b/tools/build-rpg-portable.ps1
index c28bb878e78ab7afc5854a182623401cedfa1af0..01ba9f7bce290f636dc7fe6674626fbeb10be94c 100644
--- a/tools/build-rpg-portable.ps1
+++ b/tools/build-rpg-portable.ps1
@@ -1,30 +1,57 @@
-param([string]$Version='0.58.0-rc1')
+param([string]$Version='1.0.0-rc1', [string]$RuntimeImage='')
 Set-StrictMode -Version Latest
 $ErrorActionPreference='Stop'
-if($Version -notmatch '^0[.]\d+[.]\d+-rc\d+$'){throw 'Versao invalida'}
+if($Version -notmatch '^\d+[.]\d+[.]\d+-rc\d+$'){throw 'Versao invalida'}
 $project=[IO.Path]::GetFullPath((Join-Path $PSScriptRoot '..'))
-$jdkBin=Split-Path (Get-Command java.exe).Source
+$compiled = & (Join-Path $PSScriptRoot 'compile-local.ps1') -PassThru
+$jdkBin=$compiled.JdkBin
 $buildId=$Version+'-'+[DateTime]::UtcNow.ToString('yyyyMMddTHHmmssZ')+'-'+[Guid]::NewGuid().ToString('N').Subst
ring(0,8)
 $build=Join-Path $project ('server\target\rpg-portable-'+$buildId)
 $inputDir=Join-Path $build 'input'
 $destination=Join-Path $build 'output'
 $runtimeDir=Join-Path $build 'runtime'
-$classes=Join-Path $project 'server\target\classes'
+$classes=$compiled.Classes
 New-Item -ItemType Directory -Path $inputDir,$destination | Out-Null
 function Checked([string]$tool,[string[]]$arguments){ & (Join-Path $jdkBin $tool) @arguments; if($LASTEXITCODE
-ne 0){throw "$tool falhou: $LASTEXITCODE"} }
 $jar=Join-Path $inputDir 'spirit-gate-rpg.jar'
 Checked 'jar.exe' @('--create','--file',$jar,'--main-class','com.projetok.prototype.rpg.RpgLauncher','-C',$clas
ses,'com/projetok/domain','-C',$classes,'com/projetok/prototype','-C',$classes,'com/projetok/rpg','-C',$classes,
'rpg')
 Checked 'jdeps.exe' @('--multi-release','17','--print-module-deps',$jar)
-Checked 'jlink.exe' @('--add-modules','java.base,java.desktop,java.logging,jdk.random','--strip-debug','--no-ma
n-pages','--no-header-files','--compress=2','--output',$runtimeDir)
+$modules = 'java.base,java.desktop,java.logging'
+if (@(& (Join-Path $jdkBin 'java.exe') --list-modules) -match '^jdk[.]random@') { $modules += ',jdk.random' }
+if ($RuntimeImage) {
+    # Uma imagem anterior so e reutilizada com os mesmos binarios e modulos do JDK.
+    $RuntimeImage = [IO.Path]::GetFullPath($RuntimeImage)
+    foreach ($binary in @('bin\java.exe', 'bin\server\jvm.dll')) {
+        $runtimeBinary = Join-Path $RuntimeImage $binary
+        $jdkBinary = Join-Path (Split-Path -Parent $jdkBin) $binary
+        if (-not (Test-Path -LiteralPath $runtimeBinary -PathType Leaf) -or
```

```
+           (Get-FileHash -LiteralPath $runtimeBinary -Algorithm SHA256).Hash -ne (Get-FileHash -LiteralPath
$jdkBinary -Algorithm SHA256).Hash) {
+           throw 'RuntimeImage nao corresponde ao JDK selecionado.'
+       }
+   }
+   $expectedModules = @(& (Join-Path $jdkBin 'java.exe') --limit-modules $modules --list-modules |
Sort-Object)
+   if ($LASTEXITCODE -ne 0) { throw 'Falha ao verificar os modulos do JDK.' }
+   $runtimeModules = @(& (Join-Path $RuntimeImage 'bin\java.exe') --list-modules | Sort-Object)
+   if ($LASTEXITCODE -ne 0 -or ($expectedModules -join "`n") -ne ($runtimeModules -join "`n")) {
+       throw 'RuntimeImage possui modulos diferentes dos solicitados.'
+   }
+   Copy-Item -LiteralPath $RuntimeImage -Destination $runtimeDir -Recurse
+} else {
+   Checked 'jlink.exe' @('--add-modules',$modules,'--strip-debug','--no-man-pages','--no-header-files','--comp
ress=2','--output',$runtimeDir)
+}
 $windowsVersion=($Version -split '-')[0]
 Checked 'jpackage.exe' @('--type','app-image','--dest',$destination,'--name','SpiritGateRPG','--input',$inputDi
r,'--main-jar','spirit-gate-rpg.jar','--main-class','com.projetok.prototype.rpg.RpgLauncher','--runtime-image',$
runtimeDir,'--app-version',$windowsVersion,'--vendor','Spirit Gate','--description','Spirit Gate RPG offline -
teste privado')
 $image=Join-Path $destination 'SpiritGateRPG'
 foreach($file in Get-ChildItem -LiteralPath $build -Recurse
-File){if($file.IsReadOnly){$file.IsReadOnly=$false}}
-$notice="SPIRIT GATE RPG OFFLINE $Version`r`nExtraia todo o ZIP. Abra SpiritGateRPG.exe. Java integrado; sem
banco ou rede.`r`nWASD mover; F interagir/portal; SPACE atacar; Q habilidade; 1 pocao; F5 salvar.`r`nArena
historica continua disponivel separadamente. Arte geometrica provisoria.`r`nBuild usa classes da ultima suite
completa; execute qualidade.py antes de empacotar."
+$notice="SPIRIT GATE RPG OFFLINE $Version`r`nCandidato local: extraia todo o ZIP e abra SpiritGateRPG.exe. Java
 integrado; sem banco ou rede.`r`nWASD mover; F interagir/portal; SPACE atacar; Q habilidade; 1 pocao; F5
salvar.`r`nRacas escolhidas na criacao; dominio no nivel 15 e Shunpo no nivel 20. Arte geometrica
provisoria.`r`nNao ha compra de cash ou servidor de pagamento neste cliente local. Nenhum save pessoal foi
incluido."
 [IO.File]::WriteAllText((Join-Path $image 'LEIA-ME.txt'),$notice,(New-Object Text.UTF8Encoding($false)))
 Copy-Item -LiteralPath (Join-Path $project 'docs\ROTEIRO-RPG-OFFLINE.md') -Destination (Join-Path $image
'ROTEIRO-RPG-OFFLINE.md')
+Copy-Item -LiteralPath (Join-Path $project 'docs\SISTEMA-DE-RACAS-E-DOMINIOS.md') -Destination (Join-Path
$image 'RACAS-E-DOMINIOS.md')
+Copy-Item -LiteralPath (Join-Path $project 'docs\Spirit_Gate_Racas_e_Dominios.pdf') -Destination (Join-Path
$image 'RACAS-E-DOMINIOS.pdf')
+Copy-Item -LiteralPath (Join-Path $project 'server\target\launcher\build.json') -Destination (Join-Path $image
'source-build.json')
 $manifest=@(Get-ChildItem -LiteralPath $image -Recurse -File | Sort-Object FullName | ForEach-Object {
[ordered]@{path=$_.FullName.Substring($image.Length+1).Replace('\','/');bytes=$_.Length;sha256=(Get-FileHash
-LiteralPath $_.FullName -Algorithm SHA256).Hash.ToLowerInvariant()} })
 $zip=Join-Path $project ('dist\SpiritGate-RPG-'+$buildId+'.zip')
+New-Item -ItemType Directory -Path (Split-Path -Parent $zip) -Force | Out-Null
 Compress-Archive -LiteralPath $image -DestinationPath $zip -CompressionLevel Optimal
 $manifest | ConvertTo-Json -Depth 5 | Set-Content -LiteralPath ($zip+'.manifest.json') -Encoding UTF8
 (Get-FileHash -LiteralPath $zip -Algorithm SHA256).Hash.ToLowerInvariant() | Set-Content -LiteralPath
($zip+'.sha256') -Encoding ASCII
```

### 75. [M] tools/build-rpg-sprites-portable.ps1

```
diff --git a/tools/build-rpg-sprites-portable.ps1 b/tools/build-rpg-sprites-portable.ps1
index e829642cdd6deed95af2519554b8975b564af25a..a210219ea3c9474e42bf2db00c68de15f2a054de 100644
--- a/tools/build-rpg-sprites-portable.ps1
+++ b/tools/build-rpg-sprites-portable.ps1
@@ -1,21 +1,24 @@
-param([string]$Version='0.78.0-rc1')
+param([string]$Version='1.0.0-rc1')
 Set-StrictMode -Version Latest
 $ErrorActionPreference='Stop'
-if($Version -notmatch '^0[.]\d+[.]\d+-rc\d+$'){throw 'Versao invalida'}
+if($Version -notmatch '^\d+[.]\d+[.]\d+-rc\d+$'){throw 'Versao invalida'}
 $project=[IO.Path]::GetFullPath((Join-Path $PSScriptRoot '..'))
-$jdkBin=Split-Path (Get-Command java.exe).Source
+$compiled = & (Join-Path $PSScriptRoot 'compile-local.ps1') -PassThru
+$jdkBin=$compiled.JdkBin
 $buildId=$Version+'-'+[DateTime]::UtcNow.ToString('yyyyMMddTHHmmssZ')+'-'+[Guid]::NewGuid().ToString('N').Subst
ring(0,8)
 $build=Join-Path $project ('server\target\rpg-portable-'+$buildId)
 $inputDir=Join-Path $build 'input'
 $destination=Join-Path $build 'output'
 $runtimeDir=Join-Path $build 'runtime'
-$classes=Join-Path $project 'server\target\classes'
+$classes=$compiled.Classes
 New-Item -ItemType Directory -Path $inputDir,$destination | Out-Null
 function Checked([string]$tool,[string[]]$arguments){ & (Join-Path $jdkBin $tool) @arguments; if($LASTEXITCODE
-ne 0){throw "$tool falhou: $LASTEXITCODE"} }
 $jar=Join-Path $inputDir 'spirit-gate-rpg.jar'
 Checked 'jar.exe' @('--create','--file',$jar,'--main-class','com.projetok.prototype.rpg.RpgLauncher','-C',$clas
ses,'com/projetok/domain','-C',$classes,'com/projetok/prototype','-C',$classes,'com/projetok/rpg','-C',$classes,
'rpg')
 Checked 'jdeps.exe' @('--multi-release','17','--print-module-deps',$jar)
-Checked 'jlink.exe' @('--add-modules','java.base,java.desktop,java.logging,jdk.random','--strip-debug','--no-ma
n-pages','--no-header-files','--compress=2','--output',$runtimeDir)
+$modules = 'java.base,java.desktop,java.logging'
+if (@(& (Join-Path $jdkBin 'java.exe') --list-modules) -match '^jdk[.]random@') { $modules += ',jdk.random' }
+Checked 'jlink.exe' @('--add-modules',$modules,'--strip-debug','--no-man-pages','--no-header-files','--compress
=2','--output',$runtimeDir)
 $windowsVersion=($Version -split '-')[0]
 Checked 'jpackage.exe' @('--type','app-image','--dest',$destination,'--name','SpiritGateRPG','--input',$inputDi
r,'--main-jar','spirit-gate-rpg.jar','--main-class','com.projetok.prototype.rpg.RpgLauncher','--runtime-image',$
runtimeDir,'--app-version',$windowsVersion,'--vendor','Spirit Gate','--description','Spirit Gate RPG offline -
sprites experimentais','--java-options','-Dspiritgate.rpg.experimentalSprites=true')
 $image=Join-Path $destination 'SpiritGateRPG'
@@ -39,6 +42,7 @@ $notice="SPIRIT GATE RPG OFFLINE $Version`r`nExtraia todo o ZIP. Abra SpiritGate
 Copy-Item -LiteralPath (Join-Path $project 'docs\ROTEIRO-RPG-OFFLINE.md') -Destination (Join-Path $image
'ROTEIRO-RPG-OFFLINE.md')
 $manifest=@(Get-ChildItem -LiteralPath $image -Recurse -File | Sort-Object FullName | ForEach-Object {
[ordered]@{path=$_.FullName.Substring($image.Length+1).Replace('\','/');bytes=$_.Length;sha256=(Get-FileHash
-LiteralPath $_.FullName -Algorithm SHA256).Hash.ToLowerInvariant()} })
 $zip=Join-Path $project ('dist\SpiritGate-RPG-Sprites-Teste-'+$buildId+'.zip')
+New-Item -ItemType Directory -Path (Split-Path -Parent $zip) -Force | Out-Null
 Compress-Archive -LiteralPath $image -DestinationPath $zip -CompressionLevel Optimal
 $manifest | ConvertTo-Json -Depth 5 | Set-Content -LiteralPath ($zip+'.manifest.json') -Encoding UTF8
 (Get-FileHash -LiteralPath $zip -Algorithm SHA256).Hash.ToLowerInvariant() | Set-Content -LiteralPath
($zip+'.sha256') -Encoding ASCII
```

### 76. [M] tools/build-world-studio.ps1

```
diff --git a/tools/build-world-studio.ps1 b/tools/build-world-studio.ps1
index 625c0e4b548e5c2e3a970932ff4e6eb01974368f..7ed88ecfb4bf935067f03bcfe5a9bc4067a7c77d 100644
--- a/tools/build-world-studio.ps1
+++ b/tools/build-world-studio.ps1
@@ -3,15 +3,18 @@ Set-StrictMode -Version Latest
 $ErrorActionPreference='Stop'
 if($Version -notmatch '^0[.]\d+[.]\d+-rc\d+$'){throw 'Versao invalida'}
 $project=[IO.Path]::GetFullPath((Join-Path $PSScriptRoot '..'))
-$jdk=Split-Path (Get-Command java.exe).Source
+$compiled = & (Join-Path $PSScriptRoot 'compile-local.ps1') -PassThru
+$jdk=$compiled.JdkBin
 $buildId=$Version+'-'+[DateTime]::UtcNow.ToString('yyyyMMddTHHmmssZ')+'-'+[Guid]::NewGuid().ToString('N').Substring(0,8)
 $build=Join-Path $project ('server/target/studio-'+$buildId)
 $inputDir=Join-Path $build 'input'; $output=Join-Path $build 'output'; $runtime=Join-Path $build 'runtime'
 New-Item -ItemType Directory -Path $inputDir,$output | Out-Null
 function Checked([string]$tool,[string[]]$parameters){& (Join-Path $jdk $tool) @parameters;if($LASTEXITCODE -ne 0){throw "$tool falhou"}}
-$classes=Join-Path $project 'server/target/classes'
+$classes=$compiled.Classes
 Checked 'jar.exe' @('--create','--file',(Join-Path $inputDir 'world-studio.jar'),'--main-class','com.projetok.studio.WorldStudio','-C',$classes,'com/projetok/studio','-C',$classes,'com/projetok/domain','-C',$classes,'com/projetok/prototype','-C',$classes,'com/projetok/rpg','-C',$classes,'rpg')
-Checked 'jlink.exe' @('--add-modules','java.base,java.desktop,java.logging,jdk.random','--strip-debug','--no-man-pages','--no-header-files','--compress=2','--output',$runtime)
+$modules = 'java.base,java.desktop,java.logging'
+if (@(& (Join-Path $jdk 'java.exe') --list-modules) -match '^jdk[.]random@') { $modules += ',jdk.random' }
+Checked 'jlink.exe' @('--add-modules',$modules,'--strip-debug','--no-man-pages','--no-header-files','--compress=2','--output',$runtime)
 Checked 'jpackage.exe' @('--type','app-image','--dest',$output,'--name','SpiritGateWorldStudio','--input',$inputDir,'--main-jar','world-studio.jar','--main-class','com.projetok.studio.WorldStudio','--runtime-image',$runtime,'--app-version',($Version -split '-')[0],'--vendor','Spirit Gate')
 $image=Join-Path $output 'SpiritGateWorldStudio'
 foreach($file in Get-ChildItem -LiteralPath $build -Recurse -File){if($file.IsReadOnly){$file.IsReadOnly=$false}}
@@ -29,6 +32,7 @@ foreach($name in @('WORLD-STUDIO-TUTORIAL.md','WORLD-STUDIO-FUNCIONALIDADES.md',
 [IO.File]::WriteAllText((Join-Path $image 'LEIA-ME.txt'),"Spirit Gate World Studio $Version`r`nExtraia todo o ZIP e abra SpiritGateWorldStudio.exe. Java integrado, sem PostgreSQL ou multiplayer.`r`nAssets de trabalho experimentais; nenhuma aprovacao artistica presumida. Projetos novos em studio-projects. Manual em MANUAL.md. Exemplo em assets/maps/studio_tutorial.sgmap.",[Text.Encoding]::UTF8)

 $manifest=@(Get-ChildItem -LiteralPath $image -Recurse -File | Sort-Object FullName | ForEach-Object {[ordered]@{path=$_.FullName.Substring($image.Length+1).Replace('\','/');bytes=$_.Length;sha256=(Get-FileHash -LiteralPath $_.FullName -Algorithm SHA256).Hash.ToLowerInvariant()}})
 $zip=Join-Path $project ('dist/SpiritGate-WorldStudio-'+$buildId+'.zip')
+New-Item -ItemType Directory -Path (Split-Path -Parent $zip) -Force | Out-Null
 Compress-Archive -LiteralPath $image -DestinationPath $zip -CompressionLevel Optimal
 $manifest | ConvertTo-Json -Depth 5 | Set-Content -LiteralPath ($zip+'.manifest.json') -Encoding UTF8
 (Get-FileHash -LiteralPath $zip -Algorithm SHA256).Hash.ToLowerInvariant() | Set-Content -LiteralPath ($zip+'.sha256') -Encoding ASCII
```

### 77. [A] tools/compile-local.ps1

```
diff --git a/tools/compile-local.ps1 b/tools/compile-local.ps1
new file mode 100644
index 0000000000000000000000000000000000000000..7ff73bfc50b93c94b8402d81bb1c6967af684ab6
--- /dev/null
+++ b/tools/compile-local.ps1
@@ -0,0 +1,112 @@
+<# Compila o cliente desktop e o World Studio a partir das fontes locais, sem Maven ou rede. #>
+[CmdletBinding()]
+param([string]$JavaHome = '', [switch]$Force, [switch]$PassThru)
+
+Set-StrictMode -Version Latest
+$ErrorActionPreference = 'Stop'
+$project = [IO.Path]::GetFullPath((Join-Path $PSScriptRoot '..'))
+$buildRoot = Join-Path $project 'server\target\launcher'
+$classes = Join-Path $buildRoot 'classes'
+$sourceRoot = Join-Path $project 'server\src\main\java'
+$resourceRoot = Join-Path $project 'server\src\main\resources'
+
+function Assert-BuildPath([string]$path) {
+    $absolute = [IO.Path]::GetFullPath($path)
+    $prefix = [IO.Path]::GetFullPath($buildRoot).TrimEnd('\') + '\'
+    if (-not $absolute.StartsWith($prefix, [StringComparison]::OrdinalIgnoreCase)) {
+        throw "Saida de compilacao fora da pasta permitida: $absolute"
+    }
+    if (Test-Path -LiteralPath $absolute) {
+        if ((Get-Item -LiteralPath $absolute -Force).Attributes -band [IO.FileAttributes]::ReparsePoint) {
+            throw "Saida de compilacao nao pode ser um link: $absolute"
+        }
+    }
+    return $absolute
+}
+
+if (-not $JavaHome) { $JavaHome = $env:JAVA_HOME }
+if ($JavaHome) {
+    $jdkBin = Join-Path ([IO.Path]::GetFullPath($JavaHome)) 'bin'
+} else {
+    $compiler = Get-Command javac.exe -ErrorAction SilentlyContinue
+    if (-not $compiler) { throw 'Instale um JDK 17 ou superior e configure JAVA_HOME ou javac no PATH.' }
+    $jdkBin = Split-Path -Parent $compiler.Source
+}
+$java = Join-Path $jdkBin 'java.exe'
+$javac = Join-Path $jdkBin 'javac.exe'
+if (-not (Test-Path -LiteralPath $java -PathType Leaf) -or -not (Test-Path -LiteralPath $javac -PathType Leaf))
 {
+    throw "JDK incompleto em $jdkBin. Sao necessarios java.exe e javac.exe (17 ou superior)."
+}
+$compilerVersion = (& $javac --version | Out-String).Trim()
+if ($LASTEXITCODE -ne 0 -or $compilerVersion -notmatch '^javac (\d+)') { throw 'Nao foi possivel verificar a
versao do JDK.' }
+if ([int]$Matches[1] -lt 17) { throw 'O launcher exige um JDK 17 ou superior.' }
+
+$sources = @(Get-ChildItem -LiteralPath $sourceRoot -Recurse -File -Filter '*.java' | Sort-Object FullName)
+$resources = @(Get-ChildItem -LiteralPath $resourceRoot -Recurse -File | Sort-Object FullName)
+if (-not $sources.Count) { throw 'Nenhuma fonte Java encontrada.' }
+$fingerprintLines = @('compile-local-v2', '--release=17', $javac, $compilerVersion)
+foreach ($file in @($sources) + @($resources)) {
+    $relative = $file.FullName.Substring($project.Length + 1).Replace('\', '/')
+    $fingerprintLines += $relative + ':' + (Get-FileHash -LiteralPath $file.FullName -Algorithm SHA256).Hash
+}
+$sha = [Security.Cryptography.SHA256]::Create()
+try {
+    $fingerprint = [BitConverter]::ToString($sha.ComputeHash([Text.Encoding]::UTF8.GetBytes(($fingerprintLines
-join "`n")))).Replace('-', '').ToLowerInvariant()
+} finally { $sha.Dispose() }
+$manifestFile = Join-Path $buildRoot 'build.json'
+$needsBuild = $true
+if (-not $Force -and (Test-Path -LiteralPath $manifestFile -PathType Leaf) -and
+    (Test-Path -LiteralPath (Join-Path $classes 'com\projetok\prototype\rpg\RpgLauncher.class') -PathType
Leaf)) {
+    try {
+        $manifest = Get-Content -LiteralPath $manifestFile -Raw | ConvertFrom-Json
+        $needsBuild = $manifest.fingerprint -ne $fingerprint
+        if (-not $needsBuild) {
+            foreach ($entry in $manifest.files) {
+                $output = Assert-BuildPath (Join-Path $classes $entry.path)
+                if (-not (Test-Path -LiteralPath $output -PathType Leaf) -or
```

```
+                    (Get-FileHash -LiteralPath $output -Algorithm SHA256).Hash -ne $entry.sha256) {
+                    $needsBuild = $true
+                    break
+                }
+            }
+        }
+    } catch { $needsBuild = $true }
+}
+if ($needsBuild) {
+    New-Item -ItemType Directory -Path $buildRoot -Force | Out-Null
+    $staging = Assert-BuildPath (Join-Path $buildRoot ('compile-' + [Guid]::NewGuid().ToString('N')))
+    New-Item -ItemType Directory -Path $staging | Out-Null
+    $argumentsFile = Assert-BuildPath (Join-Path $buildRoot ('sources-' + [Guid]::NewGuid().ToString('N') +
'.args'))
+    try {
+        $sourceArguments = @($sources | ForEach-Object { '"' + $_.FullName.Replace('\', '/') + '"' })
+        [IO.File]::WriteAllLines($argumentsFile, $sourceArguments, (New-Object Text.UTF8Encoding($false)))
+        Write-Host "[Spirit Gate] Compilando $($sources.Count) fontes para Java 17..."
+        & $javac --release 17 -encoding UTF-8 -d $staging ('@' + $argumentsFile)
+        if ($LASTEXITCODE -ne 0) { throw 'Compilacao falhou; o launcher nao usara classes antigas.' }
+        foreach ($file in $resources) {
+            $destination = Join-Path $staging $file.FullName.Substring($resourceRoot.Length + 1)
+            New-Item -ItemType Directory -Path (Split-Path -Parent $destination) -Force | Out-Null
+            Copy-Item -LiteralPath $file.FullName -Destination $destination
+        }
+        $inventory = @(Get-ChildItem -LiteralPath $staging -Recurse -File | Sort-Object FullName |
ForEach-Object {
+            [ordered]@{ path = $_.FullName.Substring($staging.Length + 1).Replace('\', '/'); sha256 =
(Get-FileHash -LiteralPath $_.FullName -Algorithm SHA256).Hash }
+        })
+        $previous = Assert-BuildPath (Join-Path $buildRoot ('previous-' + [Guid]::NewGuid().ToString('N')))
+        $safeClasses = Assert-BuildPath $classes
+        if (Test-Path -LiteralPath $safeClasses) { Move-Item -LiteralPath $safeClasses -Destination $previous }
+        try { Move-Item -LiteralPath $staging -Destination $safeClasses }
+        catch {
+            if (Test-Path -LiteralPath $previous) { Move-Item -LiteralPath $previous -Destination $safeClasses
}
+            throw
+        }
+        if (Test-Path -LiteralPath $previous) { Remove-Item -LiteralPath (Assert-BuildPath $previous) -Recurse
-Force }
+        $manifest = [ordered]@{ schemaVersion = 1; fingerprint = $fingerprint; java = $compilerVersion; sources
 = $sources.Count; files = $inventory }
+        [IO.File]::WriteAllText($manifestFile, ($manifest | ConvertTo-Json -Depth 5), (New-Object
Text.UTF8Encoding($false)))
+    } finally {
+        if (Test-Path -LiteralPath $argumentsFile) { Remove-Item -LiteralPath (Assert-BuildPath $argumentsFile)
 -Force }
+        if (Test-Path -LiteralPath $staging) { Remove-Item -LiteralPath (Assert-BuildPath $staging) -Recurse
-Force }
+    }
+}
+if ($PassThru) {
+    [pscustomobject]@{ Classes = $classes; Java = $java; JdkBin = $jdkBin; Fingerprint = $fingerprint; Rebuilt
= $needsBuild }
+} else { Write-Host "[Spirit Gate] Cliente atualizado: $classes" }
```

### 78. [A] tools/gerar_pdf_dominios.py

```
diff --git a/tools/gerar_pdf_dominios.py b/tools/gerar_pdf_dominios.py
new file mode 100644
index 0000000000000000000000000000000000000000..920e11d7354ce59b0f0ea090adb153ca81a6ac4c
--- /dev/null
+++ b/tools/gerar_pdf_dominios.py
@@ -0,0 +1,100 @@
+"""Gera o PDF das decisões de raças e domínios a partir do Markdown correspondente."""
+from pathlib import Path
+import re
+from xml.sax.saxutils import escape
+
+from reportlab.lib import colors
+from reportlab.lib.enums import TA_LEFT
+from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
+from reportlab.lib.pagesizes import A4
+from reportlab.platypus import (
+    SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle, PageBreak,
+)
+
+ROOT = Path(__file__).resolve().parents[1]
+SOURCE = ROOT / 'docs/SISTEMA-DE-RACAS-E-DOMINIOS.md'
+OUTPUT = ROOT / 'docs/Spirit_Gate_Racas_e_Dominios.pdf'
+INK = colors.HexColor('#18283B')
+ACCENT = colors.HexColor('#167A87')
+
+
+def generate():
+    styles = getSampleStyleSheet()
+    styles.add(ParagraphStyle(name='BodySG', fontName='Helvetica', fontSize=9.4,
+                              leading=13.4, textColor=INK, spaceAfter=7))
+    styles.add(ParagraphStyle(name='TitleSG', parent=styles['BodySG'],
+                              fontName='Helvetica-Bold', fontSize=31, leading=36,
+                              spaceAfter=13))
+    styles.add(ParagraphStyle(name='HeadingSG', parent=styles['BodySG'],
+                              fontName='Helvetica-Bold', fontSize=18, leading=23,
+                              textColor=ACCENT, spaceAfter=13))
+    styles.add(ParagraphStyle(name='SubheadingSG', parent=styles['BodySG'],
+                              fontName='Helvetica-Bold', fontSize=11, leading=15,
+                              spaceBefore=7, spaceAfter=6, keepWithNext=True))
+    styles.add(ParagraphStyle(name='CellSG', parent=styles['BodySG'],
+                              fontSize=8.6, leading=11.6, spaceAfter=0))
+    styles.add(ParagraphStyle(name='HeaderCellSG', parent=styles['CellSG'],
+                              fontName='Helvetica-Bold', textColor=colors.white))
+    story = []
+    lines = SOURCE.read_text(encoding='utf-8').splitlines()
+    i = 0
+    while i < len(lines):
+        line = lines[i].strip()
+        i += 1
+        if not line:
+            continue
+        if line == '---':
+            story.append(PageBreak())
+        elif line.startswith('|'):
+            rows = [line]
+            while i < len(lines) and lines[i].strip().startswith('|'):
+                rows.append(lines[i].strip())
+                i += 1
+            cells = [[c.strip() for c in row.strip('|').split('|')] for row in rows]
+            cells = [row for row in cells if not all(re.fullmatch(r':?-+:?', cell) for cell in row)]
+            data = [[Paragraph(escape(cell), styles['HeaderCellSG' if n == 0 else 'CellSG'])
+                     for cell in row] for n, row in enumerate(cells)]
+            widths = [110, 389] if len(cells[0]) == 2 else [68, 210, 221]
+            table = Table(data, colWidths=widths, repeatRows=1, hAlign='LEFT')
+            table.setStyle(TableStyle([
+                ('BACKGROUND', (0, 0), (-1, 0), INK),
+                ('ROWBACKGROUNDS', (0, 1), (-1, -1),
+                 [colors.HexColor('#EDF4F5'), colors.HexColor('#F7F9FB')]),
+                ('VALIGN', (0, 0), (-1, -1), 'TOP'),
+                ('LEFTPADDING', (0, 0), (-1, -1), 8),
+                ('RIGHTPADDING', (0, 0), (-1, -1), 8),
+                ('TOPPADDING', (0, 0), (-1, -1), 7),
+                ('BOTTOMPADDING', (0, 0), (-1, -1), 7),
+            ]))
+            story.extend([table, Spacer(1, 9)])
+        elif line.startswith('# '):
```

```
+            story.append(Paragraph(escape(line[2:]), styles['TitleSG']))
+        elif line.startswith('## '):
+            story.append(Paragraph(escape(line[3:]), styles['HeadingSG']))
+        elif line.startswith('### '):
+            story.append(Paragraph(escape(line[4:]), styles['SubheadingSG']))
+        else:
+            story.append(Paragraph(escape(line), styles['BodySG']))
+
+    def decorate(canvas, doc):
+        canvas.saveState()
+        canvas.setStrokeColor(ACCENT)
+        canvas.setLineWidth(2)
+        canvas.line(48, 806, 547, 806)
+        canvas.setFont('Helvetica', 8)
+        canvas.setFillColor(INK)
+        canvas.drawString(48, 819, 'SPIRIT GATE  /  DOCUMENTO DE DESIGN')
+        canvas.drawString(48, 27, 'Raças, estilos e domínios • v1.1 • 21/10/2026')
+        canvas.drawRightString(547, 27, str(doc.page))
+        canvas.restoreState()
+
+    doc = SimpleDocTemplate(str(OUTPUT), pagesize=A4, rightMargin=48,
+                            leftMargin=48, topMargin=52, bottomMargin=48,
+                            title='Spirit Gate — Raças, estilos de combate e domínios',
+                            author='Spirit Gate', subject='Decisões de design — versão 1.1')
+    doc.build(story, onFirstPage=decorate, onLaterPages=decorate)
+    print(f'PDF gerado: {OUTPUT} ({OUTPUT.stat().st_size} bytes)')
+
+
+if __name__ == '__main__':
+    generate()
```

### 79. [A] tools/launch-local.ps1

```
diff --git a/tools/launch-local.ps1 b/tools/launch-local.ps1
new file mode 100644
index 0000000000000000000000000000000000000000..744c3d81b0fab45a3937e88c9310eb4fab0a2a23
--- /dev/null
+++ b/tools/launch-local.ps1
@@ -0,0 +1,40 @@
+[CmdletBinding()]
+param(
+    [ValidateSet('Rpg', 'Sprites', 'Studio', 'Map')][string]$Mode = 'Rpg',
+    [string]$MapFolder = '',
+    [switch]$Geometric,
+    [switch]$BuildOnly
+)
+Set-StrictMode -Version Latest
+$ErrorActionPreference = 'Stop'
+try {
+    $project = [IO.Path]::GetFullPath((Join-Path $PSScriptRoot '..'))
+    Set-Location -LiteralPath $project
+    $build = & (Join-Path $PSScriptRoot 'compile-local.ps1') -PassThru
+    if ($BuildOnly) { Write-Host '[Spirit Gate] Compilacao concluida.'; exit 0 }
+    $javaArguments = @()
+    switch ($Mode) {
+        'Studio' { $entry = 'com.projetok.studio.WorldStudio'; $clientArguments = @('--project', $project) }
+        'Map' {
+            if (-not $MapFolder -or -not (Test-Path -LiteralPath $MapFolder -PathType Container)) {
+                throw 'Informe a pasta de mapa exportada: testar-mapa-rpg.cmd "pasta-exportada"'
+            }
+            $entry = 'com.projetok.studio.StudioPlaytest'
+            $clientArguments = @([IO.Path]::GetFullPath($MapFolder), $project)
+        }
+        default {
+            $entry = 'com.projetok.prototype.rpg.RpgLauncher'
+            $clientArguments = @()
+            if ($Mode -eq 'Sprites' -and -not $Geometric) {
+                $scale = if ($env:SPIRIT_GATE_SPRITE_SCALE) { $env:SPIRIT_GATE_SPRITE_SCALE } else { '0.5' }
+                $frame = if ($env:SPIRIT_GATE_FRAME_MS) { $env:SPIRIT_GATE_FRAME_MS } else { '250' }
+                $javaArguments += @("-Dspiritgate.rpg.spriteScale=$scale",
"-Dspiritgate.rpg.spriteFrameMillis=$frame")
+                $clientArguments = @('--experimental-player-sprites')
+            }
+        }
+    }
+    Write-Host '[Spirit Gate] Abrindo cliente...'
+    & $build.Java @javaArguments -cp $build.Classes $entry @clientArguments
+    if ($LASTEXITCODE -ne 0) { throw "O cliente terminou com codigo $LASTEXITCODE." }
```

```
+    exit 0
+} catch { Write-Host "[Spirit Gate] ERRO: $($_.Exception.Message)" -ForegroundColor Red; exit 1 }
```

### 80. [A] tools/run-local-tests.py

```
diff --git a/tools/run-local-tests.py b/tools/run-local-tests.py
new file mode 100644
index 0000000000000000000000000000000000000000..cd058e45966311137e4a94e213ffc73a94865971
--- /dev/null
+++ b/tools/run-local-tests.py
@@ -0,0 +1,137 @@
+"""Compile and run the Java suite with cached JUnit, without Maven or network access."""
+from __future__ import annotations
+
+import os
+from pathlib import Path
+import shutil
+import subprocess
+import sys
+import tempfile
+import zipfile
+
+ROOT = Path(__file__).resolve().parents[1]
+CACHE = Path.home() / ".m2" / "repository"
+ARTIFACTS = (
+    "org/junit/jupiter/junit-jupiter-api/5.11.4/junit-jupiter-api-5.11.4.jar",
+    "org/junit/jupiter/junit-jupiter-engine/5.11.4/junit-jupiter-engine-5.11.4.jar",
+    "org/junit/jupiter/junit-jupiter-params/5.11.4/junit-jupiter-params-5.11.4.jar",
+    "org/junit/platform/junit-platform-commons/1.11.4/junit-platform-commons-1.11.4.jar",
+    "org/junit/platform/junit-platform-engine/1.11.4/junit-platform-engine-1.11.4.jar",
+    "org/opentest4j/opentest4j/1.3.0/opentest4j-1.3.0.jar",
+    "org/apiguardian/apiguardian-api/1.1.2/apiguardian-api-1.1.2.jar",
+)
+
+RUNNER = r'''
+import java.util.*;
+import org.junit.jupiter.engine.JupiterTestEngine;
+import org.junit.platform.engine.*;
+import org.junit.platform.engine.discovery.DiscoverySelectors;
+
+public final class LocalJunitRunner {
+    public static void main(String[] args) {
+        var configuration = new ConfigurationParameters() {
+            public Optional<String> get(String key) { return Optional.empty(); }
+            public Optional<Boolean> getBoolean(String key) { return Optional.empty(); }
+            public int size() { return 0; }
+            public Set<String> keySet() { return Set.of(); }
+        };
+        var selector = DiscoverySelectors.selectPackage(args.length == 0 ? "com.projetok" : args[0]);
+        var request = new EngineDiscoveryRequest() {
+            public <T extends DiscoverySelector> List<T> getSelectorsByType(Class<T> type) {
+                return type.isInstance(selector) ? List.of(type.cast(selector)) : List.of();
+            }
+            public <T extends DiscoveryFilter<?>> List<T> getFiltersByType(Class<T> type) { return List.of(); }
+            public ConfigurationParameters getConfigurationParameters() { return configuration; }
+        };
+        var engine = new JupiterTestEngine();
+        var descriptor = engine.discover(request, UniqueId.forEngine(engine.getId()));
+        int[] results = new int[4];
+        var listener = new EngineExecutionListener() {
+            public void executionFinished(TestDescriptor test, TestExecutionResult result) {
+                if (test.isTest()) results[0]++;
+                switch (result.getStatus()) {
+                    case SUCCESSFUL -> { if(test.isTest()) results[1]++; }
+                    case ABORTED -> { if(test.isTest()) results[2]++; }
+                    case FAILED -> {
+                        results[3]++;
+                        System.err.println("FAILED: " + test.getUniqueId());
+                        result.getThrowable().ifPresent(error -> {
+                            System.err.println(error);
+                            var trace=error.getStackTrace();
+                            for(int i=0;i<Math.min(8,trace.length);i++) System.err.println("  at " + trace[i]);
+                            if(error.getCause()!=null) System.err.println("  caused by: " + error.getCause());
+                        });
+                    }
+                }
+            }
+            public void executionSkipped(TestDescriptor test, String reason) {
```

```
+                if(test.isTest()) results[2]++;
+            }
+        };
+        engine.execute(new ExecutionRequest(descriptor, listener, configuration));
+        System.out.printf("JUnit: %d executed, %d passed, %d skipped/aborted, %d failures%n",
+            results[0], results[1], results[2], results[3]);
+        System.exit(results[3] == 0 && results[0] > 0 ? 0 : 1);
+    }
+}
+'''
+
+
+def main() -> int:
+    java = shutil.which("java")
+    javac = shutil.which("javac")
+    if not java or not javac:
+        raise RuntimeError("Java e javac 17+ devem estar no PATH.")
+    jars = [CACHE / artifact for artifact in ARTIFACTS]
+    missing = [str(jar) for jar in jars if not jar.is_file()]
+    if missing:
+        raise RuntimeError("JUnit nao esta no cache local:\n" + "\n".join(missing))
+
+    build = ROOT / "server" / "target" / "local-tests"
+    build.mkdir(parents=True, exist_ok=True)
+    run = Path(tempfile.mkdtemp(prefix="run-", dir=build))
+    dependencies = run / "dependencies"
+    dependencies.mkdir()
+    classpath = []
+    # Directory classpaths avoid Windows sandbox ZipFS failures seen when javac opens cached JARs.
+    for jar in jars:
+        destination = dependencies / jar.stem
+        destination.mkdir()
+        with zipfile.ZipFile(jar) as archive:
+            for member in archive.infolist():
+                target = (destination / member.filename).resolve()
+                if not target.is_relative_to(destination.resolve()):
+                    raise RuntimeError(f"Caminho invalido no JAR: {member.filename}")
+            archive.extractall(destination)
+        classpath.append(str(destination))
+
+    classes = run / "classes"
+    classes.mkdir()
+    runner = run / "LocalJunitRunner.java"
+    runner.write_text(RUNNER, encoding="utf-8")
+    sources = sorted((ROOT / "server" / "src" / "main" / "java").rglob("*.java"))
+    tests = sorted((ROOT / "server" / "src" / "test" / "java").rglob("*.java"))
+    arguments = run / "sources.args"
+    arguments.write_text("\n".join('"' + path.as_posix() + '"' for path in sources + tests + [runner]),
encoding="utf-8")
+    print(f"Compilando {len(sources)} fontes e {len(tests)} arquivos de teste para Java 17...", flush=True)
+    compiled = subprocess.run([javac, "--release", "17", "-encoding", "UTF-8", "-cp",
os.pathsep.join(classpath), "-d", str(classes), "@" + str(arguments)], cwd=ROOT)
+    if compiled.returncode:
+        return compiled.returncode
+    shutil.copytree(ROOT / "server" / "src" / "main" / "resources", classes, dirs_exist_ok=True)
+    print(f"Resultados e classes: {run}", flush=True)
+    with (run / "test-results.log").open("w", encoding="utf-8") as log:
+        process = subprocess.Popen([java, "-Djava.awt.headless=true", "-cp", os.pathsep.join([str(classes)] +
classpath), "LocalJunitRunner", *sys.argv[1:]], cwd=ROOT / "server", stdout=subprocess.PIPE,
stderr=subprocess.STDOUT, text=True, encoding="utf-8", errors="replace")
+        assert process.stdout is not None
+        for line in process.stdout:
+            print(line, end="", flush=True)
+            log.write(line)
+            log.flush()
+        return process.wait()
+
+
+if __name__ == "__main__":
+    try:
+        raise SystemExit(main())
+    except (OSError, RuntimeError) as error:
+        print(error, file=sys.stderr)
+        raise SystemExit(1)
```

### 81. [M] world-studio.cmd

```
diff --git a/world-studio.cmd b/world-studio.cmd
index fb63bb135d1d277e0fe3f057c455a990f993497b..014f0319f76676d2f56b11cb299fc29e809feac1 100644
--- a/world-studio.cmd
+++ b/world-studio.cmd
@@ -1,8 +1,4 @@
 @echo off
 setlocal
-cd /d "%~dp0"
-if not exist "server\target\classes\com\projetok\studio\WorldStudio.class" (
-  echo Compile primeiro: python tools/qualidade.py --offline --task studio-local
-  exit /b 1
-)
-java -cp "server\target\classes" com.projetok.studio.WorldStudio --project "%~dp0."
+powershell.exe -NoProfile -ExecutionPolicy Bypass -File "%~dp0tools\launch-local.ps1" -Mode Studio %*
+exit /b %ERRORLEVEL%
```